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Chapter 1

Introduction to latent variable models

1.1 Context

Remark 1. Latent variable models provide an important tool/framework for the analysis of multivariate data, within which many methods can be unified.

Important remark 1 (Reason introduction). Reasons why a latent variable should be introduced into a model

1. **reduce dimensionality**: large-scale statistical enquires generate much more information than can be easily absorbed. Idea (like in *factor analysis*) is to reduce the number of variable while keeping the information

2. in several discipline exists the concept of **not observable/latent variables**, due to difference reasons: examples are theoretical constructs not observable/measurable (intelligence, satisfaction racial prejudice) or measurement errors (age asked to a female, salary).

Important remark 2 (Aim of the analysis). Our main aim will be to:

1. **identify the latent factors** that explain the relationships among observed variables, by fitting a specific latent variable model.
   The aim is to find the *smallest* number q (for parsimony) for which model 1.4 is *adequate* (has a goodness of fit to the data).[1]

2. **scale individuals** on the latent factor space (posterior mean or component score), that is obtain scores for latent/unobserved variables for the individuals

Important remark 3 (Main aim).

[1] Eg as other statistical model, once fitted we look at residuals we check if model assumptions are met; here we look at residuals and check if the number of latent variable should be increased.
The starting point is setting $q = 1$ and then we increase the number of latent variables up to when it's adequate.

		Manifest variables	
		Continuous	Categorical
Latent variables	Continuous	Factor analysis	Latent trait analysis
	Categorical	Latent profile analysis	Latent class analysis

Table 1.1: Proper methods for latent and observed variable

Important remark 4 (Prerequisite of the analysis). **Condition needed** to perform this kind of analysis is that the *observed variables are associated/correlated*; if they're not, it means there are no latent variables underlying the observed ones that "moves" these latters. So the first step in this kind of analysis is to assess matrix of correlation/association before moving on.

Important remark 5 (Classification of/proper latent variable models). In table 1.1 latent variable models are classified according to the nature of latent and observed/manifest variables

- factor analysis was the first model developed/reputted in this framework;
- in case categorical variables are manifest of continuous underlying ones, the proper model to apply is latent trait;
- in the contrary case latent profile analysis is used but we don't see it in this course (less used)
- Finally latent class analysis is used when both observed and latent variable are categorical (and the idea is to profile units based on latent variables).

1.2 Theoretical framework

1.2.1 Notation and framework

Definition 1.2.1 (Observed and latent variables)**.** We denote as:

- $x_1, x_2, \ldots, x_p$, the p manifest/**observed variables**[2]
- $y_1, y_2, \ldots, y_q$ the q **latent variables**

Remark 2. Differently from regression models,

- here both manifest and latent variables are considered as *random variables*. In regression model the only random variable are the y and x are considered fixed;
- we know the number of manifest variables p but don't know the number q of latent variables;

Remark 3. Being random variables, the relationship between them must be expressed in terms of *probability distributions*. We use some tools/idea of bayesian statistics (because this simplifies)

[2] Also said items/indicators

Definition 1.2.2 (Distribution of interest)**.** Let R_x, R_y be the *range spaces* of $\mathbf{x}$ and $\mathbf{y}$; we use:

- h to denote *prior distribution* (pmf or pdf) of the latent variables:

$$h(\mathbf{y}), \quad \mathbf{y} = (y_1, y_2, \ldots, y_q) \in R_y$$

- g to denote *conditional distribution* (pmf or pdf) of the observed variables $\mathbf{x}$ given the latent variables $\mathbf{y}$:

$$g(\mathbf{x}|\mathbf{y}), \quad \mathbf{x} = (x_1, x_2, \ldots, x_p) \in R_x$$

  We have that

$$g(\mathbf{x}|\mathbf{y}) = g(x_1, \ldots, x_p|\mathbf{y})$$

  This is a joint distribution of the x, conditioned to the value assumed by the latent variables $\mathbf{y}$.

  Example 1.2.1. If the x are two categorical variables we are thinking a 3d bar-diagram, one for each covariate pattern of the latent variable.

- f as the *joint distribution* (pmf of pdf) of the observed and latent variables $\mathbf{x}$ and $\mathbf{y}$, can be written as product of the previous two from probability theory:

$$f(\mathbf{x}, \mathbf{y}) = h(\mathbf{y})g(\mathbf{x}|\mathbf{y})$$

- since $\mathbf{y}$ is not observable, only $\mathbf{x}$ can be observed; so any inference must be based on the *joint marginal distribution* of $\mathbf{x}$ that can be written as

$$f(\mathbf{x}) = \int_{R_y} f(\mathbf{x}, \mathbf{y}) = \int_{R_y} h(\mathbf{y})g(\mathbf{x}|\mathbf{y})d\mathbf{y} \tag{1.1}$$

  In the formula above we just connect what we observe with what we don't observe. Note that if $q = 2$ (we have two latent variables) the integrals above are double integrals, for $q = 3$ triple etc.

Important remark 6 (Toward posterior of y with Bayes' theorem). Our aim is to obtain information on $\mathbf{y}$ after $\mathbf{x}$ has been observed $h(\mathbf{y}|\mathbf{x})$, *the posterior distribution* of the latent variables given the observed variables. To derive it we can use the Bayes' theorem:

$$h(\mathbf{y}|\mathbf{x}) = \frac{h(\mathbf{y})g(\mathbf{x}|\mathbf{y})}{f(\mathbf{x})} \tag{1.2}$$

Important remark 7. Thus

- in order to find $h(\mathbf{y}|\mathbf{x})$ we need to estimate $f(\mathbf{x})$, $g(\mathbf{x}|\mathbf{y})$ and $h(\mathbf{y})$;
- however while $f(\mathbf{x})$ can be estimated from the observed variables, $h(\mathbf{y})$ and $g(\mathbf{x}|\mathbf{y})$ are not uniquely determined (h and g could be any distribution);
- so to avoid this problem we *we must introduce further restrictions* on the classes of functions considered, that allows us to estimate h, g.

1.2.2 Assumptions

1.2.2.1 Assumption of conditional independence

Remark 4. The first assumption we do is the **local/conditional independence** of the conditional distribution/likelihood $g(\mathbf{x}|\mathbf{y})$

Important remark 8 (Conditional independence and distribution of x given y). If the dependencies among the observed x's are induced by a set of latent variables (working behind the scene), a reasonable assumption is that when all y's are accounted for (eg if all the y's are held fixed) the x's will be independent.
So assuming q latent variable exists/are chosen, we have that the joint distribution is the product of the distributions/marginals for each single observed variable x_i:

$$g(\mathbf{x}|\mathbf{y}) = g(x_1, \dots, x_p|\mathbf{y}) = \prod_{i=1}^{p} g_i(x_i|\mathbf{y}) \tag{1.3}$$

The assumption means that $\mathbf{y}$ is sufficient to explain the dependencies among the $\mathbf{x}$ variables.
It allows us to work with marginal simpler x_i's conditional distribution instead of joint $g(\mathbf{x}|\mathbf{y})$

Important remark 9 (Marginal distribution of $\mathbf{x}$). Under conditional independence of $\mathbf{x}$ given $\mathbf{y}$, if we replace equation 1.3 in equation 1.1, the marginal distribution of $\mathbf{x}$ becomes:

$$f(\mathbf{x}) = \int_{R_y} h(\mathbf{y}) g(\mathbf{x}|\mathbf{y}) d\mathbf{y} = \int_{R_y} \prod_{i=1}^{p} h(\mathbf{y}) g_i(x_i|\mathbf{y}) d\mathbf{y} \tag{1.4}$$

for some small value of q.

1.2.2.2 Assumption of General Linear Latent Variable Model (GLLVM)

Remark 5. In order to obtain the posterior distribution of $h(\mathbf{y}|\mathbf{x})$, aside from conditional independence of $g(\mathbf{x}|\mathbf{y})$, we make some further assumptions regarding each conditional distribution $g_i(x_i|\mathbf{y})$: we assume a generalized linear model for each x_i given $\mathbf{y}$ [3].

Definition 1.2.3 (General linear latent variable model (GLLVM))**.** For the conditional distribution in equation (1.3) we assume a generalized linear model with the usual *three components*:

1. *Random component*: for each conditional distribution $g_i(x_i|\mathbf{y})$, $i = 1, \dots, p$, a convenient family of distributions is the *one-parameter exponential family*[4]

$$g_i(x_i|\mathbf{y}) = g_i(x_i|\theta_i) = F_i(x_i) G_i(\theta_i) \exp(\theta_i u_i(x_i)), \quad i = 1, \dots, p \tag{1.5}$$

   Note that above:

[3] since in general here the observed variables x_i, here as explained variable, are not necessarily normal and so is convenient to assume/adopt a GLM

[4] Note that there are various formulation of the exponential family; check con gli appunti di Galimberti

- since the $\mathbf{y}$ is considered fixed in conditioning, the only random variable is the x_i; we assume that each x_i is conditionally distributed according to a one-parameter (θ) exponential family. The exponential family (1.5) includes the Bernoulli, poisson, normal and gamma distributions as special cases; if we allow x_i and θ to be vector-valued it also includes the multinomial distribution.
- we can assume a different distribution for each x_i: eg if x is binary a binomial, a count the poisson distribution and so on

2. *Systematic component*: the latent variables $y_1, \ldots, y_q$ produce a linear predictor η_i given by:

$$\eta_i = \alpha_{i0} + \alpha_{i1} y_1 + \ldots + \alpha_{iq} y_q$$

Here, differently from GLM, we have not 1 but as many linear predictor as the observed variables x_i ($i = 1, \ldots, p$), each composed by the linear combination of covariates/latent variables with its set of coefficients $\alpha_{i0}, \alpha_{i1}, \ldots, \alpha_{iq}$;

3. A *link function* between the systematic component and the conditional mean θ_i of random one: this relates the latent variables y's to θ_i. Simplest form we can assume (for continuous variables, where *identity function* is used as link) is: θ_i is:

$$\theta_i = \alpha_{i0} + \alpha_{i1} y_1 + \ldots + \alpha_{iq} y_q$$

More in general, any monotonic differentiable function (eg logit, log etc) can be assumed as a link between the systematic component and the conditional mean of the random component.

Important remark 10 (Naming). What above is the *general(ized) linear latent variable model*:

- *latent* because of y's which acts as covariate are actually not observed
- *linear* here refers to its linearity in the α's

Important remark 11 (Diff vs GLM). The main differences with the classical generalized linear model (GLM) are two:

1. there's a set of dependent variables $x_1, \ldots, x_p$ (while in the classical GLM only one);
2. the covariates $y_1, \ldots, y_q$ are unobservable.

1.2.2.3 Properties of the GLLVM

By replacing the assumptions made so far in Bayes thm equation (1.2) we get that the posterior $h(\mathbf{y}|\mathbf{x})$ can be rewritten/simplified as follows:x

$$\begin{aligned} h(\mathbf{y}|\mathbf{x}) &= \frac{h(\mathbf{y}) g(\mathbf{x}|\mathbf{y})}{f(\mathbf{x})} = \frac{h(\mathbf{y}) \prod_{i=1}^{p} g_i(x_i|\mathbf{y})}{f(\mathbf{x})} \\ &= \frac{h(\mathbf{y}) [\prod_{i=1}^{p} F_i(x_i) G_i(\theta_i)] \cdot \exp\left(\sum_{i=1}^{p} u_i(x_i) \cdot \theta_i\right)}{f(\mathbf{x})} \end{aligned}$$

Ignoring the denominator [5] and collecting all the terms that do not depend on $\mathbf{y}$ (putting them in a proportionality constant) we obtain

$$
\begin{aligned}
h(\mathbf{y}|\mathbf{x}) &\propto h(\mathbf{y}) \left[\prod_{i=1}^{p} G_i(\theta_i)\right] \exp\left(\sum_{i=1}^{p} u_i(x_i)\cdot\theta_i\right) \\
&\overset{(1)}{\propto} h(\mathbf{y}) \left[\prod_{i=1}^{p} G_i(\theta_i)\right] \exp\left[\sum_{i=1}^{p} u_i(x_i)\left(\alpha_{i0}+\sum_{j=1}^{q}\alpha_{ij}y_j\right)\right] \\
&\overset{(2)}{\propto} h(\mathbf{y}) \left[\prod_{i=1}^{p} G_i(\theta_i)\right] \exp\left[\sum_{i=1}^{p} u_i(x_i)\sum_{j=1}^{q}\alpha_{ij}y_j\right] \\
&\propto h(\mathbf{y}) \left[\prod_{i=1}^{p} G_i(\theta_i)\right] \exp\left[\sum_{j=1}^{q} y_j \sum_{i=1}^{p}\alpha_{ij}\cdot u_i(x_i)\right] \\
&\propto h(\mathbf{y}) \left[\prod_{i=1}^{p} G_i(\theta_i)\right] \exp\left(\sum_{j=1}^{q} y_j X_j\right)
\end{aligned}
\tag{1.6}
$$

where

- in (1) we did the substitution $\theta_i = \alpha_{i0} + \sum_{j=1}^{q} a_{ij}y_y$ which holds in the simplest case (identity link, I guess for illustration purposes)
- in (2) we put α_{i0} in the proportionality constant (does not depends on $\mathbf{y}$)
- we named

$$X_j = \sum_{i=1}^{p}\alpha_{ij}\cdot u_i(x_i) \quad (j=1,\ldots,q)$$

  a linear combination of the observed variables (a function $u_i(x_i)$ of the xs, which in simplest case can be just x_i) and some coefficients a_{ij}; these are very similars to components of the principal components analysis

We simplified a lot the posterior distribution $h(\mathbf{y}|\mathbf{x})$ which is now $\propto$:

- the prior distribution of $\mathbf{y}$
- the mid term depending on thetas/
- a linear combination of unobserved and observed variables (last term) on a reduction of the observed data (the X_j)

Sufficiency principle It can be proved that X_j is a sufficient statistics[6] for y_j (we can see the number of them coincides and is q). We have that:

- $h(\mathbf{y}|\mathbf{x})$ depends on $\mathbf{x}$ only through the q-dimensional vector $\mathbf{X} = (X_1, X_2, \ldots, X_q)$. The elements of $\mathbf{X}$ are called *components*; they play a similar role to principal components.

[5] It's just a normalization constant that does not change shape, just the scale, and we're interested in shape of distribution eg for optimization

[6] Summarizes all the information of the data about a parameter/useful for inference

- $\mathbf{X}$ is a *minimal sufficient statistic* for $\mathbf{y}$: it represents a reduction in dimensionality from the p-vector $\mathbf{x}$ to the q-vector $\mathbf{X}$.
- equation (1.6) does not depend on the prior distribution of the latent variables $h(\mathbf{y})$; whichever is chosen, $\mathbf{X}$ represents the best reduction of information on the posterior distribution of the latent variables.

1.2.3 Example: Bernoulli random variable

We see the simpler model of GLLVM where all observed variables x_i are binary (items for questionaire with 1=agree, 0=disagree). We can assume these are bernoulli-distributed variable with pi_i the probability of agreeing (success).
The conditional distribution of x_i is, formulated[7] in terms of an exponential family, is

$$g(x_i|\mathbf{y}) = \pi_i^{x_i}(1-\pi_i)^{1-x_i} = (1-\pi_i)\exp(x_i \text{logit}(\pi_i))$$

Regarding the exponential family components, we have that:

$$\begin{aligned} \theta_i &= \text{logit}(\pi_i) = \log[\frac{\pi_i}{1-\pi_i}] \\ G_i(\theta_i) &= 1-\pi_i \\ u_i(x_i) &= x_i \end{aligned}$$

The GLLVM is a system of following p equations

$$\text{logit}(\pi_i) = \alpha_{i0} + \sum_{j=1}^{q} \alpha_{ij} y_j$$

with logit being the *link function*. The components are the linear combinations

$$X_j = \sum_{i=1}^{p} \alpha_{ij} x_i$$

if we can estimate α_{ij} we can compute this quantities which are sufficient statistics for y_j

Example 1.2.2. If $q = 1$ (one latent variable) we obtain the logistic latent trait model, which by doing calculation can be expressed with *response function*:

$$\pi_i(y) = \frac{1}{(1+\exp(-\alpha_{i0} - \alpha_{i1} y))}$$

here we expressed the probability of a positive response on the i-th observed variable in terms of the latent variable and the alpha coefficients/weights

[7]

$$\begin{aligned} g(x_i|\mathbf{y}) &= \pi_i^{x_i}(1-\pi_i)^{1-x_i} = \exp\left[\log\left(\pi_i^{x_i}(1-\pi_i)^{1-x_i}\right)\right] = \exp\left[\log \pi_i^{x_i} + \log(1-\pi_i)^{1-x_i}\right] \\ &= \exp\left[x_i \log \pi_i + (1-x_i)\log(1-\pi_i)\right] = \exp\left[x_i \log \pi_i + \log(1-\pi_i) - x_i \log(1-\pi_i)\right] \\ &= \exp\left[x_i \log \frac{\pi_i}{1-\pi_i} + \log(1-\pi_i)\right] = (1-\pi_i)\exp\left[x_i \log \frac{\pi_i}{1-\pi_i}\right] \\ &= (1-\pi_i)\exp\left[x_i \text{logit}(\pi_i)\right] \end{aligned}$$

Remark 6. This was just the specification of the model; now we need to introduce the tools for estimating the model, that is estimating the alphas, and thus obtaining and estimates of the latent variables (y_j through X_j, I guess).

1.2.4 Estimation

In this framework we can consider two approaches:

1. Maximum likelihood estimation using EM (expectation maximization) algorithm
2. Bayesian methods

We'll refer to the *first one* (used by software packages we'll use), seen in detail for the specific models.

1.3 Rotation and interpretation

Two more general themes before going through details of the models.

1.3.1 Rotation

We actually don't know which kind of variable are the $\mathbf{y}$ (eg normal, binomial etc).However if $\mathbf{y}$ is continuous variable:

- **no latent variable model is unique**. I could choose different prior $h(\mathbf{y})$ and end up with the same conclusion/ end up with the same joint distribution $f(\mathbf{x})$ of the observed variables

$$f(\mathbf{x}) = \int_{\mathbb{R}^q} \prod_{i=1}^{p} h(\mathbf{y}) g(x_i|\mathbf{y}) \, \mathrm{d}\mathbf{y}$$

- this occurs because any transformation of the latent variable generally changes both $h(\cdot)$ and $g(\cdot)$ but (this double change elides and) leaves $f(\mathbf{x})$ unchanged
- the irrelevance of the prior for final results, is a pro: I can choose the easiest one. Usually what is done is that, if $\mathbf{y}$ is a continuous variable, to assume it's *normal*.
- another pro is that I can choose a trasformation of the prior such that it makes the latent variable estimates to rotate. *Rotation* has an important role in the interpretation of latent variables.

NB: this should be an example on why changing the prior does not change the posterior

Example 1.3.1 (Logit model for binary data)**.** If in the logit models which relates the probability of success on x_i to the only latent variable y ...

$$\text{logit}(\pi_i(y)) = \alpha_{i0} + \alpha_{i1} y, \qquad i = 1, 2, \ldots, p$$

we can assume that the prior $h(y)$ is normal (assuming y is quantitative too btw).
If we make the transform $z = H(y)$ where H is the cumulative distribution

function of the prior distribution (normal), then (instead a model with prior $h(y)$) we have one with a uniform prior on (0,1) and response function (where the same transformation is applied to the logit):

$$\text{logit}(\pi_i(z)) = \alpha_{i0} + \alpha_{i1} H^{-1}(z), \qquad i = 1, 2, \ldots, p$$

Both models lead to exactly the same distribution of x's.

Remark 7. There are two types of rotations.

Definition 1.3.1 (Orthogonal rotations)**.** Transformations that leave the form of the prior distribution and of the natural parameter unchanged.

Example 1.3.2. An example of orthogonal transformation is $y = -z$.

Example 1.3.3. Consider the following GLLVM where the linear predictor in matrix form is generated by $\mathbf{A}$ (alpha/loadings matrix) and $\mathbf{y}$ (think $\text{logit}(\pi_i) = \alpha_i y$ in system/matrix form) I can decide to apply a (non-singular) transformation of latent variables $\mathbf{y}$ into $\mathbf{z}$ through an orthogonal[8] matrix $\mathbf{M}$ ($\mathbf{M}\mathbf{M}' = \mathbf{I}$):

$$\begin{aligned} \boldsymbol{\theta} &= \mathbf{A}\mathbf{y} \\ \mathbf{z} &= \mathbf{M}\mathbf{y} \end{aligned}$$

then we have that

$$\begin{aligned} \mathbf{y} &= \mathbf{M}^{-1}\mathbf{z} \\ \boldsymbol{\theta} &= \mathbf{A}\mathbf{M}^{-1}\mathbf{z} \end{aligned}$$

Now

- if $\mathbf{y} \sim \mathrm{N}(\mathbf{0}, \mathbf{I})$ then $\mathbf{z} \sim \mathrm{N}\left(\mathbf{0}, \mathbf{M}\mathbf{M}'\right)$, since being a linear combination of multivariate normal is normal, with expected value $\mathbf{0}$ and for variance

$$\text{Var}[\mathbf{y}] = \text{Var}[\mathbf{M}\mathbf{y}] = \mathbf{M}\underbrace{\mathbf{y}\mathbf{y}^{\mathsf{T}}}_{\mathbf{I}}\mathbf{M}^{\mathsf{T}} = \mathbf{M}\mathbf{M}^{\mathsf{T}}$$

- if the transformation is orthogonal ($\mathbf{M}\mathbf{M}' = \mathbf{I}$) then $\mathbf{y}$ and $\mathbf{z}$ have the same joint distribution $\mathrm{N}(\mathbf{0}, \mathbf{I})$.

Thus we cannot distinguish between a GLLVM with *loadings matrix* $\mathbf{A}$ and one with $\mathbf{A}\mathbf{M}'$.

Remark 8. So from the practical point of view I can obtain different matrixes of loadings which are all correct, by applying any orthogonal rotation; thus I can choose among them the one that is most useful for interpretation/simpler. purposes.

Definition 1.3.2 (Oblique rotation)**.** If we assume that $\mathbf{y}$ is normal but not standard normal and has correlation (varcov is not $\mathbf{I}$), that is $\mathbf{y} \sim N(\mathbf{0}, \boldsymbol{\Phi})$, any non-singular transformation (not necessarily orthogonal) would produce $\mathbf{z} \sim N(\mathbf{0}, \mathbf{M}\boldsymbol{\Phi}\mathbf{M}') \Rightarrow$ which is distributed differently from $\mathbf{z}$

[8] matrix needs to be orthogonal for the transformation to be orthogonal as well

1.3.2 Interpretation

Interpretation of the latent variables is basically *naming the latent variables* that are computed (eg think latent trait in sociology). There are two approaches:

- we interpret the **matrix of the loadings A**[9] for which (with rotations) we aim at a **simple structure** like the following, where + means large positive loadings and . small loadings:

$$\begin{bmatrix} + & . & . \\ + & . & . \\ + & . & . \\ . & + & . \\ . & + & . \\ . & . & + \\ . & . & + \\ . & . & + \\ . & . & + \end{bmatrix}$$

  We have one observed variable per row and one latent variable per column: we want the latent variable to influence separatedly different observed variables. The interpretation of what are the latent variables then becomes easier.

- we use the loadings to compute and look at **components** (sufficient statistics for the latent variables); the j-th component is

$$X_j = \sum_{i=1}^{p} \alpha_{ij} u_i(x_i)$$

  However due to rotations that can change α_{ij}, the set of sufficient statistics $\mathbf{X}$ is not unique. Thus

  1. I may rotate until I find a simple matrix of loadings and then use its coefficients to generate this
  2. interpretation is I guess better with matrix of loadings instead of numeric variables per se
  3. however we can use the components (which interpretation is now clear) for further analyses (since in the end we probably done a dimension reduction from the original variables to the latent ones)

[9] From the tradition of the factor analysis α_{ij} are known as *factor loadings*, which express the load of latent y on observed x

Chapter 2

Normal linear factor model

Remark 9. The linear factor model (NLFM) is the oldest and most widely used latent variable model (Spearman, 1904, which wanted to study intelligence). In the 1980's Bartholomew put it in the GLM-like framework.

2.1 Introduction

We assume that

- both observed and latent are multivariate normal variables
- specifically the observed value are normal and correlated[1], that is

$$f(\mathbf{x}) \sim \mathrm{N_p}\left(\boldsymbol{\mu}, \boldsymbol{\Sigma}\right)$$

and ask whether the multivariate normal distribution admits a representation in terms of $h(\mathbf{y})$ and $g(\mathbf{x}|\mathbf{y})$. We start considering the linear factor model in the regression-form specification (classical model by Spearman)

$$\underset{p\times 1}{\mathbf{x}} = \underset{p\times 1}{\boldsymbol{\mu}} + \underset{p\times q}{\boldsymbol{\Lambda}}\,\underset{q\times 1}{\mathbf{y}} + \underset{p\times 1}{\mathbf{e}} \tag{2.1}$$

where the observed variable are put in a model like framework with

- $\boldsymbol{\mu}$ a vector of mean
- $\boldsymbol{\Lambda}$ is the $p \times q$ matrix of the *factor loadings* which multiplies
- vector of latent[2] variables $\mathbf{y} \sim \mathrm{N_q}\left(\mathbf{0}, \mathbf{I}\right)$
- $\mathbf{e} \sim \mathrm{N_p}\left(\mathbf{0}, \boldsymbol{\Psi}\right)$, is an error term (one for each observed variable) with $\boldsymbol{\Psi}$ being the $p \times p$ *diagonal* matrix of *specific variances*[3] ($\mathrm{Var}\left[e_1\right] = \psi_{11}$, no

[1] Correlation is a requirement, to be checked, and expression of the underlying operation of latent variables. If not correlated they can't be expression of common underlying variables.

[2] Only difference wrt to regression model is that here $\mathbf{y}$ is not considered fixed but a random variable with distribution

[3] Specific for, one for, each observed variable

correlation between errors similar to classical regression model);

$$\mathbf{\Psi} = \begin{bmatrix} \psi_{11} & 0 & \dots & 0 \\ 0 & \psi_{22} & \dots & 0 \\ \dots & & & \\ 0 & 0 & \dots & \psi_{pp} \end{bmatrix}$$

- $\mathbf{y}$ and $\mathbf{e}$ independent

This above is the classical factor model; but an equivalent[4] way of writing the model in probabilistic form is by assuming the following:

$$\mathbf{y} \sim \mathrm{N_q}\left(\mathbf{0}, \mathbf{I}\right) \tag{2.2}$$
$$\mathbf{x}|\mathbf{y} \sim \mathrm{N_p}\left(\boldsymbol{\mu} + \mathbf{\Lambda}\mathbf{y}, \mathbf{\Psi}\right) \tag{2.3}$$

We've rewrited it in terms of:

- prior distribution $h(\mathbf{y})$ (first equation) which is standard normal (for simplicity, since prior is actually irrelevant for estimation)
- conditional distribution $g(\mathbf{x}|\mathbf{y})$ (second equation), normal again[5], and thus belonging to the exponential family. Applying the assumption of latent models (indipendence especially) we have that

$$g(\mathbf{x}|\mathbf{y}) = \prod_{i=1}^{p} g_i(x_i|\mathbf{y})$$

where marginal conditional distribution of x_i is normal

$$g_i(x_i|\mathbf{y}) = \frac{1}{\sqrt{2\pi\psi_{ii}}} \exp\left[-\frac{1}{2}\frac{(x_i - (\mu_i + \sum_j \lambda_{ij} y_j))^2}{\psi_{ii}}\right] \tag{2.4}$$

the linear predictor/systematic component (relate the expected value of the observed values to the latent variables) is

$$\eta_i = \mu_i + \sum_j \lambda_{ij} y_j \quad i = 1, \dots, p \tag{2.5}$$

where μ_i has role of intercept (prediction with null latent variables).
Replacing (2.5) in (2.4) we can see that the normal distribution is member of the exponential family:

$$\begin{aligned} g_i(x_i|\mathbf{y}) &= \frac{1}{\sqrt{2\pi\psi_{ii}}} \exp\left[-\frac{1}{2}\frac{(x_i - \eta_i)^2}{\psi_{ii}}\right] \\ &= \frac{1}{\sqrt{2\pi\psi_{ii}}} \exp\left[-\frac{1}{2}\left(\frac{x_i^2}{\psi_{ii}} - \frac{2x_i\eta_i}{\psi_{ii}} + \frac{\eta_i^2}{\psi_{ii}}\right)\right] \end{aligned} \tag{2.6}$$

[4] Whether we write the model in terms of equation in random variables as in eq 2.1, or as probability distributions (eqs 2.3/2.2) is equivalent/a matter of taste.

[5] looking at 2.1 once conditioned on $\mathbf{y}$, assuming it's constant, $\mathbf{x}$ is a linear combination of constant and $\mathbf{e}$ which is normal, so it's normal.
Furthermore, conditioning on $\mathbf{y}$ it can be considered constant and thus

$$\mathbb{E}\left[\mathbf{x}|\mathbf{y}\right] = \mathbb{E}\left[\boldsymbol{\mu} + \mathbf{\Lambda}\mathbf{y} + \mathbf{e}\right] = \boldsymbol{\mu} + \mathbf{\Lambda}\mathbf{y} + \mathbb{E}\left[\mathbf{e}\right] = \boldsymbol{\mu} + \mathbf{\Lambda}\mathbf{y}$$
$$\mathrm{Var}\left[\mathbf{x}|\mathbf{y}\right] = \mathrm{Var}\left[\boldsymbol{\mu} + \mathbf{\Lambda}\mathbf{y} + \mathbf{e}\right] = \mathrm{Var}\left[\mathbf{e}\right] = \mathbf{\Psi}$$

This last is in the form of the exponential family with components:

$$\begin{aligned}\theta &= \eta_i/\sqrt{\psi_{ii}} \\ u_i(x_i) &= x_i/\sqrt{\psi_{ii}} \\ G_i(\theta_i) &= \exp\left(-\frac{1}{2}\eta_i^2/\psi_{ii}\right) \\ F_i(x_i) &= \frac{1}{\sqrt{2\pi\psi_{ii}}}\exp\left(-\frac{1}{2}x_i^2/\psi_{ii}\right)\end{aligned}$$

The sufficient statistics are

$$X_j = \sum_i \lambda_{ij} u_u(x_i) = \sum_i \lambda_{ij} x_i/\sqrt{\psi_{ii}}, \qquad j = 1, \ldots, q$$

so once estimated lambdas and psis we can get good computable proxies of the $\mathbf{y}$ by using above formula (once estimated lambda and psis). In matrix form, sufficient statistics are:

$$\mathbf{X} = \mathbf{\Lambda}^\mathsf{T}\mathbf{\Psi}^{-1}\mathbf{x}$$

From eq 2.3 and 2.2, it then follows that the *unconditional* distribution of the observed variables (at the denominator of Bayes) is still normal and particularly[6]:

$$\mathbf{x} \sim \mathrm{N_p}\left(\boldsymbol{\mu}, \mathbf{\Sigma}\right), \quad \text{with } \mathbf{\Sigma} = \mathbf{\Lambda}\mathbf{\Lambda}^\mathsf{T} + \mathbf{\Psi}$$

Finally (and only in the normal case) it's possible to obtain an equation for the posterior distribution of the latent variables:

$$\mathbf{y}|\mathbf{x} \sim \mathrm{N_q}\left(\mathbf{\Lambda}^\mathsf{T}(\mathbf{\Lambda}\mathbf{\Lambda}^\mathsf{T} + \mathbf{\Psi})^{-1}(\mathbf{x} - \boldsymbol{\mu}), (\mathbf{\Lambda}^\mathsf{T}\mathbf{\Psi}^{-1}\mathbf{\Lambda} + \mathbf{I})^{-1}\right)$$

that is still normal (it's a combination of normal distributions) with

- posterior mean depends on the parameters of the model ($\mathbf{\Lambda}$ and $\boldsymbol{\mu}$)
- posterior variance/covariance $(\mathbf{\Lambda}^\mathsf{T}\mathbf{\Psi}^{-1}\mathbf{\Lambda} + \mathbf{I})^{-1}$ is generally not diagonal so $\mathbf{y}$ are correlated

This is the only case in which we have a formal expressio for the posterior distribution of the latent variables; it is incidentally useful for estimation because one of the possible way of estimating the $\mathbf{y}$ is to take the expected value of the posterior distribution (in this case other than calculating the components X_j we can take the expected value $\mathbf{\Lambda}^\mathsf{T}(\mathbf{\Lambda}\mathbf{\Lambda}^\mathsf{T} + \mathbf{\Psi})^{-1}(\mathbf{x} - \boldsymbol{\mu})$ of the distribution above for having an estimate of $\mathbf{y}$).

[6]Normal because if we dont condition the sum is between a linear combination of normals $\mathbf{y}$ and other normals $\mathbf{e}$; regarding then

$$\mathbb{E}\left[\mathbf{x}\right] = \mathbb{E}\left[\boldsymbol{\mu} + \mathbf{\Lambda}\mathbf{y} + \mathbf{e}\right] = \boldsymbol{\mu}$$

$$\mathrm{Var}\left[\mathbf{x}\right] = \mathrm{Var}\left[\boldsymbol{\mu} + \mathbf{\Lambda}\mathbf{y} + \mathbf{e}\right] = \mathrm{Var}\left[\mathbf{\Lambda}\mathbf{y} + \mathbf{e}\right] = \mathrm{Var}\left[\mathbf{\Lambda}\mathbf{y}\right] + \mathrm{Var}\left[\mathbf{e}\right] = \mathbf{\Lambda}\underbrace{\mathbf{y}\mathbf{y}^\mathsf{T}}_{\mathbf{I}}\mathbf{\Lambda}^\mathsf{T} + \mathbf{\Psi} = \mathbf{\Lambda}\mathbf{\Lambda}^T + \mathbf{\Psi}$$

2.2 Properties

Some properties which are convenient in interpreting the factor model. From (2.1) it follows:

- the variance of each observed variable x_i (which we can view from the data) can be decomposed in a sum of two components (in the matrix form think the diagonal of $\mathbf{\Lambda\Lambda}^\mathsf{T} + \mathbf{\Psi}$)

$$\mathrm{Var}\left[x_i\right] = \sum_{j=1}^{q} \lambda_{ij}^2 + \psi_i$$

  which are called

  - *communality* $\sum_{j=1}^{q} \lambda_{ij}^2$ (part of the variance of x that depends/explained by factors/latent variables)
  - variance *specific to* observed variable x_i, that is ψ_i (variance coming from the error term)

  What one would expect is that most part of the variance is explained by the common factors for all the observed variables and psi-s are very small.

- about covariances between observed variables $\mathbf{x}$, looking at distribution of $\mathbf{x} \sim \mathrm{N_p}\left(\boldsymbol{\mu}, \mathbf{\Lambda\Lambda}^\mathsf{T} + \mathbf{\Psi}\right)$, the covariance of observed variable is *assumed to depend only* on Λ (that is on latent factor contribution to the variable via the loadings) and nothing else, since $\mathbf{\Psi}$ is diagonal (and does not contribute to covariances).
  So latent variable explain all the association between observed variables: this is another way to express the conditional independence. My model is a good model if all the correlations/elements out of main diagonal are explained by the latent variables through the loadings. This is an assumption; if the decomposition a posteriori is not that good the model isn't either

- the covariance between observed and latent variables, $\mathbf{x}$ and $\mathbf{y}$, is

$$\begin{aligned}\mathrm{Cov}\left(\mathbf{x}, \mathbf{y}\right) &= \mathbb{E}\left[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{y} - \mathbf{0})^\mathsf{T}\right] = \mathbb{E}\left[(\mathbf{x} - \boldsymbol{\mu})\mathbf{y}^\mathsf{T}\right] \\ &\overset{(1)}{=} \mathbb{E}\left[\mathbb{E}\left[(\mathbf{x} - \boldsymbol{\mu})|\mathbf{y}\right]\mathbf{y}^\mathsf{T}\right] = \mathbb{E}\left[\mathbb{E}\left[\mathbf{\Lambda}\boldsymbol{y} + \mathbf{e}\right]\mathbf{y}^\mathsf{T}\right] = \mathbb{E}\left[\mathbf{\Lambda}\mathbf{y}\mathbf{y}^\mathsf{T}\right] \\ &= \mathbf{\Lambda}\mathbf{I} = \mathbf{\Lambda}\end{aligned}$$

  where in (1) we used the law of iterated expectation: it's convenient to condition on $\mathbf{y}$, and not having both variables as random variables, in order to use results from the model which we're assuming holds ($(\mathbf{x}-\boldsymbol{\mu})|\mathbf{y} = \mathbf{\Lambda}\boldsymbol{y}+\mathbf{e}$)
  So the factor loadings matrix $\mathbf{\Lambda}$ can be interpreted as covariances between the manifest variables and latent variables (it's a $p \times q$ matrix, just covariances between the two sets of variables, not variances).

- the correlations between $\mathbf{x}$ and $\mathbf{y}$ are obtained dividing covariance for standard deviations, that is by premultiplying the diagonal of $\mathbf{\Sigma}$ (putting its element to denominator under square root):

$$\left(\mathbf{Diag}(\mathbf{\Sigma})\right)^{-1/2}\mathbf{\Lambda} \quad \left(= \frac{covariance}{\sigma}\right)$$

- we can even derive covariance between the components and the factors

$$\begin{aligned}\text{Cov}(\mathbf{X},\mathbf{y}) &= \mathbb{E}\left[(\mathbf{X}-\mathbb{E}[\mathbf{X}])(\mathbf{y}-\mathbf{0})^{\mathsf{T}}\right] = \mathbb{E}\left[(\mathbf{X}-\mathbb{E}[\mathbf{X}])\mathbf{y}^{\mathsf{T}}\right] \\ &\overset{(1)}{=} \mathbb{E}\left[(\boldsymbol{\Lambda}^{\mathsf{T}}\boldsymbol{\Psi}^{-1}\mathbf{x}-\mathbb{E}\left[\boldsymbol{\Lambda}^{\mathsf{T}}\boldsymbol{\Psi}^{-1}\mathbf{x}\right])\mathbf{y}^{\mathsf{T}}\right] \\ &= \mathbb{E}\left[(\boldsymbol{\Lambda}^{\mathsf{T}}\boldsymbol{\Psi}^{-1}\mathbf{x}-\boldsymbol{\Lambda}^{\mathsf{T}}\boldsymbol{\Psi}^{-1}\mathbb{E}[\mathbf{x}])\mathbf{y}^{\mathsf{T}}\right] \\ &\overset{(2)}{=} \boldsymbol{\Lambda}^{\mathsf{T}}\boldsymbol{\Psi}^{-1}\underbrace{\mathbb{E}\left[(\mathbf{x}-\boldsymbol{\mu})\mathbf{y}^{\mathsf{T}}\right]}_{\text{Cov}(\mathbf{x},\mathbf{y})} \\ &= \boldsymbol{\Lambda}^{\mathsf{T}}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}\end{aligned}$$

  where in

  - (1) I use the matrix definition of the components in the factor model ($\mathbf{X} = \boldsymbol{\Lambda}^{\mathsf{T}}\boldsymbol{\Psi}^{-1}\mathbf{x}$)
  - (2) collect constants out of expectation and note that the covariance between observed and latent variables

  Some considerations:

  - If $\boldsymbol{\Lambda}^{\mathsf{T}}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$ is diagonal then latent variable and the components are uncorrelated.
  - we note that the variance/covariance of posterior distribution of $\mathbf{y}|\mathbf{x}$ depends on this matrix $\boldsymbol{\Lambda}^{\mathsf{T}}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$ (+ the term $\mathbf{I}$). So if this matrix is diagonal, we assume that a posteriori the $\mathbf{y}$ given $\mathbf{x}$ are uncorrelated between them. In general it's not diagonal and there's correlation

2.3 Constraints on the model

2.3.1 Introduction

Going back in the general framework we said that GLLVMs with $\mathbf{y}$ continuous have the property that we can transform $\mathbf{y}$ and we get always the same solution (uninformative prior): any one-to-one transformation from $\mathbf{y}$ to $\mathbf{v}$ has no effect on $f(\mathbf{x})$. In particular, both the functions h and g will be changed but $f(\mathbf{x})$ does not change.
The indeterminacy of h leave us free to adopt a distribution for $\mathbf{y}$ such that h has a convenient form; the more convenient is the standard normal distribution. Among the transformation if we apply orthogonal rotation[7] to $\mathbf{y}$ we end up with the same solution. Considering the orthogonal rotation $\mathbf{v} = \mathbf{M}\mathbf{y}$ (that is with $\mathbf{M}$ orthogonal matrix $\mathbf{M}^{\mathsf{T}}\mathbf{M} = \mathbf{I}$), it follows that being $\mathbf{y}$ normal the rotated latent variables are still standard normal

$$\mathbf{v} = \mathbf{M}\mathbf{y} \sim \text{N}\left(\mathbf{0}, \mathbf{M}\mathbf{M}^{\mathsf{T}}\right) = \text{N}(\mathbf{0}, \mathbf{I})$$

In terms of conditional distribution what happens if we rotate the latent variable? Starting from

$$\mathbf{x}|\mathbf{y} \sim \text{N}_{\text{p}}(\boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{y}, \boldsymbol{\Psi})$$

[7] Again, rotation has an important role in the interpretation of latent variables.

if we rotate with $\mathbf{v} = \mathbf{M}\mathbf{y}$ then we can rewrite it as $\mathbf{M}^\mathsf{T}\mathbf{v} = \mathbf{y}$ and plugging in we have

$$\mathbf{x}|\mathbf{v} \sim \mathrm{N_p}\left(\boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{M}^\mathsf{T}\mathbf{v}, \boldsymbol{\Psi}\right)$$

In practice what we're doing is a rotation of the matrix of factor loadings as well; this model is undistinguishable/equivalent to the first one.
This can be also seen as follows:

NB: non c'era nelle sue slides, commentato nel LaTeX

- regarding the model in regression form we have

$$\mathbf{x} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{M}^\mathsf{T}\mathbf{v} + \mathbf{e} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{M}^\mathsf{T}\mathbf{M}\boldsymbol{\Lambda}^\mathsf{T} + \mathbf{e} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\mathsf{T} + \mathbf{e}$$

- previously we proved that if the linear factor models holds, the covariance matrix of the observed variable can be decomposed as $\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\mathsf{T} + \boldsymbol{\Psi}$. What becomes the variance of $\mathbf{x}$, $\boldsymbol{\Sigma}$, in this case if we assume the model with rotated latent variables $\boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{M}^\mathsf{T}\mathbf{v} + \mathbf{e}$ holds? By squaring the constant $\Lambda\mathbf{M}^\mathsf{T}$ and adding variance of $\mathbf{e}$ we have

$$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\mathbf{M}^\mathsf{T}\mathbf{M}\boldsymbol{\Lambda}^\mathsf{T} + \boldsymbol{\Psi} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\mathsf{T} + \boldsymbol{\Psi}$$

  So we obtain the very same decomposition of the first model; thus we cannot distinguish between a model with loading matrix $\boldsymbol{\Lambda}$ and one with $\boldsymbol{\Lambda}\mathbf{M}^\mathsf{T}$.

Since we have infinite orthogonal matrix we can use, we have infinite indistinguishible solutions that will give the same result. This property is convenient for us because we can choose the matrix that we want and in particular that makes $\boldsymbol{\Lambda}$ easy to interpret.

2.3.2 Constraints for real

There are however cases in which I would like to remove the indeterminancy because for example I want a unique solution; I do this by placing constraints on the parameters of the model, that is on $\boldsymbol{\Lambda}$ and $\boldsymbol{\Psi}$. Some constraints possible:

- if $\boldsymbol{\Lambda}^\mathsf{T}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$ (in the variance of posterior $\mathbf{y}|\mathbf{x}$) is diagonal (thus by imposing all elements out of main diagonal to be 0) then $\mathbf{y}$'s will be a posteriori uncorrelated (and being normal, independent).
  - the relation between $\mathbf{y}$'s and components will be particularly simple;
  - for some application it may be good/reasonable/desiderable to have independent latent variables, in some other no;
  - this constraint removes the freedom to arbitrarily rotate $\boldsymbol{\Lambda}$ (by pre-multiplying by $\mathbf{M}$, since then $\boldsymbol{\Lambda}^\mathsf{T}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda}$ is not diagonal anymore);
- in **confirmatory factor analysis**, which is *a factor model in which we know the structure* of $\boldsymbol{\Lambda}$, we constrain some values of $\boldsymbol{\Lambda}$ to zero (eg we know for sure that some variables we have are not affected by one of the latent variables, we fix some zeros there/do not estimate them)[8]; this usually remove the rotation freedom as well

[8] On the other hand the classical factor model where we leave the lambdas to vary as needed is called exploratory factor analysis because we're exploring the factors underlying the data and do not impose any constraint on the parameter.

- if we standardize $\mathbf{x}$ (which removes the arbitrariness of the scale of the manifest variables) the new standard scale will be reflected in the covariances and hence in the parameter estimates and their interpretation. Standardizing the $\mathbf{x}$'s has the effect to make all the diagonal elements of $\boldsymbol{\Sigma}$ equal to one and hence the variance decomposition becomes constrainted as below

$$\sum_{j=1}^{q} \lambda_{ij}^2 + \psi_i = 1$$

  and thus the lambdas estimated accordingly (can't be chosen freely but the communality now have to sum up with individual variance to 1)

Remark 10. Rotation is something we do with exploratory factor models: this is useful because if I do not know anything about my model, I can rotate until I find a good solution. But if we know a priori something about the relation between observed and latent variable, in that case rotation is not anymore a good solution because we want a specific solution, so one put constraint and fit its unique solution
We have several possibility of constraints; what we use depends on the kind of analysis (if it's preliminary or if we already know something).
Typically with big data the analysis adopted is exploratory because one does not know anything about my data and want to reduce them as much as possible

2.4 Estimation methods

We need to estimate parameter of the models and then end up to estimation of the latent variables (using components or posterior distribution of $\mathbf{y}|\mathbf{x}$).
We've seen that parameter to be estimated are $\boldsymbol{\mu}$, $\boldsymbol{\Lambda}$ and $\boldsymbol{\Psi}$ that are unknown (similarly to regression model where we estimate intercepts, beta and variance of the error term).
We have that:

- different estimation methods are available, based on different assumptions on the observed data;

- all methods search for estimates such that the *discrepancy criterion* between the observed covariance matrix $\mathbf{S}$ (only information available from the data[9]) and the theoretical matrix implied by the model

$$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^{\mathsf{T}} + \boldsymbol{\Psi}$$

  is as smaller as possible. So basically search for Λ and Ψ such that then $\boldsymbol{\Sigma}$ is then closest to $\mathbf{S}$.
  Differences among estimation methods depends on the different *discrepancy measures* between observed $\mathbf{S}$ and the theoretical covariance matrix $\boldsymbol{\Sigma}$.

[9] Or better, we actually don't use individual information but start already with a covariance matrix.

- for linear factor model we see the *maximum likelihood method* and *maximum likelihood method by the EM algorithm*. Both are based on the normality assumption of the observed variables.[10]. Good things is that other model we'll see will be estimable with EM as well.

2.4.1 Maximum likelihood estimation

If by assumption the data are normally distributed (in population), that is $\mathbf{x} \sim \mathrm{N_p}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and given a random sample $\mathbf{x}_1, \ldots, \mathbf{x}_n$, likelihood can be written as follows:

$$L(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \left[(2\pi)^{p/2}|\boldsymbol{\Sigma}|^{1/2}\right]^{-n} \exp\left[-1/2 \sum_{h=1}^{n} (\mathbf{x}_h - \boldsymbol{\mu})^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} (\mathbf{x}_h - \boldsymbol{\mu})\right]$$

We're interested (among other things) in expressing this formula in terms of $\mathbf{S}$ and $\boldsymbol{\Sigma}$. Log-likelihood becomes

$$l = \log L = -\frac{n}{2} \log\left[(2\pi)^p |\boldsymbol{\Sigma}|\right] - \frac{1}{2} \sum_{h=1}^{n} (\mathbf{x}_h - \boldsymbol{\mu})^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} (\mathbf{x}_h - \boldsymbol{\mu}) \tag{2.7}$$

$$= -\frac{n}{2} \log\left[(2\pi)^p\right] - \frac{n}{2} \log|\boldsymbol{\Sigma}| - \frac{1}{2} \sum_{h=1}^{n} (\mathbf{x}_h - \boldsymbol{\mu})^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} (\mathbf{x}_h - \boldsymbol{\mu}) \tag{2.8}$$

$$\overset{(1)}{=} \text{constant} + \frac{n}{2} \log|\boldsymbol{\Sigma}|^{-1} - \frac{n}{2} \operatorname{trace}\left[\boldsymbol{\Sigma}^{-1}\mathbf{S}\right] \tag{2.9}$$

$$\overset{(2)}{=} \text{constant} + \frac{n}{2} \left[\log|\boldsymbol{\Sigma}^{-1}| - \operatorname{trace}\left(\boldsymbol{\Sigma}^{-1}\mathbf{S}\right)\right] \tag{2.10}$$

with $\mathbf{S} = \sum_{h=1}^{n} (\mathbf{x}_h - \boldsymbol{\mu})(\mathbf{x}_h - \boldsymbol{\mu})^{\mathsf{T}}/n$. where

- in (1)
  - the quantity $\sum_{h=1}^{n} (\mathbf{x}_h - \boldsymbol{\mu})^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} (\mathbf{x}_h - \boldsymbol{\mu})$ is actually a scalar; this because $(\mathbf{x}_h - \boldsymbol{\mu})^{\mathsf{T}}$ is $1 \times p$, $\boldsymbol{\Sigma}^{-1}$ is $p \times p$, while $(\mathbf{x}_h - \boldsymbol{\mu})$ is $p \times 1$.
  - remembering properties of the trace (the last holding if dimension of permutations allow product)

$$\begin{aligned}
&\operatorname{trace}(c) = c, \quad c \in \mathbb{R} \\
&\operatorname{trace}(c\mathbf{A}) = c \operatorname{trace}(\mathbf{A}), \quad c \in \mathbb{R} \\
&\operatorname{trace}(\mathbf{A} + \mathbf{B}) = \operatorname{trace}(\mathbf{A}) + \operatorname{trace}(\mathbf{B}) \\
&\operatorname{trace}(\mathbf{XYZ}) = \operatorname{trace}(\mathbf{ZXY}) = \operatorname{trace}(\mathbf{YZX})
\end{aligned}$$

    we substitute $\sum_{h=1}^{n} (\mathbf{x}_h - \boldsymbol{\mu})^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} (\mathbf{x}_h - \boldsymbol{\mu})$ with its trace (are the same, being a constant) and rewrite the trace of the sum above using

[10] Other estimation methods *in absence of normality* of the observed variables are *least squares methods* and the *principal axis method* (see Bartholomew, Knott, Moustaki, 2011).

trace's properties:

$$\begin{aligned}\sum_{h=1}^{n}(\mathbf{x}_h-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{x}_h-\boldsymbol{\mu}) &= \text{trace}\left[\sum_{h=1}^{n}(\mathbf{x}_h-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{x}_h-\boldsymbol{\mu})\right]\\ &\overset{(a)}{=} \text{trace}\left[\sum_{h=1}^{n}\boldsymbol{\Sigma}^{-1}(\mathbf{x}_h-\boldsymbol{\mu})(\mathbf{x}_h-\boldsymbol{\mu})^\mathsf{T}\right]\\ &\overset{(b)}{=} \text{trace}\left[\boldsymbol{\Sigma}^{-1}\sum_{h=1}^{n}(\mathbf{x}_h-\boldsymbol{\mu})(\mathbf{x}_h-\boldsymbol{\mu})^\mathsf{T}\right]\\ &\overset{(c)}{=} \text{trace}\left[\boldsymbol{\Sigma}^{-1}n\mathbf{S}\right]\\ &= n\,\text{trace}\left[\boldsymbol{\Sigma}^{-1}\mathbf{S}\right]\end{aligned}$$

where in:

(a) just move the terms of the product respecting trace properties

(b) take $\boldsymbol{\Sigma}^{-1}$ out of summation since not depending on h

(c) see that $(\mathbf{x}_h-\boldsymbol{\mu})(\mathbf{x}_h-\boldsymbol{\mu})^\mathsf{T}$ is the numerator of covariance matrix $\mathbf{S}$. We can obtain it by dividing and multiplying by n

Thus

$$-\frac{1}{2}\sum_{h=1}^{n}(\mathbf{x}_h-\boldsymbol{\mu})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{x}_h-\boldsymbol{\mu}) = -\frac{n}{2}(\text{trace}\,\boldsymbol{\Sigma}^{-1}\mathbf{S})$$

- in (2) regarding the determinant of $\boldsymbol{\Sigma}$, $|\boldsymbol{\Sigma}|$ (and of its inverse $|\boldsymbol{\Sigma}^{-1}|$) we used the fact that $\det(\mathbf{AB}) = \det(\mathbf{A})\cdot\det(\mathbf{B})$; thus since

$$1 = \det(\mathbf{A}\mathbf{A}^{-1}) = \det(\mathbf{A})\det(\mathbf{A}^{-1})$$

then

$$\det\left(\mathbf{A}^{-1}\right) = \frac{1}{\det\mathbf{A}} = [\det(\mathbf{A})]^{-1}$$

2.4.1.1 Maximum likelihood estimation steps

So we ended up rewriting the loglikelihood just in terms of $\boldsymbol{\Sigma}$ and $\mathbf{S}$, which are the two quantities I want to use to estimate the parameter of the model (basically comparing them to make them as more likely as possible)

$$l = \text{constant} + \frac{n}{2}\left[\log|\boldsymbol{\Sigma}^{-1}| - \text{trace}\left(\boldsymbol{\Sigma}^{-1}\mathbf{S}\right)\right]$$

Maximum likelihood estimation then follows these steps:

1. start by computing the maximum likelihood estimate of $\boldsymbol{\mu}$ ($\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$) and replace the estimate in $\mathbf{S}$.

2. maximize loglikelihood with respect to $\boldsymbol{\Lambda}$ and $\boldsymbol{\Psi}$ ($\mathbf{S}$ is now known) considering that $\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\mathsf{T} + \boldsymbol{\Psi}$:

$$\log L(\mathbf{x},\boldsymbol{\Lambda},\boldsymbol{\Psi}) = +\frac{n}{2}\log|(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\mathsf{T}+\boldsymbol{\Psi})^{-1}| - \frac{n}{2}\text{trace}[(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\mathsf{T}+\boldsymbol{\Psi})^{-1}\mathbf{S}]$$

By imposing the usual constraing on first order derivatives

$$\frac{\partial \log L(\mathbf{x}, \mathbf{\Lambda}, \mathbf{\Psi})}{\partial \mathbf{\Psi}} = 0$$
$$\frac{\partial \log L(\mathbf{x}, \mathbf{\Lambda}, \mathbf{\Psi})}{\partial \mathbf{\Lambda}} = 0$$

analytical solutions do not exist for the above equations, but at the end of derivation we obtain the following equations suited for iterative numerical procedures in order to obtain the maximum likelihood estimates of $\hat{\mathbf{\Lambda}}$ and $\hat{\mathbf{\Psi}}$:

$$\hat{\mathbf{\Psi}} = diag(\mathbf{S} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}^{\mathsf{T}}})$$
$$\mathbf{S}\hat{\mathbf{\Psi}}^{-1}\hat{\mathbf{\Lambda}} = \hat{\mathbf{\Lambda}}(\mathbf{I} + \hat{\mathbf{\Lambda}}^{\mathsf{T}}\hat{\mathbf{\Psi}}\hat{\mathbf{\Lambda}}),$$

What we do is to give initial values to $\hat{\mathbf{\Lambda}}$ and $\mathbf{\Psi}$ then

- update $\hat{\mathbf{\Psi}}$ given the initial values (using the first equation
- update $\hat{\mathbf{\Lambda}}$ using the second one
- repeat until some criterion of convergence is met (eg we calculate the loglikelihood using these estimates and it does not increase too much)

Important remark 12. Note that the number of latent variables q here (and always) is fixed in advance

- we start by considering $q = 1$
- we do estimation and check the model; if it's not good we increase q by 1

2.4.1.2 Maximum likelihood estimation by the E-M algorithm

The other common estimation method is the maximum likelihood method by the E-M algorithm, which is an iterative technique used for maximum likelihood estimation for models/situations *where there is some missing information in the data*. In our case the missing informations are the latent variables. In section 2.15 a refresher of EM estimation.
It consists of two steps which are repeated iteratively until convergence:

- E step: compute the expected value of the joint/complete log-likelihood of $(\mathbf{x}_h, \mathbf{y}_h)$ with $h = 1, \ldots, n$ (index of observation) conditional on the observed data $\mathbf{x}_h$
- M step: maximize the expected value of complete log-likelihood with respect to the parameters of the model.

In practice, instead of maximizing the (expected value) of the conditional log-likelihood to perform maximization it is easier to compute the *score function* of complete likelihood (by partial derivative) and set it to zero.
Anyway to start we need the complete log-likelihood; remembering that in our model

$$f(x, y) = g(x|y) \cdot h(y)$$

the complete log likelohood becomes

$$l_c = \log f(x,y) = \log(g(x|y)) + \log(h(y)))$$

Now for the factor model we know that g is normal $(\mathbf{x}|\mathbf{y} \sim \mathrm{N}(\boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{y}; \boldsymbol{\Psi}))$ while h is again normal (that is $\mathbf{y} \sim \mathrm{N}(\mathbf{0}, \mathbf{I})$. So after substituting the multivariate densities in the formula we can write down the complete loglikelihood of $(\mathbf{x}_h, \mathbf{y}_h)$ with $h = 1, \ldots, n$ which comes out as:

$$l_c = \text{constant} - \frac{n}{2}\Big\{ \log|\boldsymbol{\Psi}| + \text{trace}\boldsymbol{\Psi}^{-1}\left[\frac{1}{n}\sum_{h=1}^{n}(\mathbf{x}_h - \boldsymbol{\mu} - \boldsymbol{\Lambda}\mathbf{y}_h)(\mathbf{x}_h - \boldsymbol{\mu} - \boldsymbol{\Lambda}\mathbf{y}_h)^{\mathsf{T}}\right] + \text{trace}\left(\frac{1}{n}\sum_h \mathbf{y}_h\mathbf{y}_h^{\mathsf{T}}\right)\Big\}$$

where within square brackets we have the part due to g(x|y) while the last trace is due to h(y).
Now instead of computing expectation and then optimize, we derive and then compute the expectation (Bartholomew showed this is ok). We obtain the conditioned expected score functions

- wrt $\boldsymbol{\mu}$ (the first parameter we want to find):

$$\mathbb{E}[Sc_{\boldsymbol{\mu}}] = \mathbb{E}\left[\frac{\partial l}{\partial \boldsymbol{\mu}}|\mathbf{x}_h\right] = n\,\mathbb{E}\left[\boldsymbol{\Psi}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu} - \boldsymbol{\Lambda}\bar{\mathbf{y}})|\mathbf{x}_h\right] \tag{2.11}$$

  the second part of the loglikelihood disappear because it does not depend on $\boldsymbol{\mu}$ and the part $\frac{1}{n}\sum_h \mathbf{x}_h$ becomes the sample mean $\overline{\mathbf{x}}x$, and the same occurs for $\overline{\mathbf{y}}$

- wrt $\boldsymbol{\Lambda}$

$$\mathbb{E}[Sc_{\boldsymbol{\Lambda}}] = \mathbb{E}\left[\frac{\partial l}{\partial \boldsymbol{\Lambda}}|\mathbf{x}_h\right] = n\,\mathbb{E}\left[\boldsymbol{\Psi}^{-1}(\mathbf{S}_{xy}^{\mathsf{T}} - \boldsymbol{\mu}\bar{\mathbf{y}} - \boldsymbol{\Lambda}\mathbf{S}_{yy}^{\mathsf{T}})|\mathbf{x}_h\right] \tag{2.12}$$

- wrt $\boldsymbol{\Psi}$

$$\begin{aligned}\mathbb{E}[Sc_{\boldsymbol{\Psi}}] = E\left[\frac{\partial l}{\partial \boldsymbol{\Psi}}|\mathbf{x}_h\right] = -n/2E[\boldsymbol{\Psi}^{-1} - \boldsymbol{\Psi}^{-1}(\mathbf{S}_{xx}^{\mathsf{T}} - \boldsymbol{\mu}\bar{\mathbf{x}}^{\mathsf{T}} - \bar{\mathbf{x}}\boldsymbol{\mu}^{\mathsf{T}} - \mathbf{S}_{xy}^{\mathsf{T}}\boldsymbol{\Lambda}^{\mathsf{T}}\ldots \\ - \boldsymbol{\Lambda}\mathbf{S}_{yx}^{\mathsf{T}} + \boldsymbol{\mu}\bar{\mathbf{y}}^{\mathsf{T}}\boldsymbol{\Lambda}^{\mathsf{T}} + \boldsymbol{\Lambda}\bar{\mathbf{y}}\boldsymbol{\mu}^{\mathsf{T}} + \boldsymbol{\mu}\boldsymbol{\mu}^{\mathsf{T}} + \boldsymbol{\Lambda}\mathbf{S}_{yy}\boldsymbol{\Lambda}^{\mathsf{T}})\boldsymbol{\Psi}^{-1}|\mathbf{x}_h]\end{aligned} \tag{2.13}$$

In the equations above, the only random part are the quantities that one way or another depend on the random variables $\mathbf{y}$'s ($\mathbf{x}$ is conditioned other are constant) and are the sufficient statistics (respectively the mean of unobserved variables, the covariance between observed and unobserved, the variance/covariance of unobserved):

$$\bar{\mathbf{y}} = \frac{1}{n}\sum_h \mathbf{y}_h$$

$$\mathbf{S}_{xy}^{\mathsf{T}} = \frac{1}{n}\sum_h \mathbf{x}_h\mathbf{y}_h^{\mathsf{T}}$$

$$\mathbf{S}_{yy}^{\mathsf{T}} = \frac{1}{n}\sum_h \mathbf{y}_h\mathbf{y}_h^{\mathsf{T}}$$

Thus, to obtan estimators, it is enough to compute[11] the conditional expected values for the sufficient statistics.

$$E[\bar{\mathbf{y}}|\mathbf{x}_h] = \boldsymbol{\Lambda}^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) = \hat{\bar{\mathbf{y}}} \tag{2.14}$$

$$\begin{aligned} E[\mathbf{S}_{xy}^\mathsf{T}|\mathbf{x}_h] &= \frac{1}{n}\sum_h \mathbf{x}_h E[\mathbf{y}_h^\mathsf{T}|\mathbf{x}_h] = \frac{1}{n}\sum_h \mathbf{x}_h(\boldsymbol{\Lambda}^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{x}_h - \boldsymbol{\mu}))^\mathsf{T} \\ &= (\mathbf{S}_{xx} - \mathbf{x}\boldsymbol{\mu}^\mathsf{T})\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} = \hat{\mathbf{S}}_{xy}^\mathsf{T} \end{aligned} \tag{2.15}$$

$$\begin{aligned} E[\mathbf{S}_{yy}^\mathsf{T}|\mathbf{x}_h] &= (\mathbf{I}_q + \boldsymbol{\Lambda}^\mathsf{T}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda})^{-1} + \boldsymbol{\Lambda}^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{S}_{xx}^\mathsf{T} - \boldsymbol{\mu}\bar{\mathbf{x}}^\mathsf{T} - \bar{\mathbf{x}}\boldsymbol{\mu}^\mathsf{T} + \boldsymbol{\mu}\boldsymbol{\mu}^\mathsf{T}))\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} \\ &= \hat{\mathbf{S}}_{yy}^\mathsf{T} \end{aligned} \tag{2.16}$$

After obtaining the expression for $\hat{\bar{\mathbf{y}}}$, $\hat{\mathbf{S}}_{xy}^\mathsf{T}$ and $\hat{\mathbf{S}}_{yy}^\mathsf{T}$ we replace them in formulas (2.11), (2.12) and (2.13).
Then the M-step consist into setting the conditioned score functions to zero $E(S_{\boldsymbol{\mu}}) = 0$, $E(S_{\boldsymbol{\Lambda}}) = 0$ and $E(S_{\boldsymbol{\Psi}}) = 0$; by that we obtain

$$\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}} - \boldsymbol{\Lambda}\hat{\bar{\mathbf{y}}} \tag{2.17}$$

$$\hat{\boldsymbol{\Lambda}} = (\hat{\mathbf{S}}_{xy}^\mathsf{T} - \bar{\mathbf{x}}\hat{\bar{\mathbf{y}}}^\mathsf{T})(\hat{\mathbf{S}}_{yy}^\mathsf{T} - \hat{\bar{\mathbf{y}}}\hat{\bar{\mathbf{y}}}^\mathsf{T})^{-1} \tag{2.18}$$

$$\hat{\boldsymbol{\Psi}} = diag(\mathbf{S}_{xx}^\mathsf{T} - \hat{\boldsymbol{\mu}}\bar{\mathbf{x}}^\mathsf{T} - \bar{\mathbf{x}}\hat{\boldsymbol{\mu}}^\mathsf{T} - \hat{\mathbf{S}}_{xy}^\mathsf{T}\hat{\boldsymbol{\Lambda}}^\mathsf{T} - \hat{\boldsymbol{\Lambda}}\hat{\mathbf{S}}_{yx}^\mathsf{T} + \hat{\boldsymbol{\mu}}\hat{\mathbf{y}}^\mathsf{T}\hat{\boldsymbol{\Lambda}}^\mathsf{T} + \hat{\boldsymbol{\mu}}\hat{\boldsymbol{\mu}}^\mathsf{T} + \hat{\boldsymbol{\Lambda}}\hat{\mathbf{S}}_{yy}^\mathsf{T}\hat{\boldsymbol{\Lambda}}^\mathsf{T}) \tag{2.19}$$

Even in the EM algorithm we don't have close formula equation

Important remark 13. Summarizing, the E-M algorithm consists of the following steps:

1. choose starting values for $\boldsymbol{\mu}$, $\boldsymbol{\Lambda}$ and $\boldsymbol{\Psi}$
2. E-step: compute the conditioned expected score function (2.11), (2.12), (2.13)
3. M-step compute the maximum likelihood estimates (2.17), (2.18), (2.19)
4. iterate steps 2 and 3 until convergence

Remark 11. ML and EM based estimation converges to the same estimates but EM is usually faster

[11] Regarding:

- $E[\bar{\mathbf{y}}|\mathbf{x}_h]$ we remember that

  $$f(y|x) \sim \mathrm{N_q}\left(\boldsymbol{\Lambda}^\mathsf{T}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\mathsf{T} + \boldsymbol{\Psi})^{-1}(\mathbf{x} - \boldsymbol{\mu}), (\boldsymbol{\Lambda}^\mathsf{T}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda} + \mathbf{I})^{-1}\right)$$

  so the expected value of the random variable sample mean, conditioned, is

  $$\mathbb{E}\left[\bar{\mathbf{y}}|\mathbf{x}_h\right] = \boldsymbol{\Lambda}^\mathsf{T}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\mathsf{T} + \boldsymbol{\Psi})^{-1}(\mathbf{x} - \boldsymbol{\mu}) = \boldsymbol{\Lambda}^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$$

  where and the last passage substituted $(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\mathsf{T} + \boldsymbol{\Psi})^{-1} = \boldsymbol{\Sigma}^{-1}$ by the model
- similary occurs for the expected value of covariance between observed and unobserved
- the variance/covariance

  $$\begin{aligned} \mathbb{E}\left[\mathbf{y}\mathbf{y}^\mathsf{T}|\mathbf{x}_h\right] &= \mathrm{Var}\left[\mathbf{y}^\mathsf{T}|\mathbf{x}_h\right] + \mathbb{E}\left[\mathbf{y}|\mathbf{x}_h\right]\mathbb{E}\left[\mathbf{y}|x_h\right]^\mathsf{T} \\ &= (\mathbf{I}_q + \boldsymbol{\Lambda}^\mathsf{T}\boldsymbol{\Psi}^{-1}\boldsymbol{\Lambda})^{-1} + \boldsymbol{\Lambda}^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{S}_{xx}^\mathsf{T} - \boldsymbol{\mu}\bar{\mathbf{x}}^\mathsf{T} - \bar{\mathbf{x}}\boldsymbol{\mu}^\mathsf{T} + \boldsymbol{\mu}\boldsymbol{\mu}^\mathsf{T}))\boldsymbol{\Sigma}^{-1}\boldsymbol{\Lambda} \end{aligned}$$

2.5 Goodness of fit and choice of q

Important remark 14. Once estimated the parameter of the model we want to evaluate if our is a good model; in practice in factor analysis we evaluate if the number of factor q are enough to explain the association among the variable. If not we change the model by increasing q and running estimation again.
Different way into which evaluate goodness of fit; LRT is the first.

2.5.1 Likelihood ratio test

If q is specified a priori, we can use the likelihood ratio statistic comparing the following hypotheses:

$$H_0 : \mathbf{\Sigma} = \mathbf{\Lambda}\mathbf{\Lambda}^\mathsf{T} + \mathbf{\Psi}$$
$$H_1 : \mathbf{\Sigma} \text{ is unconstrained}$$

In this case we don't want to refuse the null:

- the null hypothesis states that the theoretical covariance matrix of the observed variables is well decomposed as the model supposes.
- the alternative does not assume any model, and one does not assume any decomposition for $\mathbf{\Sigma}$

The likelihood ratio statistic is:

$$\begin{aligned} LRT &= -2[l(H_0) - l(H_1)] \\ &= n[\log|\hat{\mathbf{\Sigma}}| + \text{trace}\hat{\mathbf{\Sigma}}^{-1}\mathbf{S} - \log|\mathbf{S}| - p] \\ &= n[\text{trace}\hat{\mathbf{\Sigma}}^{-1}\mathbf{S} - \log|\hat{\mathbf{\Sigma}}^{-1}\mathbf{S}| - p] \end{aligned}$$

where

- $l(H_0)$ is the loglikelihood of the model under null hypothesis (we already derived loglik under the model) while $l(H_1)$ under alternative[12]
- we have $\hat{\mathbf{\Sigma}} = \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\mathsf{T} + \hat{\mathbf{\Psi}}$

If $\mathbf{\Psi} > 0$ then $LRT \approx \chi^2$ with

$$df = \frac{1}{2}p(p+1) - \left[pq + p - \frac{1}{2}q(q-1)\right] = \frac{1}{2}[(p-q)^2 - (p+q)]$$

If we reject the null we have to increase the number of latent variables q by 1 and restart estimation.

[12] that is

$$\begin{aligned} l(H_1) &= \text{constant} + \frac{n}{2}\left[\log\left|\mathbf{\Sigma}^{-1}\right| - \text{trace}(\mathbf{\Sigma}^{-1}\mathbf{S})\right] \\ &\overset{(1)}{=} \text{constant} + \frac{n}{2}\left[\log\left|\mathbf{S}^{-1}\right| - \text{trace}(\mathbf{I})\right] = \text{constant} + \frac{n}{2}\left[\log\left|\mathbf{S}^{-1}\right| - p\right] \end{aligned}$$

where in (1) saying that it's unconstrainted is equivalent to say that $\mathbf{\Sigma} = \mathbf{S}$

2.6 Comparing models: model selection criteria

Otherwise we can compare models with:

- Akaike's information criterion for model selection.

$$AIC = -2l + 2\nu$$

- Bayesian information criterion (BIC)

$$BIC = -2l + \nu \log n$$

where ν is the number of free parameters in the model, l is the log-likelihood function.

2.7 Identifiability

Remark 12 (Idea). In general a model is identified if I have enough observations/information to estimate the model parameters (I have a system of equations that allows me to the estimation of parameters).

A necessary condition for identifiability of the model and consistency of the parameter estimators is that there are at least as many sample statistics as the parameters.:

- the number of observation/information we have is $p(p+1)/2$, that is the number of variances and covariances from the sample covariance matrix $\mathbf{S}$;
- the number of parameters to estimate in a q-factor model is $pq+p$ (pq are the number of factor loadings, while p are the number of error variances); however the number of parameter to be estimated may be greater that the observation availables $p(p+1)/2$
- it the above occurs (and if the model is unidentifiable, other than reducing the number of latent variables) we can set constraints. If $\mathbf{\Lambda\Psi}^{-1}\mathbf{\Lambda}$ is diagonal (latent variables uncorrelated) we introduce $\frac{1}{2}q(q-1)$ constraints; this makes the number of free parameters to be estimated to decrease too $pq + p - \frac{1}{2}q(q-1)$.[13]

A necessary but not sufficient condition for consistent estimation is to have positive degrees of freedom on LRT (that should be verified), that is:

$$\frac{1}{2}p(p+1) - pq - p + 1/2q(q-1) = 1/2[(p-q)^2 - (p+q)] \geq 0$$

This condition implies that there is an upper bound to the number of factors which can be fitted that is

$$q \leq [2p + 1 - (8p+1)^{1/2}]$$

with as always q the latent variables and p the observed/available variables.

[13] This is not the only way to put constraints on the loadings but a common one especially in exploratory factor models

2.8 Scale-Invariant Estimation

This is another convenient propertise of the linear factor model
The factor model can be fitted using the sample *correlation matrix* rather than the *covariance* matrix. Using correlations standardized variables x's, but changing the scale of the x's has *no* effect on the analysis.
Considering the standard model

$$\mathbf{x} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{y} + \mathbf{e}$$

we can transform $\mathbf{x}^* = \mathbf{C}\mathbf{x}$ where $\mathbf{C}$ is diagonal with positive elements (which acts as multiplier for $\mathbf{x}$ elements).
We obtain that the models becomes (by pre-multiplying both members by $\mathbf{C}$)

$$\mathbf{x}^* = \mathbf{C}\boldsymbol{\mu} + \mathbf{C}\boldsymbol{\Lambda}\mathbf{y} + \mathbf{C}\mathbf{e}$$

and its variance

$$\begin{aligned}\operatorname{Var}[\mathbf{x}^*] &= \operatorname{Var}[\mathbf{C}\boldsymbol{\mu} + \mathbf{C}\boldsymbol{\Lambda}\mathbf{y} + \mathbf{C}\mathbf{e}] \\ &\overset{(1)}{=} \mathbf{C}\boldsymbol{\Lambda}\underbrace{\mathbf{y}\mathbf{y}^\mathsf{T}}_{\mathbf{I}}\boldsymbol{\Lambda}^\mathsf{T}\mathbf{C} + \mathbf{C}\underbrace{\mathbf{e}\mathbf{e}^\mathsf{T}}_{\boldsymbol{\Psi}}\mathbf{C} \\ &= \mathbf{C}\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\mathsf{T}\mathbf{C} + \mathbf{C}\boldsymbol{\Psi}\mathbf{C}\end{aligned}$$

where in (1) $\mathbf{C}\boldsymbol{\mu}$ is a constant and $\mathbf{C}^\mathsf{T} = \mathbf{C}$.
If $\mathbf{C} = \operatorname{diag}(\boldsymbol{\Sigma})^{-1/2}$ (by multiplying we divide for the the standard deviations of observed variables) we obtain

$$\mathbf{x}^* = \boldsymbol{\mu}^* + \boldsymbol{\Lambda}^*\mathbf{y} + \mathbf{e}^*$$

where

$$\begin{aligned}\boldsymbol{\mu}^* &= \mathbf{C}\boldsymbol{\mu} \\ \boldsymbol{\Lambda}^* &= \mathbf{C}\boldsymbol{\Lambda} \\ \mathbf{e}^* &= \mathbf{C}\mathbf{e} \\ \boldsymbol{\Psi}^* &= \mathbf{C}\boldsymbol{\Psi}\mathbf{C}\end{aligned}$$

and

$$\operatorname{Var}[\mathbf{x}^*] = \boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\mathsf{T}} + \boldsymbol{\Psi}^*$$

So I can go from the solution on raw data to standardized data, by just applying the transformation (dividing by std dev the various matrices) to the estimations done (it's no needed to use correlation matrix). The two solutions are different but mappable.

2.9 Heywood case

Remark 13. There's a problem that we can encounter when we estimate the factor model, which is called Heywood case

Remembering that $\mathbf{\Psi}$ is the diagonal variance/covariance matrix of the error term containing specific variances:

- being variances, the parameter space is restricted by the condition $\mathbf{\Psi} \geq 0$, cannot be negative
- however in some cases we can have estimates *negative or zero* ψ: this is what we call the Heywood case.[14]

Possible *causes*:

- result of sampling error. A key factor is the sample size and number of observed variables used for the model:
  - Heywood can occur with small sample sizes;
  - for a given sample size the risk deceases as p increases.
- high correlations between variables.
- attempt to extract more factors than are present.

Possible *remedies*. If we end up with a solution with negative variances

- choose a large sample with a good number of variables.
- avoid to introduce new variables which little to those already there. This will create high correlations without contributing significantly to the information about latent variables.
- avoid over-factoring using adequate criteria
- consider a Bayesian approach by using a prior distribution for the ψ's which assigns zero probability to negative values.
- stop the iteration at some arbitrary small value of ψ_i such as 0.05 or 0.01.

2.10 Rotation and related matters

As illustrated above, one of the property of the factor model is that it is invariant to orthogonal rotation of the factor loadings. It means that we can get infinite solutions all equivalent.
In practice, factor rotation consists in finding a matrix $\mathbf{M}$ such that the factor loadings $\mathbf{\Lambda}^* = \mathbf{\Lambda}\mathbf{M}^\mathsf{T}$ can be more easily interpreted than the original factor loadings $\mathbf{\Lambda}$.
Is it a good idea to look for such rotations?

- Cons: one can keep rotating the factors until one finds an interpretation that one likes.
- Pros: factor rotation does not change the overall structure of a solution. It only changes how the solution is described, and finds the simplest description.

[14] We look for a minimum that will lie on a boundary of the admissible region (one or more of the ψ's is zero).

What do we look for Factor loadings can often be easily interpreted if they show a *simple structure*:

- each variable is highly loaded on at most one factor
- all factor loadings are either large and positive, or close to zero.

In table 2.1 the unrotated and rotated factor loadings from $\mathbf{\Lambda}$ and $\mathbf{\Lambda}^*$, where the last two columns (rotated solution) are easy interpretable (the first four observed variable are affected only by the second latent variable while the last three observed by the first only).

	$\hat{\lambda}_{i1}$	$\hat{\lambda}_{i2}$	$\hat{\lambda}^*_{i1}$	$\hat{\lambda}^*_{i2}$
x_1	0.2	0.3	0.0	0.4
x_2	0.4	0.5	0.0	0.6
x_3	0.6	0.7	0.0	0.9
x_4	0.7	0.7	0.0	1.0
x_5	0.5	-0.5	0.7	0.0
x_6	0.7	-0.6	0.9	0.0
x_7	0.3	-0.2	0.4	0.0

Table 2.1: Unrotated ($\hat{\lambda}_{i1}$, $\hat{\lambda}_{i2}$) and rotated ($\hat{\lambda}^*_{i1}$, $\hat{\lambda}^*_{i2}$) solution

Two types of rotations In general we have different kind of rotation

1. *Orthogonal rotation*: the factors obtained are restricted to be *uncorrelated*.
2. *Oblique rotation*: the factors obtained may be correlated.

I guess the choice depends on the specific application/theoretical model behind. With:

- orthogonal rotation, with uncorrelated factors, the standardized factor loadings could be interpreted as the correlation coefficients of the manifest variables and the factors.
- oblique rotation this is no longer true. Indeed the covariance becomes

$$\mathbb{E}\left[(\mathbf{x}-\boldsymbol{\mu})\mathbf{y}^{\mathsf{T}}\right] = \mathbf{\Lambda}\mathbf{\Phi}$$

  where $\mathbf{\Phi}$ is the correlation matrix of the y's. With oblique rotations the matrix $\mathbf{\Phi}$ of correlation is called *structured loading matrix*, while matrix $\mathbf{\Lambda}$ is called *pattern loading matrix*.

Advantage of:

- orthogonal rotation: for orthogonal rotation (based on standardized variables), the factor loadings represent correlations between factors and observed variables. This is not the case for oblique rotations.
- oblique rotation: may be unrealistic to assume that factors are uncorrelated. One may obtain a *better fit* by dropping this assumption.

2.11 Types of rotation

The most used:

- orthogonal rotations:
  - Varimax: aims at factors with a few large loadings, and many near-zero loadings (look for simple structure).
  - Quartimax: aims to evidence the correspondence between observed and latent factors
- oblique rotations:
  - Promax: aims at simple structure with low correlation between factors.
  - Oblimin: tends to produce varimax-looking factors, but which are oblique

Take home message (credo): provarle un po tutte e vedere quella che rende la matrice piu interpretabile

2.12 Factor scores

Finally, once we found the estimation of the factor loadings and specific variances we can estimate the factor scores/latent variables for each individual (that can be used for other analyses).
Different **estimation methods** exist:

- *Bartlett's method*: it is based on the maximum likelihood estimation hence requires the normality distribution of the observed variables. For the h-th individual they are given by:

$$\hat{\mathbf{y}}_h = (\hat{\mathbf{\Lambda}}\hat{\mathbf{\Psi}}^{-1}\hat{\mathbf{\Lambda}})\hat{\mathbf{\Lambda}}^{\mathsf{T}}\hat{\mathbf{\Psi}}^{-1}\mathbf{x}_h$$

- *Thompson's method*: the estimates of the factor scores are obtained in a Bayesian framework (using posterior distribution of $y|x$)

$$\hat{\mathbf{y}}_h = (\mathbf{I} + \hat{\mathbf{\Lambda}}^{\mathsf{T}}\hat{\mathbf{\Psi}}^{-1}\hat{\mathbf{\Lambda}})^{-1}\hat{\mathbf{\Lambda}}^{\mathsf{T}}\hat{\mathbf{\Psi}}^{-1}\mathbf{x}_h$$

Both methods have advantages and disadvantages, no clear favorite.

2.13 Example

In the Table 2.2 the correlation matrix between marks obtained in 6 subjects by a sample of 220 students is reported (from Lawley e Maxwell, 1971, a benchmark example of factor analysis), the first three of which are humanistic subjects (Gaelic, English and History) while the last three scientific (Arithmetic, Algebra, Geometry).
As it should be done we look at the correlation matrix:

	Gaelic	English	History	Arithmetic	Algebra	Geometry
Gaelic	1.00					
English	0.44	1.00				
History	0.41	0.35	1.00			
Arithmetic	0.29	0.35	0.16	1.00		
Algebra	0.33	0.32	0.19	0.59	1.00	
Geometry	0.25	0.33	0.18	0.47	0.46	1.00

Table 2.2: Correlation coefficients

Subject	$\hat{\lambda}_{i1}$	$\hat{\lambda}_{i2}$
Gaelic	0.56	0.43
English	0.57	0.29
History	0.39	0.45
Arithmetic	0.74	-0.28
Algebra	0.72	-0.21
Geometry	0.60	-0.13

Table 2.3: 2 factor model

Subjects	Communality
Gaelic	0.49 $(= 0.56^2 + 0.43^2)$
English	0.41
History	0.36
Arithmetic	0.62
Algebra	0.56
Geometry	0.37

Table 2.4: Communalities

1. if observed variables are not correlated it does make sense to make factor analysis; just to note, 0.3-0.4 correlation are good for factor analysis;

2. correlations can gives one an indea of the number q of latent variables (here there are two groups of variables).

In this case we can observe the correlations are all positive: students are good or not in all the subjects. The scientific subjects present high correlation and the humanistic subjects as well. It seems there are two groups.
We assume to have two latent variables and decide to estimate a 2 factor model with the MLE method: in the Table 2.3 the estimates of the unrotated factor loadings are reported.
If we want to interpret the latent factors we have to look at the magnitude of the loadings:

- we obtain a first latetent component with all positive and quite high loading associated to each subject. We can interpret the factor loadings/coefficients can be interpreted as correlation between marks and latent variables; so for example, eg 0.56 can be interpreted as the correlation between Gaelic and the first factor.
  For interpretation we could call the first factor a "general ability" (high value in this factor tend to be associated with good marks in all the subjects)

- the second factor is a contrast between humanistic and science subjects; an high value here is associated with best performances in humanistic subjects and less ability with science subjects

Looking now at **communalities** (the $\sum_j \lambda_{ij}^2$ in $\text{Var}[x_i] = (\sum_j \lambda_{ij}^2) + \psi$) in tab 2.4: working with correlation matrix, so having standardized variables (with variance 1), the communality of a manifest variable represents the *percentage of the variability* explained by the common factor:

- for example, the 49% of the variance of the mark in Gaelic is explained by the 2 factors. This communality is given by $0.49 = 0.56^2 + 0.43^2$ where 0.56 and 0.43 are the factor loadings associated to the 2 factors;

- If an observed variable presents a high communality, it will be a pure indicator of the common factor;

- The sum of the communality represents the variance explained by the model. In this case this value is 2.18 that is the 47% of 6 (total variance, which is trace of the variance covariance matrix with 6 variables);

- to give an idea with 50% explained variance we're good to go; there's always trade off in the sense that if we want to increase the explained variance we have to increase the number of latent variables estimated, but this can lead to more difficult to interpret results

To evaluate if our model is good we have to look further at godness of fit test.

Goodness of fit and choice of the number of factors q. Actually it turns out we have two way to evaluate goodness of fit:

- the first is to *Reproduce the correlation matrix*: One way to test the goodness of the estimated model is to compare the estimated (reproduced) and the observed correlation matrix of $\mathbf{x}$.
  The elements of the reproduced matrix (for example between Gaelic and English) are computed as

$$\begin{aligned} corr(x_1, x_2) &= \sum_j \lambda_{1j}\lambda_{2j} = \hat{\lambda}_{11}\hat{\lambda}_{21} + \hat{\lambda}_{12}\hat{\lambda}_{22} \\ &= (0.57 \times 0.56) + (0.29 \times 0.43) = 0.44 \end{aligned}$$

  and these are the elements out of main diagonal in the $\mathbf{\Sigma} = \Lambda\Lambda^{\mathsf{T}} + \Psi$ matrix (being varcov of standardized variables it's a correlation matrix). Notice that on the diagonal we have the communalities (explained variance).
  If the model is good we expect that element out of diagonal (reproduced correlation) are equal or near the actual correlation.
  In table 2.5 the reproduced correlations and differences with data correlation matrix from which we started; given the low differences, the 2 factor model seems to be a good model

- *goodness of fit test*: under the normality assumption of the data we test the hypothesis $H_0 : \mathbf{\Sigma} = \mathbf{\Lambda\Lambda}^{\mathsf{T}} + \mathbf{\Psi}$, that is the covariance/correlation matrix has the specified form. We use the test W that is distributed as a chi square with $[(p-q)^2 - (p+q)]/2$ degrees of freedom.
  In the example W is 2.18 with 4 degrees of freedom (number of varcov - n params), leading to a p-value=0.70; this suggests that the fit of the model to the data is good.

Correlation	Gaelic	English	History	Arithmetic	Algebra	Geometry
Gaelic	0.49	0.44	0.41	0.29	0.31	0.28
English	0.44	0.41	0.35	0.34	0.35	0.30
History	0.41	0.35	0.36	0.16	0.19	0.17
Arithmetic	0.29	0.34	0.16	0.62	0.59	0.48
Algebra	0.31	0.35	0.19	0.59	0.56	0.46
Geometry	0.28	0.30	0.17	0.48	0.46	0.37
Discrepancy	**Gaelic**	**English**	**History**	**Arithmetic**	**Algebra**	**Geometry**
Gaelic		0.00	0.00	0.00	0.02	- 0.03
English	0.00		0.00	0.01	- 0.03	0.03
History	0.00	0.00		0.00	0.00	0.00
Arithmetic	0.00	0.01	0.00		0.00	0.00
Algebra	0.00	-0.03	0.00	0.00		0.00
Geometry	-0.03	0.03	0.00	0.00	0.00	0.00

Table 2.5: Reproduced correlation and difference (residuals) wrt raw correlation

Solutions under different rotations Finally looking at different rotations, in table 2.6 we have two orthogonal rotations (varimax/quartimax) and an oblique rotation (oblimin).

- in all the rotations we have simple structure with a first factor associated to/namable "ability in scientific subjects", while second "ability with humanities");
- oblimin maybe has the simplest structure (non bold coefficients are the lowest) but there is correlation between latent factor (which can be reasonable in this application); particularly with oblimin the correlation between the factors ability in scientific subject and ability with humanities is 0.52.

Sublects								
	unrotated		Varimax		Quartimax		Oblimin	
Gaelic	0.56	0.43	0.23	**0.66**	0.32	**0.62**	0.04	**0.68**
English	0.57	0.29	0.32	**0.55**	0.39	**0.50**	0.17	**0.53**
History	0.39	0.45	0.08	**0.59**	0.16	**0.57**	-0.11	**0.65**
Arithmetic	0.74	-0.28	**0.77**	0.17	**0.79**	0.06	**0.81**	-0.04
Algebra	0.72	-0.21	**0.72**	0.22	**0.74**	0.12	**0.73**	0.03
Geometry	0.60	-0.13	**0.57**	0.22	**0.56**	0.13	**0.57**	0.07

Table 2.6: Rotations

2.14 Addendum EFA/CFA

The model we considered above is classificable as exploratory factor analysis and is contrasted to confirmatory factor analysis.
In case of **exploratory factor analysis** 2.2:

- we make no a priori hypothesis on the model, all the observed variables ($x_1, \dots x_6$) are influenced by all the factors (in this case η_1, η_2), and assume latent variables are standard normal and uncorrelated; we cannot assume latent variable are correlated otherwise the model is unadentifiable (we estimate it uncorrelated and then make them correlated, in case, with non-orthogonal rotations)
- all the parameters have to be estimated (both loadings, represented by rows, and unique variances, $\delta_1, \dots \delta_6$) they are free parameters;
- we get equivalent solutions at each rotation of the axis (invariance to orthogonal rotations)

Supposing some loadings are close to 0 in exploratory factor analysis: we want to test this using a confirmatory factor analysis.
In **confirmatory factor analysis** 2.2:

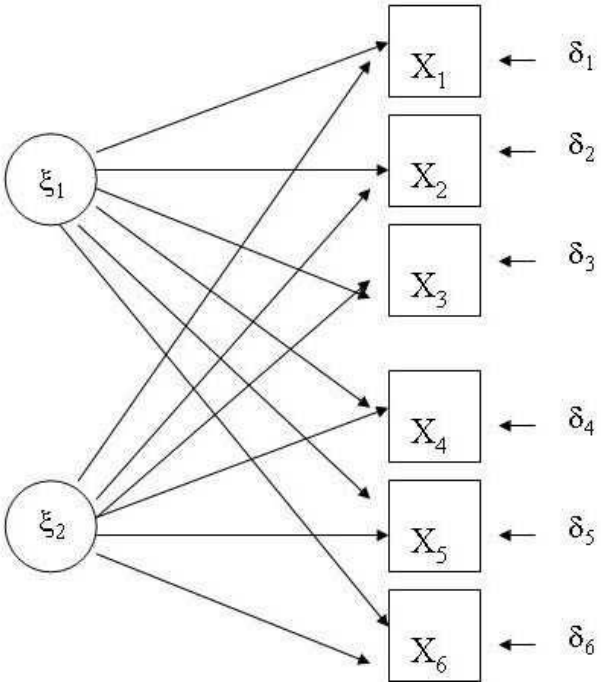


Figure 2.1: Exploratory factor analysis

- we do specific hypothesis on the number of factors, on the relations between latent factors (here we can assume they are related), on the relations between factors and variables using results from previous analysis, existent theories, or from the characteristics of the experimental design or from an inspection of the correlation matrix.
  In the figure we assume that the first latent variable influences the first three observed variables, while the second the last three
- some parameters are free and some are fixed (eg the loading of first latent variable on last three observed is fixed to 0), we choose the parameters to be estimated according to the hypothesis of the theoretical model
- the solution is unique (not invariant)

2.15 EM algorithm refresher

2.15.1 Introduction

NB: non parte dell'esame

Important remark 15. The EM algorithm (Dempster et al., 1977) is one of the most used algorithm in statistics for maximum likelihood estimation in complex problems. In particular, it is used in the cases where we want to obtain maximum likelihood estimates but the model depends on unobserved latent variables.

Remark 14. The iterative algorithm composed by two steps: E-step and M-step.

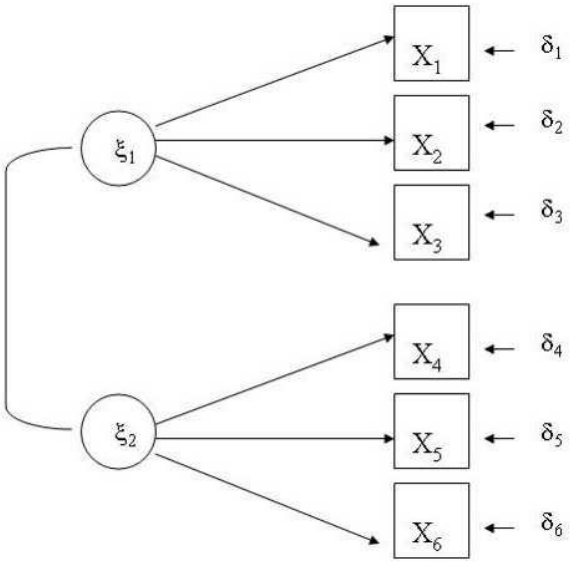


Figure 2.2: Confirmatory factor analysis

1. **E step**: Compute the conditional expected value (of the joint loglikelihood) given the current values of the parameter estimates.
2. **M step**: maximized the conditioned expected value with respect to the parameters of interest.

The two steps are applied iteratively until convergence

Example 2.15.1 (Motivating example)**.** We have two coins, Coin 1 and Coin 2. Each has its own probability of seeing 'H' on any one flip. Let

$$p_1 = \mathbb{P}\left(\text{H on Coin 1}\right)$$
$$p_2 = \mathbb{P}\left(\text{H on Coin 2}\right)$$

We want to estimate the probability of head in the two coins.
We select a coin at random and flip that one coin m times; we repeat the experiment n times. The obtained data:

$$\begin{array}{ll} X_{11}X_{12}\ldots X_{1m} & Y_1 \\ X_{21}X_{22}\ldots X_{2m} & Y_2 \\ X_{31}X_{32}\ldots X_{3m} & Y_3 \\ \ldots & \\ X_{n1}X_{n2}\ldots X_{nm} & Y_n \end{array}$$

Here

- the X_{ij} are Bernoulli random variables taking values in $\{0, 1\}$ where

$$X_{ij} = \begin{cases} 1 & \text{if the j-th flip for the ith coin chosen is H} \\ 0 & \text{if the j-th flip for the ith coin chosen is T.} \end{cases}$$

- the Y_i is random as well, is in $\{1, 2\}$ and indicates which coin was used on the n-th trial.

Note that all the X's are independent and in particular

$$X_{i1}X_{i2}\dots X_{im}|Y_i = j \sim \text{Bern}\,(p_j)$$

If we know everything about this experiment (know both the coin used and the flips results), we can write out the joint pdf of all $nm+n$ random variables and formally come up with MLE's for probability of success for the two coins p_1 and p_2. As expected they are:

$$\hat{p}_1 = \frac{\text{total number of times Coin 1 came up H}}{\text{total number of times Coin 1 was flipped}}$$
$$\hat{p}_2 = \frac{\text{total number of times Coin 2 came up H}}{\text{total number of times Coin 1 was flipped}}$$

However, now suppose that the Y_i are not observed: that is we don't know which coin was used, eg it was drawed from an urn and once done with the m flip put back into it. Thus for example we don't know denominators of the above estimators (how many times each coin was flipped). We still want MLE's for p_1 and p_2, but the data set now consists of only the X's and dataset is said to be *incomplete*. The goal of the EM Algorithm is to find MLEs for p_1 and p_2 (in this case).

2.15.2 EM algorithm (Dempster, Laird, Rubin, 1977)

Important remark 16 (Notation). Let:

- X be the *observed* data (generated by some distribution depending on some parameters). These data may or may not be iid. X will be called an **incomplete data set**;
- Y be the *hidden/latent* or *unobserved* data depending on some parameters;
- $Z = (X, Y)$ represent the **complete data set**. We say that it is a *completion* of the data given by X.

The distribution of Z (likely a big fat joint distribution) depends on some (likely high-dimensional) parameter θ ((p_1, p_2) in the example) and assume we can write the pdf for Z as:

$$f(z, \theta) = f(x, y, \theta) = f(y|x, \theta)f(x, \theta)$$

To apply ML estimation we go with likelihood definition first. In this case we can denote by:

- $L(z, \theta)$ the **complete likelihood** function defined as

  $$L_c(\theta) = L(z, \theta) = L(x, y, \theta) = f(x, y, \theta)$$

  it's the likelihood of the complete dataset

- $L(x, \theta)$ the **incomplete likelihood** functions defined as

$$L(\theta) = L(x, \theta) = f(x, \theta)$$

it's the likelihood of the observable dateaset

- the incomplete log-likelihood is

$$\log L(\theta) = \log L(x; \theta) = \log f(x; \theta)$$

So what is the idea of EM algorithm? It's that we can actually using the complete distributions/likelihood of Z without requiring the knowledge of the latent variables

- it starts with the aim of maximizing the *incomplete log-likelihood*, that is $\max_\theta \log f(x; \theta)$; this maximization is equivalent to the one of the expected value of the joint pdf divided by the conditional distribution of $y|x$, where the joint depends on θ, while the conditional on θ' (values of the parameter that are fixed). In fact the maximization can be rewritten as follows:

$$\begin{aligned}
\max_\theta \log f(x; \theta) &= \max_\theta \log \sum_y f(x, y; \theta) \\
&= \max_\theta \log \sum_y f(y|x; \theta') \frac{f(x, y; \theta)}{f(y|x; \theta')} \\
&= \max_\theta \log E_{y|x;\theta'} \left[\frac{f(x, y; \theta)}{f(y|x; \theta')} \right]
\end{aligned}$$

where $f(y|x; \theta')$ depends on a set of parameters θ' that are fixed. These are the unknown quantities of interest but are known (eg come from previous steps or by fixing them at first)

- now it is convenient to apply the Jensen inequality[15] basically exchanging the sum with the logarithm:

$$\underbrace{\log \sum_y f(y|x; \theta') \frac{f(x, y; \theta)}{f(y|x; \theta')}}_{\log L(\theta)} \geq \underbrace{\sum_y f(y|x; \theta') \log \frac{f(x, y; \theta)}{f(y|x; \theta')}}_{h(\theta|\theta')}$$

$h(\theta|\theta')$ is a lower bound function of the incomplete log-likelihood that we want to maximize. Thus in order to maximize $\log L(\theta)$ we can maximize $h(\theta|\theta')$ with respect to θ. We have that

$$h(\theta|\theta') = \sum_y f(y|x; \theta') \log f(x, y; \theta) - \sum_y f(y|x; \theta') \log f(y|x; \theta')$$

We can notice that the second term does not depend on θ but only on θ' that we suppose to be known[16]; thus maximizing $h(\theta|\theta')$ with respect to θ is equivalent to maximize only $\sum_y f(y|x; \theta') \log f(x, y; \theta)$.

[15] If g is a concave function (i.e. its second derivative is negative) then we have:

$$g(\mathbb{E}[x]) \geq \mathbb{E}[g(x)]$$

In our case $g(x) = \log x$ and $g''(x) = -\frac{1}{x^2} < 0$, thus we can apply the inequality

[16] This term is the *entropy* of the posterior $f(y|x; \theta')$.

- we want to maximize:

$$\begin{aligned}\arg\max_{\theta} h(\theta|\theta') &= \arg\max_{\theta} \sum_{y} f(y|x;\theta') \log f(x,y;\theta) \\ &= \arg\max_{\theta} \sum_{y} f(y|x;\theta') \log L_c(\theta) \\ &= \arg\max_{\theta} E_{y|x;\theta'}[\log L_c(\theta)]\end{aligned}$$

  So maximizing is nothing else that maximizing the expected value of the complete loglikelihood with respect to the posterior distribution of y given x; this is the idea.
  The estimation problem is equivalent to maximize (done in **M-STEP**) the conditional expected value (obtained in the **E-STEP**) of the complete density, instead of the incomplete one.

Important remark 17 (EM algorithm). The algorithm can be formalized in the following way:

- chose $\theta^{(0)} = \theta^{(h)}$ with $h = 0$ (*initialization*)
- repeat the following steps until convergence:
  - **E-STEP**: Compute the posterior expectation

$$Q(\theta;\theta^{(h)}) = E_{y|x;\theta^{(h)}}[\log L_c(\theta)]$$

  - **M-STEP**: Find $\theta^{(h+1)}$ that maximize the posterior expectation $Q(\theta;\theta^{(h)})$, that is the value such that

$$Q(\theta^{(h+1)};\theta^{(h)}) \geq Q(\theta;\theta^{(h)}) \quad \forall\theta \in \Omega$$

  - if convergence is not obtained (that is difference between ... is not lower than a treshold) then go the next step with $h = h + 1$

Remark 15. Summarizing:

- I start from the incomplete (?) log-likelihood
- prove that this is equivalent to a formula that is the logarithm of the expected value of the complete density divided by the posterior distribution of y|x
- instead of maximizing this quantity (which is complex to be maximized) I apply the Jensen inequality that allows me to find a lower bound of the loglikelihood, the h function, which is nothing else of the posterior expectation of the complete loglikelihood with respect to the posterior distribution of y|x
- the em consist in the E step to compute this last posterior expectation (of the complete loglikelihood with posterior distribution of y|x that depends on θ), and in the M step to maximize this quantity by finding iteratively the best thetas which will go in the next iteration
- we will go on until convergence, that is

2.15.3 Monotonicity property

NB: giusto accennata

Proposition 2.15.1. *The EM algorithm satisfies monotonicity condition of the log-likelihood, that is the algorithm guarantees that, in each iterative step, the log-likelihood does not decrease:*

$$\log L\theta^{(h+1)} \geq \log L\theta^{(h)} \quad \forall h$$

Proof. We can observe that the joint density $f(x, y; \theta)$ can be arbitrarily decomposed as $f(x, y; \theta) = f(x|y; \theta)f(x; \theta)$ or as $f(x, y; \theta) = f(y|x; \theta)f(x; \theta)$. Considering the last expression we can write the conditional expected value as:

$$\begin{aligned} Q(\theta; \theta^{(h)}) &= E_{y|x;\theta^{(h)}}[\log L_c(\theta)] \\ &= sum_y f(y|x; \theta^{(h)}) \log f(y|x; \theta) + \sum_y f(y|x; \theta^{(h)}) \log f(x; \theta) \\ &= sum_y f(y|x; \theta^{(h)}) \log f(y|x; \theta) + \log f(x; \theta) \sum_y f(y|x; \theta^{(h)}) \\ &= H(\theta; \theta^{(h)}) + \log f(x; \theta) = H(\theta; \theta^{(h)}) + \log L(\theta) \end{aligned}$$

where the first term is given in compact form as $H(\theta; \theta^{(h)})$ that is a kind of entropy. From the previous equation we have:

$$\log L(\theta) = Q(\theta; \theta^{(h)}) - H(\theta; \theta^{(h)}) \tag{2.20}$$

The monotonic property is verified if

$$\log L(\theta^{(h+1)}) - \log L(\theta^{(h)}) \geq 0$$

Let's start from the difference that we can rephrase using equation 2.20:

$$\log L(\theta^{(h+1)}) - \log L(\theta^{(h)}) = \left[Q(\theta^{(h+1)}; \theta^{(h)}) - Q(\theta^{(h)}; \theta^{(h)})\right] - \left[H(\theta^{(h+1)}; \theta^{(h)}) - H(\theta^{(h)}; \theta^{(h)})\right] \tag{2.21}$$

In the previous first difference is greater or equal to 0 because of the M-step. The monotonic property is verified if we can prove

$$H(\theta^{(h+1)}; \theta^{(h)}) - H(\theta^{(h)}; \theta^{(h)}) \leq 0$$

for every θ we have

$$\begin{aligned} H(\theta^{(h+1)}; \theta^{(h)}) - H(\theta^{(h)}; \theta^{(h)}) &= E_{y|x;\theta^{(h)}}\left[\log \frac{f(y|x; \theta^{(h+1)})}{f(y|x; \theta^{(h)})}\right] \\ &\leq \log\left(E_{y|x;\theta^{(h)}}\left[\log \frac{f(y|x; \theta^{(h+1)})}{f(y|x; \theta^{(h)})}\right]\right) \\ &= \log \sum_y f(y|x; \theta^{(h+1)}) = 0 \end{aligned}$$

where in the previous development the Jensen inequality has been applied. □

2.16 Addendum NLFA

$$\underset{p\times p}{\mathbf{\Sigma}} = \underset{p\times q}{\mathbf{\Lambda}}\underset{q\times p}{\mathbf{\Lambda}'} + \underset{p\times p}{\mathbf{\Psi}}$$

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1^2 & \dots & \sigma_{1p} \\ \dots & \sigma_1^2 & \dots \\ \dots & & \\ \sigma_{p1} & \dots & \sigma_p^2 \end{bmatrix} = [\textstyle\sum_j] + \begin{bmatrix} \psi_{11} & 0 & \dots & 0 \\ 0 & \psi_{22} & \dots & 0 \\ \dots & & & \\ 0 & \dots & \dots & \psi_{pp} \end{bmatrix}$$

da cui

$$\sigma_i^2 = \operatorname{Var}[x_i] = \sum_{j=1}^{q} \lambda_{ij}^2 + \psi_{ii}$$

where $\sum_{j=1}^{q} \lambda_{ij}^2$ is called *communality* and ψ_{ii} specific variance.

$$\mathbf{\Lambda\Lambda}' = \begin{bmatrix} \sum_j \lambda_{1j}^2 & \sum_j \lambda_{1j}\lambda_{2j} & \dots & \sum_j \lambda_{1j}\lambda_{pj} \\ \dots & \sum_j \lambda_{2j}^2 & \dots & \dots \\ \dots & \dots & \dots & \dots \\ \sum_j \lambda_{pj}^2 & \sum_j \lambda_{pj}\lambda_{2j} & \dots & \sum_j \lambda_{pj}^2 \end{bmatrix}$$

$$\operatorname{Cov}(x,y) = \mathbb{E}[(x-\mu)y'] = \mathbb{E}[\mathbb{E}[(x-\mu)|y]\,y'] = \mathbb{E}[\mathbb{E}[(\Lambda y + e)|y]\,y'] = \mathbb{E}[\Lambda y y'] = \Lambda\,\mathbb{E}[yy'] = \Lambda$$

$$X = \Lambda'\Psi^{-1}x$$

$$\operatorname{Cov}(X,y) = \mathbb{E}[(X-\mathbb{E}[X])y'] = \mathbb{E}\left[(\Lambda'\Psi^{-1}x - \Lambda'\Psi^{-1}\,\mathbb{E}[x])y'\right] = \Lambda'\Psi^{-1}\,\mathbb{E}[(x-\mathbb{E}[X])y'] = \Lambda'\Psi^{-1}\,\mathbb{E}[(x-\mu)y'] =$$

$$\nu = My \quad y = M^{-1}\nu = M'\nu$$

$$M'M = IM^{-1} = M\nu \sim \mathrm{N}(\mathbf{0},\mathbf{I})x|\nu \sim \mathrm{N}(\mu + \Lambda M)$$

2.17 Uso di R

```
library(lbdatasets)

##
## Caricamento pacchetto:  'lbdatasets'
## Il seguente oggetto è mascherato da 'package:survival':
##
##     ovarian
```

```
library(lblatent)

## Caricamento del pacchetto richiesto:  lavaan
## This is lavaan 0.7-2
## lavaan is FREE software!  Please report any bugs.
## Caricamento del pacchetto richiesto:  ltm
## Caricamento del pacchetto richiesto:  MASS
##
## Caricamento pacchetto:  'MASS'
## Il seguente oggetto è mascherato da 'package:lbdatasets':
##
##     anorexia
## Caricamento del pacchetto richiesto:  msm
##
## Caricamento pacchetto:  'msm'
## Il seguente oggetto è mascherato da 'package:boot':
##
##     cav
## Caricamento del pacchetto richiesto:  polycor
## Caricamento del pacchetto richiesto:  GPArotation

## ------------------------------------------------------------
## EXERCISE 1
## ------------------------------------------------------------

## The classic Holzinger and Swineford (1939) dataset consists of mental
## ability test scores of seventh-and eighth-grade children from two different
## schools (Pasteur and Grant-White).  In the original dataset there are scores
## for 26 tests; however a smaller subset with 9 variables is more widely used
## in the literature. We'll perform first an exploratory factor analysis and
## then a confirmatory factor analysis on this data set.

## 1) Load and display the data set HolzingerSwineford1939 (library lavaan).

library(lavaan)
data(HolzingerSwineford1939)

## The dataset is referrend to 301 students in the following variables (all
## continuous): see ?HolzingerSwineford1939
#
# x1 Visual perception
# x2 Cubes
# x3 Lozenges
# x4 Paragraph comprehension
# x5 Sentence completion
# x6 Word meaning
# x7 Speeded addition
# x8 Speeded counting of dots
# x9 Speeded discrimination straight and curved capitals
#
```

```
## The variable can be grouped in the following groups
#
# x1-x2-x3 = Visualization factor
# x4-x5-x6 = Verbal intelligence/ textual factor
# x7-x8-x9 = Speed factor

## See the first rows
head(HolzingerSwineford1939)
```

```
##   id sex ageyr agemo  school grade    x1   x2    x3    x4   x5     x6    x7
## 1  1   1    13     1 Pasteur     7 3.333 7.75 0.375 2.333 5.75 1.2857 3.391
## 2  2   2    13     7 Pasteur     7 5.333 5.25 2.125 1.667 3.00 1.2857 3.783
## 3  3   2    13     1 Pasteur     7 4.500 5.25 1.875 1.000 1.75 0.4286 3.261
## 4  4   1    13     2 Pasteur     7 5.333 7.75 3.000 2.667 4.50 2.4286 3.000
## 5  5   2    12     2 Pasteur     7 4.833 4.75 0.875 2.667 4.00 2.5714 3.696
## 6  6   2    14     1 Pasteur     7 5.333 5.00 2.250 1.000 3.00 0.8571 4.348
##     x8    x9
## 1 5.75 6.361
## 2 6.25 7.917
## 3 3.90 4.417
## 4 5.30 4.861
## 5 6.30 5.917
## 6 6.65 7.500
```

```
## We select just the scoring variable to perform factor analysis on them
vars <- paste0("x", 1:9)
data <- HolzingerSwineford1939[, vars]


## 2) Explore the correlation matrix of the data in order evaluate if a factor
## model can be fitted.
#
## It is convenient to inspect correlation matrix cause can give many info
## about the data and number of factor to include

round((cormat <- cor(data)), 2)
```

```
##      x1    x2   x3   x4   x5   x6    x7   x8   x9
## x1 1.00  0.30 0.44 0.37 0.29 0.36  0.07 0.22 0.39
## x2 0.30  1.00 0.34 0.15 0.14 0.19 -0.08 0.09 0.21
## x3 0.44  0.34 1.00 0.16 0.08 0.20  0.07 0.19 0.33
## x4 0.37  0.15 0.16 1.00 0.73 0.70  0.17 0.11 0.21
## x5 0.29  0.14 0.08 0.73 1.00 0.72  0.10 0.14 0.23
## x6 0.36  0.19 0.20 0.70 0.72 1.00  0.12 0.15 0.21
## x7 0.07 -0.08 0.07 0.17 0.10 0.12  1.00 0.49 0.34
## x8 0.22  0.09 0.19 0.11 0.14 0.15  0.49 1.00 0.45
## x9 0.39  0.21 0.33 0.21 0.23 0.21  0.34 0.45 1.00
```

```
## regarding correlation she commented as high correlated were rho > 0.3
## - x1 (visual perception) in general is correlated with x3 x4 x6 and x9: enough
```

```
##   correlated with several other items
## - x2 not negligible corr with x3
## - x3 corr with x1, x2, x9
## - x4 corr with x1 x5 x6
## - x5 corr with
## - x9 corr x1 x3 x7 and x9
#
## seems there are 3 different groups of variables x1-x3, x4-x6 and x7-x9 (but
## there is some relation even out of these groups eg with x1 since exploratory
## model it's convenient to start fitting 1,2,3-factor modelso see if one of
## them present good of fit to the data.



## 3) Estimate a factor model with 1,2 and 3 factors using the Maximum
## Likelihood method and select the best model using the Chi-square test. Use
## the function factanal.

## factanal in base pacakge for continuous variable:
##
## - first param is the formula we want to analyze
##
## - second data (or we can specify a correlation matrix using covmat) if we
##   use the covariance matrix we have to specify the number of observations
##
## - third options (factors) is the n of factors to include
##
## - by default a varimax rotation is done, to disable rotation="none"
## see ?factanal

# FACTANAL use classical ML

## f1 <- factanal(~x1+x2+x3+x4+x5+x6+x7+x8+x9, data=data, factors = 1)
## f2 <- factanal(~x1+x2+x3+x4+x5+x6+x7+x8+x9, data=data, factors = 2)
## f3 <- factanal(~x1+x2+x3+x4+x5+x6+x7+x8+x9, data=data, factors = 3)

## see which one is the best model among the three: to check this the first
## tool to use is the LRT (a chisq with df = number of obs variances and
## covariances - number of parameters to be estimated)

## ## Collect the stats of LRT for the three models
## (Chisq <- round(c(f1$STATISTIC, f2$STATISTIC, f3$STATISTIC), 3))
## (df <- c(f1$dof,f2$dof,f3$dof))
## (pvalue <- round(c(f1$PVAL,f2$PVAL,f3$PVAL),4))

(efa_choose <- lblatent::efa_preliminary(data, nfactors=1:3))

## $readable_cormatrix
##    x1 x2     x3     x4     x5     x6 x7     x8     x9
## x1  1 NA 0.4407 0.3727     NA 0.3568 NA     NA 0.3903
```

```
## x2 NA  1 0.3398     NA     NA     NA NA     NA     NA
## x3 NA NA 1.0000     NA     NA     NA NA     NA 0.3287
## x4 NA NA     NA 1.0000 0.7332 0.7045 NA     NA     NA
## x5 NA NA     NA     NA 1.0000 0.7200 NA     NA     NA
## x6 NA NA     NA     NA     NA 1.0000 NA     NA     NA
## x7 NA NA     NA     NA     NA     NA  1 0.4868 0.3406
## x8 NA NA     NA     NA     NA     NA NA 1.0000 0.4490
## x9 NA NA     NA     NA     NA     NA NA     NA 1.0000
##
## $chisq
##      chi df         p rounded_p    comment
## 1 306.56 27 3.579e-49    0.0000 bad/reject
## 2 127.64 19 4.077e-18    0.0000 bad/reject
## 3  22.38 12 3.351e-02    0.0335        meh

## the df decrease becaus with an higher number of latent variables we have
## more factor loadings.  For the first models two the p are very low: we
## reject all the three factor models at 0.05. However we don't reject the
## three-factors model at the level of 0.01.
##
## - We say that it's not convenient to consider a 4-factor models because if
##   we increase the n of factors the solutions becomes difficult to be
##   interpreted; we're already happy with the 3-factors model where we don't
##   reject the null at level of 0.01.
##
## - Another good reason for 3 groups is that in the correlation matrix we
##   basically have three groups of variables (with some exception)
##
## so we go on with the 3-factors solution

## for the exam say



(chosen <- lblatent::efa_estimate(data, nfactors=3))

## $model
##
## Call:
## factanal(x = formula, factors = nfactors, data = data, rotation = "none")
##
## Uniquenesses:
##    x1    x2    x3    x4    x5    x6    x7    x8    x9
## 0.513 0.749 0.543 0.279 0.243 0.305 0.502 0.469 0.543
##
## Loadings:
##    Factor1 Factor2 Factor3
## x1  0.488   0.314   0.389
## x2  0.244   0.173   0.402
## x3  0.272   0.407   0.466
```

```
## x4  0.835  -0.153
## x5  0.839  -0.209
## x6  0.823  -0.129
## x7  0.229   0.485  -0.459
## x8  0.270   0.622  -0.269
## x9  0.376   0.561
##
##                Factor1 Factor2 Factor3
## SS loadings      2.717   1.313   0.824
## Proportion Var   0.302   0.146   0.092
## Cumulative Var   0.302   0.448   0.539
##
## Test of the hypothesis that 3 factors are sufficient.
## The chi square statistic is 22.38 on 12 degrees of freedom.
## The p-value is 0.0335
##
## $communalities
##     x1     x2     x3     x4     x5     x6     x7     x8     x9
## 0.4875 0.2513 0.4572 0.7208 0.7571 0.6948 0.4978 0.5314 0.4568
##
## $uniqueness
##     x1     x2     x3     x4     x5     x6     x7     x8     x9
## 0.5125 0.7487 0.5428 0.2792 0.2429 0.3052 0.5022 0.4686 0.5432
##
## $cor_original
##    x1  x2   x3   x4   x5   x6    x7   x8   x9
## x1  1 0.3 0.44 0.37 0.29 0.36  0.07 0.22 0.39
## x2 NA 1.0 0.34 0.15 0.14 0.19 -0.08 0.09 0.21
## x3 NA  NA 1.00 0.16 0.08 0.20  0.07 0.19 0.33
## x4 NA  NA   NA 1.00 0.73 0.70  0.17 0.11 0.21
## x5 NA  NA   NA   NA 1.00 0.72  0.10 0.14 0.23
## x6 NA  NA   NA   NA   NA 1.00  0.12 0.15 0.21
## x7 NA  NA   NA   NA   NA   NA  1.00 0.49 0.34
## x8 NA  NA   NA   NA   NA   NA    NA 1.00 0.45
## x9 NA  NA   NA   NA   NA   NA    NA   NA 1.00
##
## $cor_reproduced
##      x1   x2   x3   x4   x5   x6    x7   x8   x9
## x1 0.49 0.33 0.44 0.35 0.31 0.37  0.09 0.22 0.37
## x2   NA 0.25 0.32 0.16 0.13 0.19 -0.04 0.07 0.20
## x3   NA   NA 0.46 0.15 0.10 0.18  0.05 0.20 0.34
## x4   NA   NA   NA 0.72 0.74 0.71  0.13 0.14 0.23
## x5   NA   NA   NA   NA 0.76 0.72  0.14 0.12 0.20
## x6   NA   NA   NA   NA   NA 0.69  0.12 0.14 0.24
## x7   NA   NA   NA   NA   NA   NA  0.50 0.49 0.35
## x8   NA   NA   NA   NA   NA   NA    NA 0.53 0.44
## x9   NA   NA   NA   NA   NA   NA    NA   NA 0.46
##
## $cor_residual
```

```
##      x1    x2   x3    x4    x5    x6    x7    x8    x9
## x1 0.51 -0.03 0.00  0.03 -0.01 -0.01 -0.02  0.00  0.02
## x2   NA  0.75 0.01 -0.01  0.01  0.01 -0.03  0.03  0.01
## x3   NA    NA 0.54  0.01 -0.02  0.02  0.03 -0.01 -0.01
## x4   NA    NA   NA  0.28  0.00  0.00  0.04 -0.03 -0.02
## x5   NA    NA   NA    NA  0.24  0.00 -0.03  0.02  0.03
## x6   NA    NA   NA    NA    NA  0.30  0.00  0.01 -0.02
## x7   NA    NA   NA    NA    NA    NA  0.50  0.00 -0.01
## x8   NA    NA   NA    NA    NA    NA    NA  0.47  0.00
## x9   NA    NA   NA    NA    NA    NA    NA    NA  0.54
##
## $loadings_raw
##    Factor1 Factor2 Factor3
## x1  0.4880  0.3135  0.3886
## x2      NA      NA  0.4019
## x3      NA  0.4071  0.4662
## x4  0.8345      NA      NA
## x5  0.8390      NA      NA
## x6  0.8234      NA      NA
## x7      NA  0.4845 -0.4590
## x8      NA  0.6217      NA
## x9  0.3765  0.5608      NA
##
## $loadings_varimax
##    Factor1 Factor2 Factor3
## x1  0.3202      NA  0.6066
## x2      NA      NA  0.4809
## x3      NA      NA  0.6619
## x4  0.8379      NA      NA
## x5  0.8667      NA      NA
## x6  0.8151      NA      NA
## x7      NA  0.6954      NA
## x8      NA  0.7036      NA
## x9      NA  0.5106  0.4089
## attr(,"covariance")
##      [,1] [,2] [,3]
## [1,]    1    0    0
## [2,]    0    1    0
## [3,]    0    0    1
##
## $loadings_quartimax
##    Factor1 Factor2 Factor3
## x1  0.3526      NA  0.5897
## x2      NA      NA  0.4740
## x3      NA      NA  0.6566
## x4  0.8441      NA      NA
## x5  0.8687      NA      NA
## x6  0.8234      NA      NA
## x7      NA  0.6923      NA
```

```
## x8      NA  0.7021      NA
## x9      NA  0.5078  0.3974
## attr(,"covariance")
##      [,1] [,2] [,3]
## [1,]    1    0    0
## [2,]    0    1    0
## [3,]    0    0    1
##
## $loadings_oblimin
##    Factor1 Factor2 Factor3
## x1      NA      NA  0.6020
## x2      NA      NA  0.5054
## x3      NA      NA  0.6893
## x4  0.8405      NA      NA
## x5  0.8882      NA      NA
## x6  0.8076      NA      NA
## x7      NA  0.7231      NA
## x8      NA  0.7015      NA
## x9      NA  0.4632  0.3661
## attr(,"covariance")
##        [,1]   [,2]   [,3]
## [1,] 1.0000 0.2164 0.3258
## [2,] 0.2164 1.0000 0.2705
## [3,] 0.3258 0.2705 1.0000
```

```
## 4) On the basis of the matrix of the factor loadings interpret the factors
## individuated
##

f3 <- factanal(~x1+x2+x3+x4+x5+x6+x7+x8+x9, data=data, factors = 3)
loadings(f3)
```

```
##
## Loadings:
##    Factor1 Factor2 Factor3
## x1  0.277   0.623   0.151
## x2  0.105   0.489
## x3          0.663   0.130
## x4  0.827   0.165
## x5  0.861
## x6  0.801   0.212
## x7                  0.696
## x8          0.162   0.709
## x9  0.132   0.406   0.524
##
##                Factor1 Factor2 Factor3
## SS loadings      2.185   1.343   1.327
## Proportion Var   0.243   0.149   0.147
## Cumulative Var   0.243   0.392   0.539
```

```
## by default R visualize loadings > 0.1 (they do not contribute to the
```

```
## intepretation of the factor): a better rule of thumb is not to consider
## loadings < 0.3 in interpreting the factors.

## These are the unrotated solutions but it's already interpretable solution
## (in realtà qui la prof si è sbagliata e factanal fa la varimax di default,
## ecco perché i loadings raw sono interpretabili:
##
## - the first factor presents high loadings associated to the x4-x6 (verbal
##   intelligence/textual factor). loadings are interpretable as correlations
##   (factanal give the solution given on correlation matrix) between latent
##   factor and observed variables. There's high correlation (over 0.8) between
##   x4-x6 and Factor1. This can be interpreted as "Verbal Intelligence"
##
## - second factor is associated to x1-x3 and x9: surely we can interpret this
##   factor as "Visualization Ability" factor. Why x9 is in? item x9 (speed
##   discrimination straight and curved capitals) in a sense also capture a
##   measure/concept of visualization
##
## - the third one is associated to items regarding speed so it can be
##   conceptualized as "Speed Ability" factor
##
## not always the results are this interpretable and we may need to do rotations

## 5) Compute the communalities and comment them. Which percentage of the variance of
## the model is explained by the three-factor model?

## SS loadings is sum of square of loadings (total variance explained by each
## factor): eg for Factor1, 2.18 is the ability that depends of factors 1 (sum
## of squares of first column of loadings). This can be reproduced by

(var_expl <- c(sum(loadings(f3)[,1]^2),
               sum(loadings(f3)[,2]^2),
               sum(loadings(f3)[,3]^2)))

## [1] 2.185 1.343 1.327

## To get proportion of variability explained by each factors we have to divide
## the variance explained by the factor by the total variability/variance,
## which in this case is 9 (9 standardized items with 1variance, since working
## with correlation). Thus proportion of variance explained (second row)
var_expl / 9

## [1] 0.2427 0.1492 0.1475

## Third row is the cumulative explained variability and reproduced as
cumsum(var_expl / 9)

## [1] 0.2427 0.3919 0.5394

## the first factor explain 24% of variability; if we take 3, overall we
## explain 53% of variability
```

```
# for the communality we have to square the loadings and sum by row
(comm <- rowSums(loadings(f3)^2)) # row sums of loading squares . it's the communalit

##     x1     x2     x3     x4     x5     x6     x7     x8     x9
## 0.4875 0.2513 0.4572 0.7208 0.7571 0.6948 0.4978 0.5314 0.4568

# comm is the percentage of item's variance which can be explained by the
# latent factors

## the uniqueness are portion of each item's variance which is specific and  for each
## term) the psi in the formula below
##
## var(x_i) = sum_j \lambda_{ij}^2 + \psi_i
##
## where sum_j \lambda_{ij}^2 is communality and psi_i are uniqueness
##
## we have the communalities in comm: being variance = 1 we have psi as 1 - comm
(uniq <- 1 - comm)

##     x1     x2     x3     x4     x5     x6     x7     x8     x9
## 0.5125 0.7487 0.5428 0.2792 0.2429 0.3052 0.5022 0.4686 0.5432

## 6) Compute the reproduced correlation matrix and the discrepancy between the obser
## and reproduced correlation.

## there are other ways to check goodness solution other than chisq/LRT. we can
## consider the reproduced correlation matrix.
##
## we have that: R = \Lambda \Lambda ' + \Psi where the reproduced the correlation ma
## is \Lambda \Lambda ' and we look at CorrMatrix - \Lambda \Lambda '
##
## if it's a good model, elements out of the main diagonal will be close to
## 0 that is our model explain all the correlation between the observed variables).
## This allows to find intems for which we don't have good of fit

repcorr <- loadings(f3) %*% t(loadings(f3)) # reproduced corr matrix \Lambda \Lambda'
(resmat <-  (round(cormat - repcorr,3)))

##        x1     x2     x3     x4     x5     x6     x7     x8     x9
## x1  0.513 -0.032 -0.001  0.026 -0.013 -0.011 -0.018  0.002  0.021
## x2 -0.032  0.749  0.015 -0.012  0.009  0.007 -0.031  0.027  0.007
## x3 -0.001  0.015  0.543  0.008 -0.021  0.018  0.026 -0.015 -0.013
## x4  0.026 -0.012  0.008  0.279 -0.002 -0.002  0.042 -0.032 -0.020
## x5 -0.013  0.009 -0.021 -0.002  0.243  0.004 -0.033  0.016  0.031
## x6 -0.011  0.007  0.018 -0.002  0.004  0.305  0.002  0.012 -0.024
## x7 -0.018 -0.031  0.026  0.042 -0.033  0.002  0.502  0.001 -0.006
## x8  0.002  0.027 -0.015 -0.032  0.016  0.012  0.001  0.469  0.005
## x9  0.021  0.007 -0.013 -0.020  0.031 -0.024 -0.006  0.005  0.543
```

```
## in general out of the main diagonal the highest residuals is -0.03. So this
## model is reasonably good.  if I find residual large, the variable may be
## excluded from the analysis.

## on the main diagonal if the model is a good model we expect residual to be closed to
## uniqueness since
##
## R- \Lambda \Lambda ' = \Psi

diag(resmat)

##    x1    x2    x3    x4    x5    x6    x7    x8    x9
## 0.513 0.749 0.543 0.279 0.243 0.305 0.502 0.469 0.543

uniq

##     x1     x2     x3     x4     x5     x6     x7     x8     x9
## 0.5125 0.7487 0.5428 0.2792 0.2429 0.3052 0.5022 0.4686 0.5432

## 7) Apply different orthogonal and oblique rotations and interpret the solutions obtained
##
## one property is invariance to rotation: here the solution is easy to be
## interpret but in some cases is convenient to apply rotations. The package to
## do them is GPArotation. it has many orthogonal and oblique rotation, we can
## apply anyone of # them and chose the one that is more interpretable.

library(GPArotation)

## here we apply varimax orthogonal that give loadings close to simple structure
(vm <- (Varimax(loadings(f3))) )

## Orthogonal rotation method Varimax converged.
## Loadings:
##    Factor1 Factor2 Factor3
## x1  0.320   0.130   0.607
## x2  0.135           0.481
## x3          0.113   0.662
## x4  0.838           0.113
## x5  0.867
## x6  0.815           0.162
## x7  0.102   0.695
## x8          0.704   0.174
## x9  0.170   0.511   0.409
##
##                Factor1 Factor2 Factor3
## SS loadings      2.291   1.286   1.279
## Proportion Var   0.255   0.143   0.142
## Cumulative Var   0.255   0.397   0.540
## AUC              0.869   0.895   0.853
## FSI              0.199   0.247   0.157
```

```
# vm$Th # rotating matrix

## we interpret similarly to loading so looking at coefficient above 0.3
##
## - first factor has a loading above 0.3 even for x1
## - second is the speed factor
## - third is again the visualization factor

## Looking the variability explained by each factor
## - even if the variance explained (SS loadings) and Proportion var are now differen
## - the total cumulative variance explained by the factor model does not change wrt

quartimax(loadings(f3)) ## quartimax, another orthogonal rotation, gives solution clo

## Orthogonal rotation method Quartimax converged.
## Loadings:
##    Factor1 Factor2 Factor3
## x1  0.353   0.124   0.590
## x2  0.158           0.474
## x3  0.115   0.114   0.657
## x4  0.844
## x5  0.869
## x6  0.823           0.122
## x7  0.116   0.692
## x8  0.104   0.702   0.167
## x9  0.202   0.508   0.397
##
##                Factor1 Factor2 Factor3
## SS loadings      2.373   1.267   1.215
## Proportion Var   0.264   0.141   0.135
## Cumulative Var   0.264   0.404   0.539
## AUC              0.860   0.899   0.861
## FSI              0.186   0.253   0.167

oblimin(loadings(f3))   ## first oblique rotation. Loading does not change so much

## Oblique rotation method Oblimin Quartimin converged.
## Loadings:
##    Factor1 Factor2 Factor3
## x1  0.191   0.602
## x2          0.505  -0.117
## x3          0.689
## x4  0.840
## x5  0.888
## x6  0.808
## x7         -0.152   0.723
## x8          0.104   0.701
## x9          0.366   0.463
##
##             Factor1 Factor2 Factor3
```

```
## SS loadings    2.237   1.337   1.278
## AUC            0.890   0.870   0.913
## FSI            0.237   0.182   0.283
##
## Phi:
##         Factor1 Factor2 Factor3
## Factor1   1.000   0.326   0.216
## Factor2   0.326   1.000   0.270
## Factor3   0.216   0.270   1.000

## In oblique rotations as above, Phi matrix is the correlation matrix among
## the factors (in oblique they can be correlated, differently from orthogonal).
## We can see the correlation of oblimin are not negligible; it could be
## reasonable to assume that factor are correlatod and adopt oblimin as rotation

##7) Compute the factor scores using the Bartlett and the Thompson method

## To compute the factor scores from the unrotated solutions we specify scores
## in factanal call. if scores= regression = Thompson method, bartlett = bartlett
##
## There are no preferences clear winner for one or the other; I can choose one
## of the two.
##
## Once obtained we can use these variables in other analyses


## Bartlett
fB <- factanal(~., data=data, factors = 3, scores = "Bartlett")$scores
head(fB, 20)

##      Factor1   Factor2 Factor3
## 1    0.17234 -1.122376  0.1041
## 2   -1.48845  0.796485  1.1498
## 3   -1.98591 -0.029408 -1.4705
## 4   -0.10607  1.297085 -1.2230
## 5   -0.03305 -0.859078  0.5780
## 6   -1.85013  0.526015  1.6542
## 7    1.18463 -2.272099  0.7066
## 8    0.05276 -0.008634 -1.1953
## 9    0.62038 -0.523807  0.3434
## 10   0.65448 -1.975034 -0.6658
## 11  -0.71711 -0.781277 -0.2671
## 12   0.10132  0.621474 -0.9938
## 13  -0.43967  0.637316  0.9091
## 14   0.37820 -0.405284 -0.2149
## 15   1.23557  1.029089 -0.9087
## 16  -0.42981 -0.412737 -0.9235
## 17  -0.67586 -0.958047  0.6750
## 18  -1.25044  0.671245 -0.2463
## 19   1.08475  1.089368 -0.6779
```

```
## 20 -0.11323  2.034852 -1.0964

# eg
# - first individual has Factor1 and 3 close to the median (0) while second
# (Visualization Ability) is  absolutely lower
# - 3rd individual is not good in everything
# - 15 is good in first two ability but not in the third (speed).

## Thompson
fT <- factanal(~., data=data, factors = 3, scores = "regression")$scores
head(fT, 20) #score changes, first individual has low score for factor 2
```

```
##      Factor1  Factor2   Factor3
## 1   0.082472 -0.72586 -0.001592
## 2  -1.212737  0.51620  0.822875
## 3  -1.768413 -0.24769 -1.088699
## 4  -0.044497  0.76669 -0.769782
## 5  -0.066667 -0.53066  0.344127
## 6  -1.530295  0.34996  1.148006
## 7   0.907039 -1.38207  0.367722
## 8   0.012681 -0.08696 -0.838373
## 9   0.515935 -0.28414  0.220987
## 10  0.427257 -1.31537 -0.589241
## 11 -0.679561 -0.58163 -0.262395
## 12  0.099806  0.34801 -0.651018
## 13 -0.317534  0.45926  0.671326
## 14  0.297566 -0.26020 -0.169204
## 15  1.113660  0.69517 -0.531450
## 16 -0.424685 -0.36579 -0.689280
## 17 -0.629028 -0.62960  0.387712
## 18 -1.051793  0.34933 -0.159611
## 19  0.992621  0.74202 -0.369257
## 20 -0.001119  1.26424 -0.628950
```

```
## ------------------------------------------------------------
##
## 8. On the basis of the previous analysis perform a confirmatory factor
## analysis using the function cfa in lavaan.

## To perform confirmatory factor analysis we use lavaan;

## interesting the fact thata we have z test for each parameter we can check if
## each estimated parameter is significant/different

## lavaan needs to have the model specified: name of latent variable and which
## observed are linked to which observed variables. Here we assume a model
## where:
##
## y1 is related to x1-x3
## y2 is related to x4-x6
```

```
## y3 is related to x7-x9
##
## y1,y2,y3 can be related/linked between them; actually lavaan will assume
## correlated latent variabile by default (one need to specify otherwise)
##
## this is an hypothetical model from the exploratory factor analysis



model <- "
visual  =~ x1 + x2 + x3
textual =~ x4 + x5 + x6
speed   =~ x7 + x8 + x9
"

## confirmatory factor analysis:
fit <- cfa(model, data=data)
summary(fit, fit.measures = TRUE)
```

```
## lavaan 0.7-2 ended normally after 35 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        21
##
##   Number of observations                           301
##
## Model Test User Model:
##
##   Test statistic                                85.306
##   Degrees of freedom                                24
##   P-value (Chi-square)                           0.000
##
## Model Test Baseline Model:
##
##   Test statistic                               918.852
##   Degrees of freedom                                36
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.931
##   Tucker-Lewis Index (TLI)                       0.896
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -3737.745
##   Loglikelihood unrestricted model (H1)      -3695.092
##
##   Akaike (AIC)                                7517.490
```

```
##   Bayesian (BIC)                                 7595.339
##   Sample-size adjusted Bayesian (SABIC)          7528.739
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.092
##   90 Percent confidence interval - lower         0.071
##   90 Percent confidence interval - upper         0.114
##   P-value H_0: RMSEA <= 0.050                    0.001
##   P-value H_0: RMSEA >= 0.080                    0.840
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.065
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.959
##   90 Percent confidence interval - lower         0.939
##   90 Percent confidence interval - upper         0.976
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)
##   visual =~
##     x1                1.000
##     x2                0.554    0.100    5.554    0.000
##     x3                0.729    0.109    6.685    0.000
##   textual =~
##     x4                1.000
##     x5                1.113    0.065   17.014    0.000
##     x6                0.926    0.055   16.703    0.000
##   speed =~
##     x7                1.000
##     x8                1.180    0.165    7.152    0.000
##     x9                1.082    0.151    7.155    0.000
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)
##   visual ~~
##     textual           0.408    0.074    5.552    0.000
##     speed             0.262    0.056    4.660    0.000
##   textual ~~
##     speed             0.173    0.049    3.518    0.000
```

```
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)
##    .x1                0.549    0.114    4.833    0.000
##    .x2                1.134    0.102   11.146    0.000
##    .x3                0.844    0.091    9.317    0.000
##    .x4                0.371    0.048    7.779    0.000
##    .x5                0.446    0.058    7.642    0.000
##    .x6                0.356    0.043    8.277    0.000
##    .x7                0.799    0.081    9.823    0.000
##    .x8                0.488    0.074    6.573    0.000
##    .x9                0.566    0.071    8.003    0.000
##     visual            0.809    0.145    5.564    0.000
##     textual           0.979    0.112    8.737    0.000
##     speed             0.384    0.086    4.451    0.000

lblatent::cfa_estimate(model=model, data = data)

## Length  Class   Mode
##      1 lavaan     S4

## by specifying fit.measures=TRUE we have several GOF criteria for evaluating
## the model (otherwise only Chi/AIC/BIC)
##
## - first info about estimation (ML) and the optimizer for the non close
##   solution (NLMINB, a netwon raphson), number of parameters (loadings + other
##   parameters). here we have 9 loadings estimated (y1 to x1 ... y3 to x9) out
##   of 27 possible estimated all the other fixed to 0
##
## - then we have gof tests: the first test is chi square stat (according to we
##   reject the model/not good).  the second one is LRT (again
##   rejected). Consider however that chisq test has a formula like chisq \cdot
##   sample_size so if n is very large (like here), chisq could be very large
##   and easily lead to refuse null.  That is, if n is large this test
##   statistics is not reliable!  This is reason why lavaaan gives other
##   indices (which are transformation of Chisq, to correct for sample size);
##   these are not test but indexes between 0 and 1.
##   If the Chisq reject the model we should check these indicators as well
##   If chisq is not rejected we're okay with the model.
##
## - Comparative fit index and tucker lewis range between 0 and 1; the rule of
##   thumb is that if CFI and TLI are > 0.95 the model is good.  Here we cannot
##   say that the model is very good (one is .93 the other .89)
##
## - AIC and BIC can be used only to compare among different models
##
## - RMSE of approximation is based on matrix of residual. Minimum is 0 and
##   there's no maximum. Rule of thumb: model is good if RMSE < 0.08. Here
##   > 0.09 and thus is not so good. There's a confidence as well: we don't
##   reject the null that is greater than 0.08
```

```
## Let's check another model: the only difference is that now in visual ability
## we include x9 (according to the exploratory analysis loadings)


model2 <- "
visual  =~ x1 + x2 + x3+ x9
textual =~ x4 + x5 + x6
speed   =~ x7 + x8 + x9
"

fit2 <- cfa(model2, data = data)
summary(fit2, fit.measures = TRUE)

## lavaan 0.7-2 ended normally after 34 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        22
##
##   Number of observations                           301
##
## Model Test User Model:
##
##   Test statistic                                52.382
##   Degrees of freedom                                23
##   P-value (Chi-square)                           0.000
##
## Model Test Baseline Model:
##
##   Test statistic                               918.852
##   Degrees of freedom                                36
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.967
##   Tucker-Lewis Index (TLI)                       0.948
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -3721.283
##   Loglikelihood unrestricted model (H1)      -3695.092
##
##   Akaike (AIC)                                7486.566
##   Bayesian (BIC)                              7568.123
##   Sample-size adjusted Bayesian (SABIC)       7498.351
##
```

```
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.065
##   90 Percent confidence interval - lower         0.042
##   90 Percent confidence interval - upper         0.089
##   P-value H_0: RMSEA <= 0.050                    0.133
##   P-value H_0: RMSEA >= 0.080                    0.158
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.045
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.981
##   90 Percent confidence interval - lower         0.964
##   90 Percent confidence interval - upper         0.993
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)
##   visual =~
##     x1                1.000
##     x2                0.578    0.098    5.918    0.000
##     x3                0.754    0.103    7.291    0.000
##     x9                0.437    0.081    5.367    0.000
##   textual =~
##     x4                1.000
##     x5                1.115    0.066   17.016    0.000
##     x6                0.926    0.056   16.685    0.000
##   speed =~
##     x7                1.000
##     x8                1.207    0.185    6.540    0.000
##     x9                0.675    0.112    6.037    0.000
##
## Covariances:
##                    Estimate  Std.Err  z-value  P(>|z|)
##   visual ~~
##     textual           0.396    0.072    5.506    0.000
##     speed             0.177    0.055    3.239    0.001
##   textual ~~
##     speed             0.136    0.051    2.675    0.007
##
## Variances:
```

```
##                    Estimate  Std.Err  z-value  P(>|z|)
##    .x1                0.576    0.100    5.731    0.000
##    .x2                1.120    0.100   11.153    0.000
##    .x3                0.830    0.087    9.515    0.000
##    .x9                0.558    0.060    9.336    0.000
##    .x4                0.373    0.048    7.800    0.000
##    .x5                0.444    0.058    7.602    0.000
##    .x6                0.357    0.043    8.285    0.000
##    .x7                0.740    0.086    8.595    0.000
##    .x8                0.375    0.094    3.973    0.000
##     visual            0.783    0.134    5.842    0.000
##     textual           0.978    0.112    8.728    0.000
##     speed             0.444    0.097    4.567    0.000
## - again chisq suggest to avoid the model, however
##
## - NOW CFI/TLI improved a lot and this is a good solution, just by adding the
##   x9
##
## - RMSE is no 0.06 which is ok (we don't reject it to be higher than 0.08)
##
## So the solution appears to be good and we can start interpreting it; from
## the column of p-value we see if the loading are significant or not
## so for example x9 in visual is not superhigh bvut nonetheless significant
##  in general all the loadings are statistically significant (or the observed variab
##  affected by the latent as specified by the model)

##
## Note that the first loading for each factor is fixed to 1 and not estimated:
## it concerns the identification of the factor model

## in the exploratory factor model we assume that latent variables are y sim
## N(0, I) (for easiness, uncorrelated).

## however this assumption is also related to the identification problem of
## latent variables. Given that latent variables are not known we dont know the
## mean and the scale; this is an indeterminancy related to latent variable. we
## need to fix them otherwise are not identified.

## In the exploratory analysis we solve the identifiability problem
## standardizing them that is by assumeing mean 0 and variance 1 (and no
## cov). This is just one way to do it this. eg there are some problem where we want
## estimate variance/covariance matrix among related variable.
##
## An alternative way to solve identification: what lavaan does is to assume y
## sim N (0, \Gamma) with \Gamma covariance matrix to be estimated. So lavaan
## by default estimate the covariance matrix between latent variables.  However
## if the estimation goes without assumptions this is not identifiable; to
## solve this lavaan fix to 1 the factor loadings for the first variables,
## assuming that the first factor has the same unit of measure and of the first
```

```
## observed variable (an alternative way to solve the identifiability). This is
## reason first loading is fixed to 1.
##
## - The covariances between latent variables are all significant/existent (all
##   p-value are <0.05) with the higher covariance between visual and textual
##
## - At the end we have the variances: first (the .x1 to .x9) are the
##   uniqueness and then the variances of the latent variables
##
## We say model is good: even if we reject model with Chisquare here the other
## reliable masures are quite good




## we can fit the model assuming standard normal y ~ N(0,1) as in exploratory
## factor model. we do this by setting std.lv = TRUE (standardize the data)
##
## the solution is equivalent GOF measures are the same as before (its a
## different parametrization on the same model); however of interest,
##
## - if we go to the matrix of loadings we don't have anymore the first loading
##   fix to 1 but it is estimated; again all the loadings are significant (good!)
##
## - in the "Covariances" section actually we have correlations (because we
##   standardized latent variables); the corr betweren visual and textual
##   ability is 0.45, statistically significant, and the highest one
##
## - furthermore the variance of the latent variables at the end are fixed to 1
##   (again imposed standardized)
fit2b <- cfa(model2, data = HolzingerSwineford1939, std.lv = TRUE)
summary(fit2b,fit.measures=TRUE)
```

```
## lavaan 0.7-2 ended normally after 20 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        22
##
##   Number of observations                           301
##
## Model Test User Model:
##
##   Test statistic                                52.382
##   Degrees of freedom                               23
##   P-value (Chi-square)                          0.000
##
## Model Test Baseline Model:
##
##   Test statistic                               918.852
```

```
##   Degrees of freedom                                36
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.967
##   Tucker-Lewis Index (TLI)                       0.948
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -3721.283
##   Loglikelihood unrestricted model (H1)      -3695.092
##
##   Akaike (AIC)                                7486.566
##   Bayesian (BIC)                              7568.123
##   Sample-size adjusted Bayesian (SABIC)       7498.351
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.065
##   90 Percent confidence interval - lower         0.042
##   90 Percent confidence interval - upper         0.089
##   P-value H_0: RMSEA <= 0.050                    0.133
##   P-value H_0: RMSEA >= 0.080                    0.158
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.045
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.981
##   90 Percent confidence interval - lower         0.964
##   90 Percent confidence interval - upper         0.993
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)
##   visual =~
##     x1                0.885    0.076   11.685    0.000
##     x2                0.511    0.076    6.698    0.000
##     x3                0.667    0.072    9.241    0.000
##     x9                0.387    0.064    6.034    0.000
##   textual =~
```

```
##     x4                0.989    0.057   17.455    0.000
##     x5                1.103    0.063   17.602    0.000
##     x6                0.916    0.054   17.073    0.000
##   speed =~
##     x7                0.666    0.073    9.134    0.000
##     x8                0.804    0.074   10.934    0.000
##     x9                0.450    0.066    6.824    0.000
##
## Covariances:
##                   Estimate  Std.Err  z-value  P(>|z|)
##   visual ~~
##     textual           0.453    0.062    7.242    0.000
##     speed             0.301    0.080    3.763    0.000
##   textual ~~
##     speed             0.206    0.070    2.937    0.003
##
## Variances:
##                   Estimate  Std.Err  z-value  P(>|z|)
##    .x1                0.576    0.100    5.731    0.000
##    .x2                1.120    0.100   11.153    0.000
##    .x3                0.830    0.087    9.515    0.000
##    .x9                0.558    0.060    9.336    0.000
##    .x4                0.373    0.048    7.800    0.000
##    .x5                0.444    0.058    7.602    0.000
##    .x6                0.357    0.043    8.285    0.000
##    .x7                0.740    0.086    8.595    0.000
##    .x8                0.375    0.094    3.973    0.000
##     visual            1.000
##     textual           1.000
##     speed             1.000
```

```
## finally if we specify orthogonal=TRUE if cfa, we impose/want orthogonal/uncorrelated
## latent varaible as in exploratory factor models.
## the first loading will be fixed = 1 but now the covariances are fixed to
## zero/uncorrelated; latent variables variances are estimated btw

fit2c <- cfa(model2, data = HolzingerSwineford1939, orthogonal = TRUE)
summary(fit2c, fit.measures = TRUE)
```

```
## lavaan 0.7-2 ended normally after 31 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        19
##
##   Number of observations                           301
##
## Model Test User Model:
##
##   Test statistic                               106.240
```

```
##   Degrees of freedom                                26
##   P-value (Chi-square)                           0.000
##
## Model Test Baseline Model:
##
##   Test statistic                               918.852
##   Degrees of freedom                                36
##   P-value                                        0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.909
##   Tucker-Lewis Index (TLI)                       0.874
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -3748.212
##   Loglikelihood unrestricted model (H1)      -3695.092
##
##   Akaike (AIC)                                7534.424
##   Bayesian (BIC)                              7604.860
##   Sample-size adjusted Bayesian (SABIC)       7544.602
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.101
##   90 Percent confidence interval - lower         0.082
##   90 Percent confidence interval - upper         0.122
##   P-value H_0: RMSEA <= 0.050                    0.000
##   P-value H_0: RMSEA >= 0.080                    0.962
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.140
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.947
##   90 Percent confidence interval - lower         0.924
##   90 Percent confidence interval - upper         0.965
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)
```

```
##   visual =~
##     x1                1.000
##     x2                0.704    0.120    5.885    0.000
##     x3                0.940    0.140    6.709    0.000
##     x9                0.554    0.092    6.022    0.000
##   textual =~
##     x4                1.000
##     x5                1.133    0.067   16.906    0.000
##     x6                0.924    0.056   16.391    0.000
##   speed =~
##     x7                1.000
##     x8                1.056    0.164    6.431    0.000
##     x9                0.697    0.104    6.679    0.000
##
## Covariances:
##                   Estimate  Std.Err  z-value  P(>|z|)
##   visual ~~
##     textual           0.000
##     speed             0.000
##   textual ~~
##     speed             0.000
##
## Variances:
##                   Estimate  Std.Err  z-value  P(>|z|)
##    .x1                0.737    0.104    7.113    0.000
##    .x2                1.073    0.101   10.604    0.000
##    .x3                0.726    0.096    7.594    0.000
##    .x9                0.525    0.063    8.359    0.000
##    .x4                0.382    0.049    7.805    0.000
##    .x5                0.416    0.059    7.038    0.000
##    .x6                0.369    0.044    8.367    0.000
##    .x7                0.676    0.091    7.404    0.000
##    .x8                0.457    0.089    5.113    0.000
##     visual            0.621    0.126    4.946    0.000
##     textual           0.969    0.112    8.640    0.000
##     speed             0.507    0.108    4.713    0.000
```

```
## if we specify both the conditions to have uncorrelated standard normal
## latent variables (that is actually y sim N(0, I) as in exploratory)
fit2d <- cfa(model2, data = HolzingerSwineford1939, std.lv = TRUE, orthogonal = TRUE)
summary(fit2d, fit.measures = TRUE)
```

```
## lavaan 0.7-2 ended normally after 20 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                        19
##
##   Number of observations                           301
##
```

```
## Model Test User Model:
##
##   Test statistic                                106.240
##   Degrees of freedom                                 26
##   P-value (Chi-square)                            0.000
##
## Model Test Baseline Model:
##
##   Test statistic                                918.852
##   Degrees of freedom                                 36
##   P-value                                         0.000
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                     0.909
##   Tucker-Lewis Index (TLI)                        0.874
##
## Loglikelihood and Information Criteria:
##
##   Loglikelihood user model (H0)              -3748.212
##   Loglikelihood unrestricted model (H1)      -3695.092
##
##   Akaike (AIC)                                7534.424
##   Bayesian (BIC)                              7604.860
##   Sample-size adjusted Bayesian (SABIC)       7544.602
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.101
##   90 Percent confidence interval - lower         0.082
##   90 Percent confidence interval - upper         0.122
##   P-value H_0: RMSEA <= 0.050                    0.000
##   P-value H_0: RMSEA >= 0.080                    0.962
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.140
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.947
##   90 Percent confidence interval - lower         0.924
##   90 Percent confidence interval - upper         0.965
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
```

```
##
## Latent Variables:
##                   Estimate  Std.Err  z-value  P(>|z|)
##   visual =~
##     x1                0.788    0.080    9.892    0.000
##     x2                0.555    0.079    7.064    0.000
##     x3                0.741    0.077    9.646    0.000
##     x9                0.436    0.060    7.317    0.000
##   textual =~
##     x4                0.984    0.057   17.280    0.000
##     x5                1.115    0.063   17.812    0.000
##     x6                0.910    0.054   16.854    0.000
##   speed =~
##     x7                0.712    0.076    9.426    0.000
##     x8                0.752    0.073   10.256    0.000
##     x9                0.497    0.060    8.224    0.000
##
## Covariances:
##                   Estimate  Std.Err  z-value  P(>|z|)
##   visual ~~
##     textual           0.000
##     speed             0.000
##   textual ~~
##     speed             0.000
##
## Variances:
##                   Estimate  Std.Err  z-value  P(>|z|)
##    .x1                0.737    0.104    7.113    0.000
##    .x2                1.073    0.101   10.604    0.000
##    .x3                0.726    0.096    7.594    0.000
##    .x9                0.525    0.063    8.359    0.000
##    .x4                0.382    0.049    7.805    0.000
##    .x5                0.416    0.059    7.038    0.000
##    .x6                0.369    0.044    8.367    0.000
##    .x7                0.676    0.091    7.404    0.000
##    .x8                0.457    0.089    5.113    0.000
##     visual            1.000
##     textual           1.000
##     speed             1.000
```

```
## ------------------------------------------------------------
## EXERCISE 2: social mobility in UK
## ------------------------------------------------------------

## The data are based on information provided by 713 male or female married
## respondents to a survey carried out in 1949. The variables relate to the
## respondent, his or his spouse, father, father-in-law, and firstborn son. The
## file socmob.txt contains the full correlation matrix.
```

```
## The 10 variables are coded as follows:
## X1=Husband's father's occupational status
## X2=Wife's father's occupational status
## X3=Husband's further education
## X4=Husband's qualifications
## X5=Husband's occupational status
## X6=Wife's further education
## X7=Wife's qualifications
## X8=Firstborn's further education
## X9=Firstborn's qualifications
## X10=Firstborn's occupational status

## 1. Load the correlation matrix socmob.txt.
## ------------------------------------------
library(lbdatasets)
socmob

##        V1    V2    V3    V4    V5    V6    V7    V8    V9   V10
## V1  1.000 0.372 0.231 0.100 0.431 0.171 0.128 0.175 0.077 0.293
## V2  0.372 1.000 0.225 0.134 0.375 0.149 0.096 0.184 0.095 0.283
## V3  0.231 0.225 1.000 0.534 0.354 0.276 0.281 0.318 0.247 0.287
## V4  0.100 0.134 0.534 1.000 0.235 0.234 0.380 0.310 0.348 0.215
## V5  0.431 0.375 0.354 0.235 1.000 0.196 0.138 0.231 0.109 0.444
## V6  0.171 0.149 0.276 0.234 0.196 1.000 0.473 0.260 0.115 0.189
## V7  0.128 0.096 0.281 0.380 0.138 0.473 1.000 0.208 0.191 0.161
## V8  0.175 0.184 0.318 0.310 0.231 0.260 0.208 1.000 0.500 0.438
## V9  0.077 0.095 0.247 0.348 0.109 0.115 0.191 0.500 1.000 0.331
## V10 0.293 0.283 0.287 0.215 0.444 0.189 0.161 0.438 0.331 1.000

## ## socmob is a correlation matrix out of the box
## cormat <- as.matrix(read.table("socmob.txt")) #if only "read.table" --> data frame
## rownames(cormat) <- colnames(cormat)
## cormat

# (arbitrarily) considering 0.3 as a cutoff:
# V1 correlated with V2 and V5
# V2 correlated with V1 and V5
# V3 correlated with V4, V5, V8
# V4 correlated with V3, V7, V8, V9
# V5 correlated with V1, V2, V3, V10
# V6 correlated with V7
# V7 correlated with V4, V6
# V8 correlated with V3, V4, V9, V10
# V9 correlated with V4, V8, V10
# V10 correlated with V5, V8, V9

# We can suspect that:
# - V1, V2 and V5 share a common latent factor;
# - V8, V9, V10 share a common latent factor
```

```
# There are other correlations that might be explained by other latent factors


## 2. Perform an exploratory factor analysis.
# ----------------------------------------
## with a correlation matrix we have to provide n




n <- 713
lblatent::efa_preliminary(cormatrix=socmob, nobs=n, nfactors=1:4)

## $readable_cormatrix
##     V1    V2 V3    V4    V5 V6    V7    V8    V9   V10
## V1   1 0.372 NA    NA 0.431 NA    NA    NA    NA    NA
## V2  NA 1.000 NA    NA 0.375 NA    NA    NA    NA    NA
## V3  NA    NA  1 0.534 0.354 NA    NA 0.318    NA    NA
## V4  NA    NA NA 1.000    NA NA 0.380 0.310 0.348    NA
## V5  NA    NA NA    NA 1.000 NA    NA    NA    NA 0.444
## V6  NA    NA NA    NA    NA  1 0.473    NA    NA    NA
## V7  NA    NA NA    NA    NA NA 1.000    NA    NA    NA
## V8  NA    NA NA    NA    NA NA    NA 1.000 0.500 0.438
## V9  NA    NA NA    NA    NA NA    NA    NA 1.000 0.331
## V10 NA    NA NA    NA    NA NA    NA    NA    NA 1.000
##
## $chisq
##      chi df         p rounded_p    comment
## 1 565.34 35 5.996e-97    0.0000 bad/reject
## 2 316.90 26 8.695e-52    0.0000 bad/reject
## 3 143.80 18 1.183e-21    0.0000 bad/reject
## 4  16.62 11 1.198e-01    0.1198         ok

## # varimax by default
## f1 <- factanal(~V1+V2+V3+V4+V5+V6+V7+V8+V9+V10, covmat = socmob, n.obs = n, 1, rotation = "n
## f2 <- factanal(~V1+V2+V3+V4+V5+V6+V7+V8+V9+V10, covmat = socmob, n.obs = n, 2, rotation = "n
## f3 <- factanal(~V1+V2+V3+V4+V5+V6+V7+V8+V9+V10, covmat = socmob, n.obs = n, 3, rotation = "n
## f4 <- factanal(~V1+V2+V3+V4+V5+V6+V7+V8+V9+V10, covmat = socmob, n.obs = n, 4, rotation = "n

## ## Chisq<-round(c(f1$STATISTIC,f2$STATISTIC,f3$STATISTIC, f4$STATISTIC),3)
## ## df<-c(f1$dof,f2$dof,f3$dof, f4$dof)                          # |(p-q)^2-(p+q)|/2 = |(10-q)^
## ## pvalue<-round(c(f1$PVAL,f2$PVAL,f3$PVAL, f4$PVAL),4)


# The 4-factors model seems a good fit for this specific correlation matrix
# Nevertheless, the communality of V6 is very close to 1
# communalities are
#
# V1        V2        V3        V4        V5        V6        V7        V8
# 0.3498912 0.2785486 0.4082032 0.9222036 0.5723136 0.9950000 0.3094194 0.5806686
```

```
# V9       V10
# 0.4754036 0.4585646
#
# Some software will raise a warning in this case,
# because during the fitting process the communality V6 exceeded 1 ("Heywood case")
# REF: "Analysis of multivariate social science data" - Bartholomew et al.


lblatent::efa_estimate(cormatrix=socmob, nobs=n, nfactors=4)
## $model
##
## Call:
## factanal(factors = nfactors, covmat = cormatrix, n.obs = nobs,     rotation = "non
##
## Uniquenesses:
##    V1    V2    V3    V4    V5    V6    V7    V8    V9   V10
## 0.650 0.721 0.592 0.078 0.428 0.005 0.691 0.419 0.525 0.541
##
## Loadings:
##     Factor1 Factor2 Factor3 Factor4
## V1    0.122   0.176   0.475  -0.281
## V2    0.157   0.154   0.423  -0.227
## V3    0.529   0.288   0.193
## V4    0.916   0.253  -0.134
## V5    0.271   0.204   0.571  -0.362
## V6            0.997
## V7    0.279   0.480
## V8    0.343   0.269   0.439   0.445
## V9    0.401   0.124   0.270   0.476
## V10   0.268   0.197   0.584
##
##                Factor1 Factor2 Factor3 Factor4
## SS loadings      1.661   1.595   1.393   0.702
## Proportion Var   0.166   0.159   0.139   0.070
## Cumulative Var   0.166   0.326   0.465   0.535
##
## Test of the hypothesis that 4 factors are sufficient.
## The chi square statistic is 16.62 on 11 degrees of freedom.
## The p-value is 0.12
##
## $communalities
##     V1     V2     V3     V4     V5     V6     V7     V8     V9    V10
## 0.3499 0.2785 0.4082 0.9222 0.5723 0.9950 0.3094 0.5807 0.4754 0.4586
##
## $uniqueness
##     V1     V2     V3     V4     V5     V6     V7     V8     V9    V10
## 0.6501 0.7215 0.5918 0.0778 0.4277 0.0050 0.6906 0.4193 0.5246 0.5414
##
## $cor_original
```

```
##     V1   V2   V3   V4   V5   V6   V7   V8   V9  V10
## V1   1 0.37 0.23 0.10 0.43 0.17 0.13 0.17 0.08 0.29
## V2  NA 1.00 0.22 0.13 0.38 0.15 0.10 0.18 0.10 0.28
## V3  NA   NA 1.00 0.53 0.35 0.28 0.28 0.32 0.25 0.29
## V4  NA   NA   NA 1.00 0.23 0.23 0.38 0.31 0.35 0.22
## V5  NA   NA   NA   NA 1.00 0.20 0.14 0.23 0.11 0.44
## V6  NA   NA   NA   NA   NA 1.00 0.47 0.26 0.12 0.19
## V7  NA   NA   NA   NA   NA   NA 1.00 0.21 0.19 0.16
## V8  NA   NA   NA   NA   NA   NA   NA 1.00 0.50 0.44
## V9  NA   NA   NA   NA   NA   NA   NA   NA 1.00 0.33
## V10 NA   NA   NA   NA   NA   NA   NA   NA   NA 1.00
##
## $cor_reproduced
##       V1   V2   V3   V4   V5   V6   V7   V8   V9  V10
## V1  0.35 0.31 0.23 0.10 0.44 0.17 0.10 0.17 0.07 0.32
## V2    NA 0.28 0.23 0.13 0.40 0.15 0.10 0.18 0.09 0.30
## V3    NA   NA 0.41 0.53 0.35 0.28 0.28 0.30 0.26 0.30
## V4    NA   NA   NA 0.92 0.23 0.23 0.38 0.31 0.35 0.21
## V5    NA   NA   NA   NA 0.57 0.20 0.15 0.24 0.12 0.42
## V6    NA   NA   NA   NA   NA 1.00 0.47 0.26 0.12 0.19
## V7    NA   NA   NA   NA   NA   NA 0.31 0.22 0.17 0.16
## V8    NA   NA   NA   NA   NA   NA   NA 0.58 0.50 0.44
## V9    NA   NA   NA   NA   NA   NA   NA   NA 0.48 0.33
## V10   NA   NA   NA   NA   NA   NA   NA   NA   NA 0.46
##
## $cor_residual
##       V1   V2   V3   V4    V5 V6    V7    V8    V9   V10
## V1  0.65 0.06 0.00 0.00 -0.01  0  0.03  0.00  0.01 -0.03
## V2    NA 0.72 0.00 0.00 -0.02  0 -0.01  0.00  0.01 -0.02
## V3    NA   NA 0.59 0.00  0.01  0  0.00  0.01 -0.01 -0.02
## V4    NA   NA   NA 0.08  0.00  0  0.00  0.00  0.00  0.00
## V5    NA   NA   NA   NA  0.43  0 -0.01 -0.01 -0.01  0.03
## V6    NA   NA   NA   NA    NA  0  0.00  0.00  0.00  0.00
## V7    NA   NA   NA   NA    NA NA  0.69 -0.01  0.02  0.00
## V8    NA   NA   NA   NA    NA NA    NA  0.42  0.00  0.00
## V9    NA   NA   NA   NA    NA NA    NA    NA  0.52  0.00
## V10   NA   NA   NA   NA    NA NA    NA    NA    NA  0.54
##
## $loadings_raw
##     Factor1 Factor2 Factor3 Factor4
## V1       NA      NA  0.4747      NA
## V2       NA      NA  0.4229      NA
## V3   0.5286      NA      NA      NA
## V4   0.9162      NA      NA      NA
## V5       NA      NA  0.5713 -0.3617
## V6       NA  0.9973      NA      NA
## V7       NA  0.4799      NA      NA
## V8   0.3434      NA  0.4389  0.4446
## V9   0.4007      NA      NA  0.4758
```

```
## V10      NA      NA  0.5842      NA
##
## $loadings_varimax
##     Factor1 Factor2 Factor3 Factor4
## V1       NA      NA  0.5737      NA
## V2       NA      NA  0.5043      NA
## V3   0.4936      NA  0.3106      NA
## V4   0.9393      NA      NA      NA
## V5       NA      NA  0.7224      NA
## V6       NA  0.9857      NA      NA
## V7   0.3293  0.4320      NA      NA
## V8       NA      NA      NA  0.6984
## V9       NA      NA      NA  0.6340
## V10      NA      NA  0.4711  0.4621
## attr(,"covariance")
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    0    1    0    0
## [3,]    0    0    1    0
## [4,]    0    0    0    1
##
## $loadings_quartimax
##     Factor1 Factor2 Factor3 Factor4
## V1       NA      NA  0.5764      NA
## V2       NA      NA  0.5076      NA
## V3   0.4965      NA  0.3176      NA
## V4   0.9417      NA      NA      NA
## V5       NA      NA  0.7259      NA
## V6       NA  0.9849      NA      NA
## V7   0.3326  0.4306      NA      NA
## V8       NA      NA      NA  0.6899
## V9       NA      NA      NA  0.6293
## V10      NA      NA  0.4852  0.4460
## attr(,"covariance")
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    0    1    0    0
## [3,]    0    0    1    0
## [4,]    0    0    0    1
##
## $loadings_oblimin
##     Factor1 Factor2 Factor3 Factor4
## V1       NA      NA  0.5977      NA
## V2       NA      NA  0.5189      NA
## V3   0.4366      NA      NA      NA
## V4   0.9576      NA      NA      NA
## V5       NA      NA  0.7480      NA
## V6       NA  1.0009      NA      NA
## V7       NA  0.3974      NA      NA
```

```
## V8       NA      NA      NA  0.7313
## V9       NA      NA      NA  0.6797
## V10      NA      NA  0.4238  0.4285
## attr(,"covariance")
##        [,1]   [,2]   [,3]   [,4]
## [1,] 1.0000 0.2586 0.2487 0.4206
## [2,] 0.2586 1.0000 0.2595 0.2512
## [3,] 0.2487 0.2595 1.0000 0.3601
## [4,] 0.4206 0.2512 0.3601 1.0000

# Therefore, we keep the 3-factors model, even if it is a worse fit

# 2 - On the basis of the matrix of the factor loadings, interpret the
#     factors individuated

mod <- lblatent::efa_estimate(cormatrix=socmob, nobs=n, nfactors=3)
mod$loadings_raw

##     Factor1 Factor2 Factor3
## V1   0.4258  0.4028      NA
## V2   0.4043  0.3432      NA
## V3   0.5918      NA      NA
## V4   0.5578      NA      NA
## V5   0.5750  0.4808      NA
## V6   0.4508      NA  0.3685
## V7   0.4770      NA  0.4623
## V8   0.6147      NA      NA
## V9   0.5190 -0.3581 -0.3806
## V10  0.6024      NA      NA

## Looking at loadings_raw
# F1: Positively correlated with all variables (not a big help in discriminating)
# F2: Occupational status of family members (V1, V2, V5) vs qualifications (V4, V7, V9)
# F3: Variables about first born (V8, V9, V10) vs educational status of the mother (V6, V7)



# 3 - Compute the communalities and comment them.
#     Which percentage of the variance of the model is explained by the three-factor model?

# percentage of variance "explained" by the latent factors, not "specific" to the
# variable
mod$communalities

##     V1     V2     V3     V4     V5     V6     V7     V8     V9    V10
## 0.3464 0.2813 0.3642 0.3830 0.5628 0.3550 0.5286 0.4976 0.5424 0.4389

## variance explained is SS laodings: that is
# 2.777631      SS loadings Factor 1
# 0.8658556     SS loadings Factor 2
```

```
# 0.6567901       SS loadings Factor 3

## with 10 variable the proportion of total variance explained by each factor is
# 0.2777631        Proportion Var Factor 1
# 0.08658556       Proportion Var Factor 2
# 0.06567901     Proportion Var Factor 3

## cumulative proportion explained
# 0.4300276  Cumulative Var  Fact1+Fact2+Fact3

# 4 - Compute the reproduced correlation matrix and the discrepancy
#     between the observed and reproduced correlation

mod$cor_reproduced

##       V1   V2   V3   V4   V5   V6   V7   V8   V9  V10
## V1  0.35 0.31 0.25 0.15 0.44 0.16 0.11 0.17 0.06 0.31
## V2    NA 0.28 0.23 0.14 0.40 0.14 0.10 0.18 0.08 0.30
## V3    NA   NA 0.36 0.35 0.33 0.31 0.34 0.34 0.27 0.33
## V4    NA   NA   NA 0.38 0.21 0.33 0.39 0.35 0.33 0.27
## V5    NA   NA   NA   NA 0.56 0.21 0.15 0.25 0.11 0.42
## V6    NA   NA   NA   NA   NA 0.35 0.42 0.19 0.14 0.17
## V7    NA   NA   NA   NA   NA   NA 0.53 0.22 0.18 0.14
## V8    NA   NA   NA   NA   NA   NA   NA 0.50 0.50 0.40
## V9    NA   NA   NA   NA   NA   NA   NA   NA 0.54 0.34
## V10   NA   NA   NA   NA   NA   NA   NA   NA   NA 0.44

mod$cor_residual

##       V1   V2    V3    V4    V5    V6    V7    V8    V9   V10
## V1  0.65 0.06 -0.02 -0.05 -0.01  0.01  0.02  0.00  0.02 -0.02
## V2    NA 0.72 -0.01 -0.01 -0.02  0.01  0.00  0.00  0.01 -0.02
## V3    NA   NA  0.64  0.18  0.02 -0.04 -0.06 -0.02 -0.03 -0.04
## V4    NA   NA    NA  0.62  0.03 -0.09 -0.01 -0.04  0.02 -0.06
## V5    NA   NA    NA    NA  0.44 -0.01 -0.01 -0.02 -0.01  0.02
## V6    NA   NA    NA    NA    NA  0.64  0.05  0.06 -0.02  0.02
## V7    NA   NA    NA    NA    NA    NA  0.47 -0.01  0.01  0.02
## V8    NA   NA    NA    NA    NA    NA    NA  0.50  0.00  0.04
## V9    NA   NA    NA    NA    NA    NA    NA    NA  0.46  0.00
## V10   NA   NA    NA    NA    NA    NA    NA    NA    NA  0.56

# In the residuals check the values outside the main diagonal: should be close
# to 0 (--> if not consider exclusion) on the main diagonal we should have
# something similar to uniquenesses (reasonably ok)

mod$uniqueness

##     V1     V2     V3     V4     V5     V6     V7     V8     V9    V10
## 0.6536 0.7187 0.6358 0.6170 0.4372 0.6450 0.4714 0.5024 0.4576 0.5611
```

```
# 5 - Apply different orthogonal and oblique rotations and interpret the solutions obtained

mod$loadings_varimax

##     Factor1 Factor2 Factor3
## V1       NA  0.5784      NA
## V2       NA  0.5172      NA
## V3   0.3004  0.3363  0.4011
## V4   0.3748      NA  0.4709
## V5       NA  0.7301      NA
## V6       NA      NA  0.5617
## V7       NA      NA  0.7136
## V8   0.6507      NA      NA
## V9   0.7268      NA      NA
## V10  0.4360  0.4931      NA
## attr(,"covariance")
##      [,1] [,2] [,3]
## [1,]    1    0    0
## [2,]    0    1    0
## [3,]    0    0    1

# Orthogonal rotation method varimax converged.
# Loadings:
#   Factor1 Factor2 Factor3
# V1   0.5784  0.0453  0.0992
# V2   0.5172  0.0881  0.0774
# V3   0.3363  0.3004  0.4011
# V4   0.1441  0.3748  0.4709
# V5   0.7301  0.1164  0.1273
# V6   0.1741  0.0957  0.5617
# V7   0.0557  0.1274  0.7136
# V8   0.2130  0.6507  0.1700
# V9   0.0205  0.7268  0.1173
# V10  0.4931  0.4360  0.0751
#
# Factor1 Factor2 Factor3
# SS loadings      1.591   1.421   1.288
# Proportion Var   0.159   0.142   0.129
# Cumulative Var   0.159   0.301   0.430

# Several variables have moderate loadings on more than one factor
# F1: Occupation (V1, V2, V5, V10)
# F2: Firstborn son (V8, V9, V10)
# F3: Parents' education and occupation (V3, V4, V6, V7)

mod$loadings_quartimax

##     Factor1 Factor2 Factor3
## V1       NA  0.5817      NA
## V2       NA  0.5209      NA
```

```
## V3       NA  0.3533  0.3867
## V4   0.3794      NA  0.4604
## V5       NA  0.7356      NA
## V6       NA      NA  0.5551
## V7       NA      NA  0.7096
## V8   0.6487      NA      NA
## V9   0.7280      NA      NA
## V10  0.4265  0.5040      NA
## attr(,"covariance")
##      [,1] [,2] [,3]
## [1,]    1    0    0
## [2,]    0    1    0
## [3,]    0    0    1

# Orthogonal rotation method Quartimax converged.
# Loadings:
#   Factor1 Factor2 Factor3
# V1   0.5817  0.0345  0.0828
# V2   0.5209  0.0783  0.0620
# V3   0.3533  0.2998  0.3867
# V4   0.1646  0.3794  0.4604
# V5   0.7356  0.1029  0.1056
# V6   0.1912  0.1012  0.5551
# V7   0.0778  0.1380  0.7096
# V8   0.2311  0.6487  0.1530
# V9   0.0389  0.7280  0.1042
# V10  0.5040  0.4265  0.0543

#                Factor1 Factor2 Factor3
# SS loadings      1.654   1.414   1.232
# Proportion Var   0.165   0.141   0.123
# Cumulative Var   0.165   0.307   0.430

# Loadings > 0.3:
# F1: V1, V2, V3, V5, V10
# F2: V4, V8, V9, V10
# F3: V3, V4, V6, V7
# Very similar to Varimax

## round(quartimax(loadings(f3))$Th, 5)

## #          [,1]    [,2]     [,3]
## # [1,]  0.60995 0.62678  0.48487
## # [2,] -0.47484 0.77895 -0.40960
## # [3,] -0.63442 0.01960  0.77274


mod$loadings_oblimin

##     Factor1 Factor2 Factor3
```

```
## V1       NA  0.6018      NA
## V2       NA  0.5332      NA
## V3       NA      NA  0.3438
## V4       NA      NA  0.4337
## V5       NA  0.7513      NA
## V6       NA      NA  0.5739
## V7       NA      NA  0.7503
## V8   0.6338      NA      NA
## V9   0.7604      NA      NA
## V10  0.3726  0.4602      NA
## attr(,"covariance")
##        [,1]   [,2]   [,3]
## [1,] 1.0000 0.3421 0.3829
## [2,] 0.3421 1.0000 0.3258
## [3,] 0.3829 0.3258 1.0000
```

```
# exploratory model assumes no correlation between factors
# but we can rotate it a posteriori to get correlated factors

# Oblique rotation method Oblimin Quartimin converged.
# Loadings:
#   Factor1  Factor2   Factor3
# V1   0.6018 -0.07079  0.023276
# V2   0.5332 -0.00938  0.000864
# V3   0.2652  0.18468  0.343805
# V4   0.0389  0.28383  0.433736
# V5   0.7513 -0.02550  0.022109
# V6   0.0967 -0.04299  0.573923
# V7  -0.0558 -0.01982  0.750328
# V8   0.1168  0.63381  0.051629
# V9  -0.0933  0.76039  0.007659
# V10  0.4602  0.37259 -0.054741

# Loadings > 0.3:
# F1: V1, V2, V5, V10
# F2: V8, V9, V10
# F3: V3, V4, V6, V7
# Very similar to varimax

#                Factor1 Factor2 Factor3
# SS loadings      1.629   1.360   1.311
# Proportion Var   0.163   0.136   0.131
# Cumulative Var   0.163   0.299   0.430
#
# Phi:
#           Factor1 Factor2 Factor3
# Factor1   1.000   0.342   0.326
# Factor2   0.342   1.000   0.383
# Factor3   0.326   0.383   1.000
```

```
# All pairs of factors show moderate correlation: reasonable interpretation

## round(oblimin(loadings(f3))$Th, 5)

## #          [,1]    [,2]     [,3]
## # [1,]  0.76982 0.77056  0.71332
## # [2,] -0.38920 0.63734 -0.35685
## # [3,] -0.50586 0.00599  0.60318


## f3.vm <- factanal(~V1+V2+V3+V4+V5+V6+V7+V8+V9+V10, covmat = socmob, n.obs = n, 3)
## # default rotation = "varimax"
## loadings(f3.vm)

## # Loadings:
## #   Factor1 Factor2 Factor3
## # V1  0.576           0.111
## # V2  0.516
## # V3  0.329   0.288   0.416
## # V4  0.135   0.360   0.485
## # V5  0.728   0.113   0.144
## # V6  0.164           0.567
## # V7          0.106   0.718
## # V8  0.209   0.645   0.194
## # V9          0.723   0.140
## # V10 0.491   0.434
## #
## # Factor1 Factor2 Factor3
## # SS loadings      1.568   1.379   1.353
## # Proportion Var   0.157   0.138   0.135
## # Cumulative Var   0.157   0.295   0.430

## varimax(loadings(f3))

## # Loadings:
## #   Factor1 Factor2 Factor3
## # V1          0.576   0.111
## # V2          0.516
## # V3  0.288   0.329   0.416
## # V4  0.360   0.135   0.485
## # V5  0.113   0.728   0.144
## # V6          0.164   0.567
## # V7  0.106           0.718
## # V8  0.645   0.209   0.194
## # V9  0.723           0.140
## # V10 0.434   0.491
## #
## # Factor1 Factor2 Factor3
## # SS loadings      1.379   1.568   1.353
```

```
## # Proportion Var   0.138   0.157   0.135
## # Cumulative Var   0.138   0.295   0.430
## #
## # $rotmat
## # [,1]          [,2]       [,3]
## # [1,]  0.5986297  0.590525147  0.5412232
## # [2,] -0.4389402  0.807018509 -0.3950350
## # [3,] -0.6700552 -0.001084975  0.7423104

## ?Varimax # package GPArotation
## ?varimax # package stats (like factanal)



# 6 - Compute the factor scores using the Bartlett and the Thompson method

# This cannot be done, since we don't have the original matrix X of observed values

# 7 - On the basis of the previous analysis, perform a confirmatory FA using cfa in lavaan

library(lavaan)

socmob_model <- "
overall =~ V1 + V2 + V3 + V4 + V5 + V6 + V7 + V8 + V9 + V10
occup_vs_quali =~ V1 + V2 + V4 + V5 + V7 + V9
firstborn_vs_edumother =~ V6 + V7 + V8 + V9 + V10
"


lblatent::cfa_estimate(model=socmob_model, cormatrix = socmob, nobs = n, orthogonal = TRUE)

## Length  Class   Mode
##      1 lavaan     S4

# fit <- cfa(socmob_model, sample.cov = socmob, sample.nobs = n, orthogonal = T)
# assuming no correlation between latent variables
# don't assume standard normal distribution

# we don't know scale and unit of measure of the latent variables.
# In exploratory analysis we solve the problem by assuming standard normal distribution
# Nevertheless, sometimes it can be useful or necessary to relax such assumption;
# In such case, one can assume that y is normally distributed with mean 0 and covariance gamma
# gamma is estimated and the first coefficient is set to 1
# which means that we assume the factor to have the same scale and unit of measure as the first

## The first chisq (202.409) can be very large if n is large, so sometimes not
## reliable) and thus it's p.value prone to type 1 error

## - we see CFI and TLI are both < 0.95 not good
```

```
## - RMSEA not good (0.10 > 0.08) and furthermore is extremely unlikely
##   (P-value H_0: RMSEA <= 0.050 = 0.000) that RMSEA <= 0.05 no good fit


## the stats are the same but the first loadings are now estimated
lblatent::cfa_estimate(socmob_model, cormatrix = socmob, nobs = n, std.lv = T, orthog

## Length  Class   Mode
##      1 lavaan     S4

## below not assuming orthogonal factors: problems in estimation out of
## Avvertimento: lavaan->lav_model_vcov():
##    Could not compute standard errors! The information matrix could not be
##    inverted. This may be a symptom that the model is not identified.
lblatent::cfa_estimate(socmob_model, cormatrix = socmob, nobs = n, std.lv = T, orthog

## Warning:  lavaan->lav_model_vcov():
##    Could not compute standard errors!  The information matrix could
not be
##    inverted.  This may be a symptom that the model is not identified.

## Length  Class   Mode
##      1 lavaan     S4
```

Chapter 3

Latent trait models

Remark 16. This when observed variable are binary and latent variables are continuous

3.1 Introduction

Binary responses

- are very common in social and educational sciences (eg agree/not agree).
- are often supposed to be indicators of more fundamental attitudes or abilities
- with p questions to which the respondent is required to give a binary response (right/wrong, agree/disagree, yes/no) we have 2^p theoretical/possible **response patterns**, that is possible string of zeros and ones. The number of actually observed response patterns depends on n.

3.1.1 Example

Example 3.1.1 (Attitude towards abortion)**.** An inquiry on $n = 379$ individuals with four agree/disagree items where questions were "Do you agree with abortion under these conditions?"

1. the woman decides on her own [WomanDecide]
2. the couple agree that they do not wish to have the child [CoupleDecide]
3. the woman is not married and does not wish to marry the man [NotMarried]
4. the couple cannot afford any more children [CannotAfford]

	Yes	No
Yes	159	7
No	66	147

Table 3.1: items1/2 cross tab

Response pattern	Frequency
1111	141
0000	103
0111	44
0011	21
0001	13
1110	12
0010	10
0110	7
1011	6
0101	6
1101	3
1100	3
1000	1
1010	0
1001	0
Total	379

Table 3.2: Response patterns freqs

Percentage of individual agreeing that abortion should be legal under circumstances described by item 1 to 4 are: 43.8, 59.4, 63.6, 61.7. Cross tabulation of items 1 and 2 is reported in 3.1.
It's probable that the response on the questions depends on the latent variable expressing the attitude toward abortion (one in favour or against tend to respond consistently). Aim of the analysis:

1. measure a single latent variable expressing the attitude toward abortion
2. ranking the individuals according to the latent variable

If $p = 4$ items (0=agree, 1=disagree) we have $2^4 = 16$ possible response patterns, of which just 14 have positive frequency; response patterns frequencies are reported in table 3.2. We see that 141+103, large majority reports consistent responses.

Remark 17. In normal model we check correlation between embarking in model estimation. Here to check correlation we look at chi square for every possible couple of variables

3.1.2 Approaches

Two approaches to introduce these models:

1. *Response function approach*: it analyzes the data as they are by imposing distributions on the observed variables (we choose a proper distribution for the binary data).
2. *Underlying variable approach* (SEM, structural equation modelling, a framework in which we specify factor models but also estimate the relation between them): it assumes that the data have been produced by *dichotomizing underlying continuous variables* (and this simplifies computation).

3.2 Response function approach

We don't know the joint probability distribution $f(\mathbf{x})$ which assigns probabilities to the 2^p response patterns (in the population).
For some purposes it is useful to specify the joint distribution in terms of set of *marginal probabilities*:

$$
\begin{aligned}
&P(x_i = 1) \quad (i = 1, 2, \ldots, p) \\
&P(x_i = 1, x_j = 1) \quad (i, j = 1, 2, \ldots, p) \quad (i \neq j) \\
&P(x_i = 1, x_j = 1, x_k = 1) \quad (i, j, k = 1, 2, \ldots, p) \quad (i \neq j \neq k) \\
&\vdots \\
&P(x_1 = 1, x_2 = 1, \ldots, x_p = 1)
\end{aligned}
$$

Total number of probabilities mass functions are $2^p - 1$

$$\binom{p}{1} + \binom{p}{2} + \cdots + \binom{p}{p} = 2^p - 1$$

where we have $\binom{p}{1}$ possible single-marginals $\binom{p}{2}$ bivariate joints, and $\binom{p}{p}$ 1 single overall joint probability

3.2.1 Logit/normal model

Retaking our general framework of

$$h(\mathbf{y}|\mathbf{x}) = \frac{h(\mathbf{y})g(\mathbf{x}|\mathbf{y})}{f(\mathbf{x})}$$

for

- $h(\mathbf{y})$: for simplicity being $\mathbf{y}$ continuous we assume (the prior) is multivariate standard normal $h(\mathbf{y}) \sim \mathrm{N}(\mathbf{0}, \mathbf{I})$ that is the latent variables are *standardized independent normal* variables;
- for $g(\mathbf{x}|\mathbf{y})$ is useful to use the conditional independence (if the dependence among the x's is wholly explained by a vector $\mathbf{y}$ of latent variables they may be regarded as mutually independent random variables). That is we assume

  $$g(\mathbf{x}|\mathbf{y}) = \prod_{i=1}^{p} g(x_i|\mathbf{y})$$

  where for notation we have

  $$g(x_i|\mathbf{y}) = \mathbb{P}(x_i = 1|\mathbf{y}) = \pi_i(\mathbf{y}) \quad (x_i = 0, 1; i = 1, \ldots, p)$$

  We assume $g(x_i|\mathbf{y})$ is bernoulli with parameter π_i, depending on latent variable.
  These probabilities $\pi_i(\mathbf{y})$ are called *item response function or item characteristic curve* with $0 \leq \pi_i(\mathbf{y}) \leq 1$ (item characteristic curve in IRT) and are not decreasing monotonic function.

Definition 3.2.1 (Logit/normal model)**.** In this model[1], the conditional distribution $g(\mathbf{x}|\mathbf{y})$ is assumed to a be a GLLVM as follows:

- Random component, we assume that $g_i(x_i|\mathbf{y}) \sim Ber(\pi_i(\mathbf{y}))$ with $\pi_i(\mathbf{y})$ probability of success, that is

$$g(x_i|\mathbf{y}) = \pi_i(\mathbf{y})^{x_i}(1-\pi_i(\mathbf{y}))^{1-x_i}$$

- Systematic component: the linear predictor is a linear combination of latent variables with an intercept which is item specific (in the factor model was the mean)

$$\eta_i = \alpha_{i0} + \sum_{j=1}^{q} \alpha_{ij} y_j \quad i = 1, \ldots, p$$

- Link function: we apply a logit link funtion that relates the canonical parameter θ_i to the systematic component

$$\theta_i = \text{logit}(\pi_i(\mathbf{y})) = \eta_i = \alpha_{i0} + \sum_{j=1}^{q} \alpha_{ij} y_j$$

  We can express equivalently the probability of positive response in terms of latent factors and various weights/logistic models

$$\pi_i(\mathbf{y}) = \frac{\exp(\alpha_{i0} + \sum_{j=1}^{q} \alpha_{ij} y_j)}{1 + \exp(\alpha_{i0} + \sum_{j=1}^{q} \alpha_{ij} y_j)}$$

Important remark 18 (Parameter interpretation). we have that

- α_{i0} (the equivalent in the factor model was the mean) determines probability of positive response for the median individual (that is an individual who has all the $y = 0$, and being y normals we have the mean/median individual), and this probability is

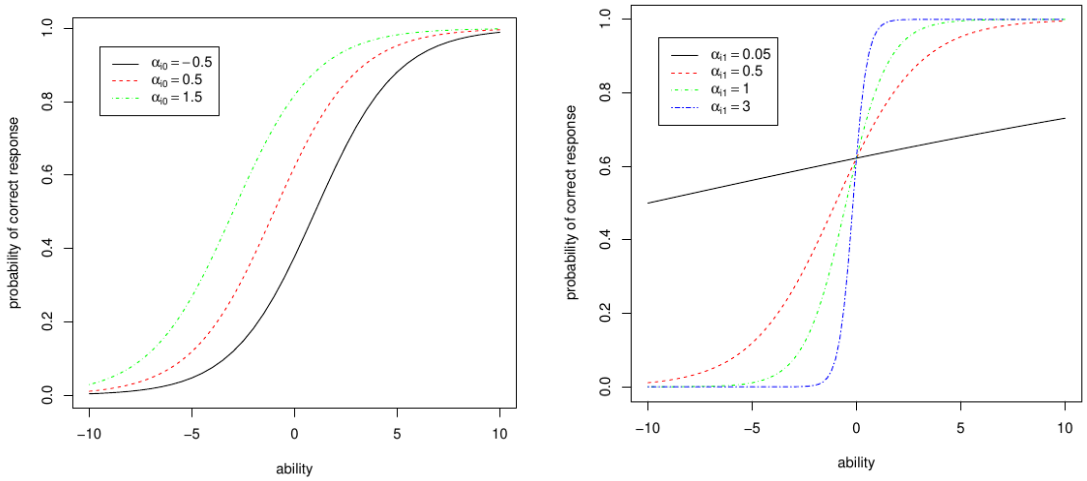
$$\pi_i(0) = \frac{1}{1 + \exp(-\alpha_{i0})}$$

- α_{ij} determines the effect of the latent variable y_j on the probability of success (if 0: no effect) with respect to x_i. A given change of y_j will produce a larger change in the probability of success of the variable x_i when this parameter is large than when it is small.

We have two interpretation of these parameters:

- IRT (Item Response Theory: educational context)
  - α_{i0} is called *difficulty parameter* (counterintuitive naming): an increasing of the value of this parameter increases the probability of positive response for all values of y_j (intuitively the naming should be the contrary, should be called ease parameter);

[1] logit comes from the link function, normal for the distributional assumption on y



(a) Item response function for different values of the **difficulty parameter** and $\alpha_{i1} = 0.5$

(b) Item response function for different values of the **discrimination parameter** and $\alpha_{i0} = 0.5$

Figure 3.1: Effect of latent variable on probability

  - α_{ij} is called *discrimination parameter*: the bigger is the value of this parameter the easier it will be to discriminate between a pair of individuals a given distance apart on the latent scale.

- Factor analysis α_{i0} intercept, α_{ij} factor loading (load of latent variable on probability of response)

Remark 18. In figure 3.1 the effect of change on probability given different values of the parameters.

- On the left if we focus on ability=0 we see that with increasing alpha, the probability of correct response increases.

- on the right the effect of latent variable through increasing values of discrimination parameter: the line for $\alpha_{i1} = 0.05$ is almost flat and there's a low impact of the ability on the probability of positive/correct response in the item (an individual with low ability will have a quite similar probability as an high able individual and the item/observed variable does not discriminate between an able individuale and a less able individual).
  On the contrary when the discrimination parameter increases, eg the highest, individual with low ability will have zero probability of correcting responding while individual with high ability will almost surely respond correctly and the considered item has highly discriminative power among the two groups of individuals

3.2.2 Component scores

We always want to obtain a reasonable estimate of the latent variables; the sufficient statistics we can use are still the component scores, described by

$$X_j = \sum_{i=1}^{p} \alpha_{ij} x_i \quad (j = 1, \ldots, q)$$

Since in latent trait models x_i are either 0 or 1, X_j is simply the sum of those α_{ij} for which $x_i = 1$.

Example 3.2.1. Given the individuals h, h' with the following response patterns

$$x_h = (0, 1, 1, 1, 0, 1, 0)$$
$$x_{h'} = (0, 0, 1, 1, 0, 1, 0)$$

The component scores for the j-th latent variable are:

$$X_{hj} = \alpha_{2j} + \alpha_{3j} + \alpha_{4j} + \alpha_{6j}$$
$$X_{h'j} = \alpha_{3j} + \alpha_{4j} + \alpha_{6j}$$

If α_{2j} (the only different element) is large the scores of the two individuals (the estimate of the latent variable) will be very different.

3.2.3 The Rasch model (Rasch, 1960)

Remark 19 (Historical note). As in factor analysis, model was born and then put in a general framework. Here the first model was the Rasch model, introduced by Rasch in 1960

A special case of the unidimensional model is obtained when all the discrimination parameters are equal to 1. The probability that the h-th individual will respond true/success is specified as function of latent variable and difficulty parameter as:

$$P(x_i = 1, \beta_{i0}, y_h) = \frac{\exp(y_h - \beta_{i0})}{1 + \exp(y_h - \beta_{i0})}$$

where

- β_{i0} remains the intercept/difficulty parameter;
- y_h's is the individual's ability (just one variable here, measured for the h-th individual, and its discrimination parameter is 1)[2];

The Rasch model is still quite popular in educational testing (to study the abilities of a particular set of individuals) because of its simplicity and its attractive theoretical properties (one does not need to estimate things?). In particular:

- the total score $\sum_{i=1}^{p} x_{ih}$ (the total number of correct responses given by individual h) is sufficient for y_h: that is, it contains all the information in the data about the β's if the model is true.
  So one sum all the correct response by an individual and has an estimate of its ability[3].

[2] In the first specification of the model the ability y was not assumed as normal

[3] this is correct only if one has all binary items and assumes a Rasch model

- $\sum_{h=1}^{n} x_{ih}$, the total number of positive/correct responses for item i (among all the individuals), is sufficient estimator for difficulty parameter β_{i0}.

3.2.4 2-Parametric Logistic model (Birnbaum, 1969)

Remark 20. It's an immediate extension of the Rasch model in the case in which the discrimination parameters are all equal is the 2-parameter logistic model (2PL)

The model is written as

$$P(x_i = 1, \beta_{i0}, \beta_{i1}, y_h) = \frac{\exp(\beta_{i1}(y_h - \beta_{i0}))}{1 + \exp(\beta_{i1}(y_h - \beta_{i0}))}$$

where the model shift/scale the ability according to the 2 parameters β_{i0},and β_{i1}:

- β_{i0} is still the difficulty parameter of the item;
- β_{i1} is the discrimination parameters of the item (we obtain the rasch by setting this to 1)

There's a relation between this model and the general latent trait model seen before, since this model represents a different parametrization (so in principle we could estimate the 2PL and and obtain the estimate of latent trait):

$$\beta_{i0} = -\alpha_{i0}/\alpha_{i1}$$
$$\beta_{i1} = \alpha_{i1}$$

the

- first parameter is -intercept/difficulty over discrimination of the latent trait model: the confusing name of "difficulty parameter" comes historically from here since here if β_{i0} increases (so α_{i0} diminish) the probability of 1 response decrease
- second just the same discrimination

In the IRT framework there are models with more parameters than two parameters (3PL model) and extensions to more than one ability.

3.2.5 Model estimation

Remark 21. Once specified the model we need to estimate it. Here we see only the maximum likelihood estimation via the E-M algorithm and just for one factor model (one latent variable y, which is considered the missing observation for the EM algorithm)

Starting from the log-likelihood l_c for the complete data (x_{ih}, y_h) (with $i = 1, \dots, p$ observed variables on $h = 1, \dots, n$ individuals):

$$l_c = \sum_{h=1}^{n} \log f(\mathbf{x}_h, y_h) = \sum_{h=1}^{n} \log \prod_{i=1}^{p} f(x_{ih}, y_h) = \sum_{h=1}^{n} \log \prod_{i=1}^{p} g(x_{ih}|y_h) \cdot \log h(y_h)$$
$$= \sum_{h=1}^{n} \left[\sum_{i=1}^{p} \log g(x_{ih}|y_h) + \log h(y_h) \right]$$

Since $h(y)$ does not depend on the parameters[4] we can replace l_c with

$$l = \sum_{h=1}^{n} \log g(\mathbf{x}_h|y_h) = \sum_{h=1}^{n}\sum_{i=1}^{p} \log g(x_{ih}|y_h)$$

where $g(x_{ih}|y_h)$ are all bernoulli; we can rewrite bernoulli as[5]:

$$g(x_{ih}|y_h) = (1 - \pi_i(y_h)) \exp(x_{ih}(\alpha_{i0} + \alpha_{i1} y_h))$$

thus maximization goes on with substituting and exchanging the two sums (and splitting the inner one):

$$\begin{aligned} l &= \sum_{h=1}^{n} \left[\sum_{i=1}^{p} \log \left((1 - \pi_i(y_h)) \exp(x_{ih}(\alpha_{i0} + \alpha_{i1} y_h)) \right) \right] \\ &= \sum_{h=1}^{n} \left[\sum_{i=1}^{p} (\log(1 - \pi_i(y_h)) + x_{ih}(\alpha_{i0} + \alpha_{i1} y_h)) \right] \\ &= \sum_{i=1}^{p} \left[\left(\sum_{h=1}^{n} \log(1 - \pi_i(y_h)) \right) + \left(\alpha_{i0} \sum_{h=1}^{n} x_{ih} \right) + \left(\alpha_{i1} \sum_{h=1}^{n} x_{ih} y_h \right) \right] \end{aligned}$$

Now for the EM algorithm we have the two steps:

1. E-step: compute $\mathbb{E}[l_c|\mathbf{x}_1, \ldots, \mathbf{x}_n]$ having

$$\mathbb{E}[l|\mathbf{x}] = \sum_{i=1}^{p} \left[\left(\sum_{h=1}^{n} \mathbb{E}[\log(1 - \pi_i(y_h))|\mathbf{x}_1, \ldots, \mathbf{x}_n] \right) + \left(\alpha_{i0} \sum_{h=1}^{n} x_{ih} \right) + \left(\alpha_{i1} \sum_{h=1}^{n} x_{ih}\, \mathbb{E}[y_h|\mathbf{x}_1, \ldots, \mathbf{x}_n \right.$$

2. M-step: maximise $E[l_c|\mathbf{x}_1, \ldots, \mathbf{x}_n]$ over the parameters. Differently from the factor model, here we don't know the posterior distribution of y given x (and so we don't know its expected value). What we can do is to "open" the expression of the expectation. The expectation of a continuous random variable (y) is an integral, so what we do is rewriting the expectation by replacing the formula of expectation, that is the integral of each term with respect to the posterior distribution of y| x.
   For each item i we maximize

$$\begin{aligned} E[l_i|\mathbf{x}] &= \int_R \left[\log(1 - \pi_i(y)) \sum_{h=1}^{n} h(y|\mathbf{x}_h) + \alpha_{i0} \sum_{h=1}^{n} x_{ih} + \alpha_{i1} \sum_{h=1}^{n} x_{ih} h(y|\mathbf{x}_h) \right] dy \\ &= \int_R \left[\log(1 - \pi_i(y)) N(y) + \alpha_{i0} \sum_{h=1}^{n} x_{ih} + \alpha_{i1} R_i(y) \right] dy \qquad (3.1) \end{aligned}$$

   where

   - in the first term (first equation) we have the log multiplied by the posterior distribution of y | x (that can be seen as the probability of the expected value)

[4] Here the parameters to be estimated are α_{i0}, α_{i1} which are not in the distribution equation of y

[5] see first lessons slides

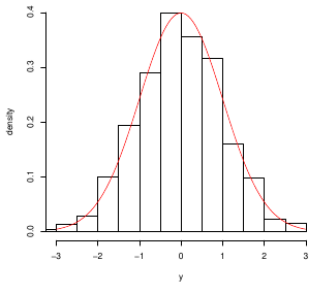


Figure 3.2: Approximating integrals

- the second one is a constant
- the third
- in the last eq for whatever reason (readability I guess) we just renamed

$$R_i(y) = \sum_{h=1}^{n} x_{ih} h(y|\mathbf{x}_h), \qquad N(y) = \sum_{h=1}^{n} h(y|\mathbf{x}_h)$$

The integral in equation (3.1) is not solvable analytically thus numerical approximation is needed

Estimation: approximation of the integral Some facts:

- It is possible to approximate any density $f(\cdot)$ having finite moments with a discrete distribution with given level of accuracy. That is: we approximate the density with histogram and do sum of rectangles (fig 3.2);
- when the density is standard normal the appropriate approximation method is called *Gauss-Hermite quadrature*;
- in our case the latent variable y is treated as discrete with values (nodes) $y_1, \dots, y_k$ and corresponding weights $h(y_1), \dots, h(y_k)$.
  A key point is the choice of the quadrature, that is to decide the number k of nodes: if too many the computation become precise but unfeasible. For 1 latent variable generally 30/40 poinst are ok[6].
  The approximated integral (with k approximation points) for each item i becomes easy to be computed (y now depends on t, becoming discrete):

$$\sum_{t=1}^{k} \left[\log(1 - \pi_i(y_t)) n_t + \alpha_{i0} \sum_{h=1}^{n} x_{ih} + \alpha_{i1} r_{it} \right]$$

with

$$n_t = \sum_{h=1}^{n} h(y_t|\mathbf{x}_h), \qquad r_{it} = \sum_{h=1}^{n} x_{ih} h(y_t|\mathbf{x}_h)$$

where

[6] If we have two latent variables $\mathbb{E}[l_i|\mathbf{x}]$ becomes a double integral and for approximation we have a $30 \cdot 30$ integral approximation problem. In general as the number of latent variables increases, the computational time/complexity does so.

- r_{it} expected number of people predicted to have $y = y_t$ and responding positively to the i-th item (x_{ih} acts as a 0-1 switch of the observed response);
- $n(t)$ expected number of individuals in latent position $y = y_t$

Once we have this computable approximated expected value $E[l_i|\mathbf{x}]$, differentiating with respect to α_{i0} and α_{i1} leads to non-linear equations in the parameters and thus Newton-Raphson iterative procedure is needed for optimization.

The E-M algorithm in this case follows these steps:

1. Step 1: choose starting values for αs
2. Step 2: compute r_{it} and $n(t)$ (E-step)
3. Step 3: obtain improved estimates of the α's treating r_{it} and $n(t)$ as given numbers (M-step)
4. Step 4: return to Step 2 until convergence

Remarks:

- it can happen that the EM algorithm does not converge. Possible causes are small sample size, small number of items, the discrimination parameters are very different;
- the number of nodes k can affect the final estimation. It is recommended to repeat the estimation procedure with increasing number of quadrature points (we should choose k as large as we can).

3.3 Underlying variable approach (UVA): Probit/normal model

Remark 22. In the model above (logit/normal) in case of 5/6 latent variables the integral in the approximation will be difficult to estimate. There's another way to analyze this kind of data (with assumptions which simplifies a lot estimation of the model).

3.3.1 Probit/normal model

Alternative model for binary data is the probit model, where instead of using logit link function we use the probit (which is the inverse is cumulative normal):

$$\Phi^{-1}(\pi_i(\mathbf{y})) = \alpha_{i0} + \sum_{j=1}^{q} \alpha_{ij} y_j \quad (i = 1, \ldots, p)$$

It follows that the probability of success is related to latent variables by:

$$\pi_i(\mathbf{y}) = P(x_i = 1|\mathbf{y}) = \Phi(\alpha_{i0} + \sum_{j=1}^{q} \alpha_{ij} y_j) \quad (i = 1, \ldots, p)$$

Probit

- is very close to the logit model since the two function are proportional

$$\text{logit}(u) = \frac{\pi}{\sqrt{3}} \cdot \Phi^{-1}(u)$$

and turns out that a parameter estimated with logit $\hat{\alpha}_{logit}$ is proportional as well to the same parameter estimated via probit link function

$$\hat{\alpha}_{logit} = \underbrace{\frac{\pi}{\sqrt{3}}}_{\approxeq 1.8} \cdot \hat{\alpha}_{probit}$$

- however the probit/normal model lacks the sufficiency property of the components X (thus we cannot combine linearly observed data and estimated parameters to have a good estimate of latent variables).
- is related to the standard normal factor model

3.3.2 Underlying variable approach

Given an the *observed binary* variable x_i, we suppose that *underlying* to this indicator there is a *continuous standard normally distributed* variable ξ, which is dichotomized according to a threshold parameter τ_i:

$$x_i = \begin{cases} 1, & \xi_i \leq \tau_i \\ 0, & \text{otherwise} \end{cases}$$

The latent variable ξ_i is assumed to be generated by a standard normal factor model; in matrix form

$$\boldsymbol{\xi} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{y} + \boldsymbol{\varepsilon}$$

with similar assumptions to that model:

- $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \boldsymbol{\Psi})$ with $\boldsymbol{\Psi}$ a diagonal matrix (as in the classical normal linear factor model);
- $\boldsymbol{\Lambda}$ is the matrix of the factor loadings
- $\boldsymbol{\xi} \sim N(\mathbf{0}, \mathbf{R}_{\boldsymbol{\xi}})$ with $\mathbf{R}_{\boldsymbol{\xi}} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}' + \boldsymbol{\Psi}$ is a unknown correlation matrix and has to be estimated. In the binary case the correlations are called *tetrachoric correlations*[7].

3.3.3 UVA estimation procedure (steps)

Remark 23. In this model there's a lot to be estimated: thresholds τ_is, tetracoric correlation and then the normal stuff from the standard factor model (loadings/specific variances).

[7] Tetrachoric correlation estimates the association between two underlying continuous variables that have been dichotomized into binary outcomes (e.g., yes/no, correct/incorrect), assuming those latent variables (which are hidden variables that can't be measured or observed directly) follow a bivariate normal distribution
The tetrachoric correlation estimates the Pearson correlation that would exist if the variables had been measured on their original continuous scale instead of being dichotomized.

The estimation procedure has these steps:

1. we maximum likelihood estimate the thresholds $\tau_1, \tau_2, \ldots, \tau_p$ by using each marginal distribution of the latent variables. The probability of a positive response is the probability of having a latent variable under treshold:

$$\pi_i = P(x_i = 1) = \int_{-\infty}^{\tau_{(i)}} \phi(\xi_i)\mathrm{d}\xi_i = \Phi(\tau_{(i)})$$

from which, we substitute the sample proportion P_i of positive response in item i (the maximum likelihood estimation of π_i) and apply the inverse/quantile to obtain:

$$\hat{\tau}_{(i)} = \Phi^{-1}(P_i)$$

2. using the same rational we estimate of $\mathbf{R}_{\boldsymbol{\xi}}$, the matrix of tetrachoric correlations using the bivariate distribution of the latent variables:

$$\pi_{ij} = P(x_i = 1, x_j = 1) = \int_{-\infty}^{\tau_{(i)}} \int_{-\infty}^{\tau_{(j)}} \phi(\xi_i, \xi_j, \rho_{ij})\mathrm{d}\xi_i$$

That is the probability of a positive answer both in i-th and j-th is the probability of mvn under both the treshold (which are known from the previous stem) which depends on the correlation of the two variables ρ_{ij}; doing similar passages to the previous step (that is substituting sample proportion to the left and applying the inverse function) we obtain $\hat{\rho}_{ij}$. And we can do this for each possible pair of variables thus reconstructing an estimate of the correlation matrix;

3. finally, once obtained treshold and correlation matrix, we estimate the loadings and the specific variances using the classical methods of the classical linear factor analysis (which has no integrals and thus we can estimate how many latent variable we want).

3.4 Equivalence between response function and UVA approach

TODO: non chiarissimo fino in fondo

UVA and response function are equivalent.
Let S be any non-empty subset of $\{1, 2, \ldots, p\}$ (index of observed variables) then we can express the joint probability of success in these subset of observed variables as the probability of intersection of each being 1 (that can be rewritten as integral on the joint distribution (on y), which is splitted in prior of y times likelihood):

$$P(x_i = 1, x_j = 1, \ldots) = P\left\{\bigcap_{i \in S}(x_i = 1)\right\} = \int \ldots \int P\left\{\bigcap_{i \in S} h(\mathbf{y})(x_i = 1|\mathbf{y})\right\} d\mathbf{y}$$
$$= \int \ldots \int \prod_{i \in S} h(\mathbf{y})\, \mathbb{P}\,(x_i = 1|\mathbf{y}) d\mathbf{y}$$

But at the same time the probability of joint 1 can be written as the probability of all latent variables being less than their threshold (and thus similarly):

$$P\left\{\bigcap_{i\in S}(x_i=1)\right\}=P\left\{\bigcap_{i\in S}(\xi_i\leq\tau_i)\right\}=\int\dots\int\prod_{i\in S}h(\mathbf{y})\,\mathbb{P}\left(\xi_i\leq\tau_i|\mathbf{y}\right)d\mathbf{y}$$

Remembering that ξ_i is assumed to be generated through a linear factor model ($\boldsymbol{\xi}=\mu+\boldsymbol{\Lambda}y+\boldsymbol{\varepsilon}$, then for the i-th component $\xi_i=\mu_i+\sum_j\lambda_{ij}y_j+\varepsilon_i$) we can develop further (substituting and then solving for the only random variable remaining, wince we're conditioning on y, which is error e_i (we standardize it too, obtaining the probability of a standard normal distribution):

$$\begin{aligned}\mathbb{P}\left(\xi_i\leq\tau_i|\mathbf{y}\right)&=\mathbb{P}\left(\mu_i+e_i+\sum_{j=1}^{q}\lambda_{ij}y_j\leq\tau_i|\mathbf{y}\right)=\mathbb{P}\left(\frac{e_i}{\psi_i^{1/2}}\leq\frac{\tau_i-\mu_i-\sum_{j=1}^{q}\lambda_{ij}y_j}{\psi_i^{1/2}}|\mathbf{y}\right)\\&=R\left(\frac{\tau_i-\mu_i-\sum_{j=1}^{q}\lambda_{ij}y_j}{\psi_i^{1/2}}\right)\end{aligned}$$

where at the last passage $R(\cdot)$ is the distribution function of $e_i/\psi_i^{1/2}$ (standard normal but we dont use Φ as usual because here we can assume the error term e_i is not normal, eg it's logistical distributed and R is the inverse of the logit.. check).
We can see the equivalence of the two approaches recalling that the probability of positive response for the response function model is

$$P(x_i=1|\mathbf{y})=G\left(\alpha_{i0}+\sum_{j=1}^{q}\alpha_{ij}y_j\right)$$

where G is tipically inverse of logit or of the probit[8]. In both cases we're modelling the conditional probability of a positive response. The two formulation are equivalent (parameters in the two formulatins) Thus the equivalence exists if $G\equiv R$ (same CDF) and I substitute below i get the stuff from the response function approach)

$$\begin{cases}\alpha_{i0}=\frac{\tau_i-\mu_i}{\sqrt{\psi_i}}\\\alpha_{ij}=-\frac{\lambda_{ij}}{\sqrt{\psi_i}}\end{cases}$$

Above 2 equation with 4 unknown parameters. The parameters τ_i, μ_i and λ_{ij} are not individually estimable because we have no information about the standard

[8]So that for example

$$logit(P(x_i=1|\mathbf{y}))=\alpha_{i0}+\sum_{j=1}^{q}\alpha_{ij}y_j$$

deviation of ξ_i. To procede we introduce constraint of unitary variance:

$$\begin{aligned}\text{Var}\left[\xi_i\right] = 1 &= \sum_{j=1}^{q} \lambda_{ij}^2 + \psi_i \\ \alpha_{ij} &= -\frac{\lambda_{ij}}{\sqrt{1 - \sum_{j=1}^{q} \lambda_{ij}^2}} \\ st\alpha_{ij} &= \frac{\alpha_{ij}}{\sqrt{1 + \sum_{j=1}^{q} \alpha_{ij}^2}}\end{aligned}$$

where

1. in first equation fixing unitary variance allows to say that

$$\psi_i = 1 - \sum_{j=1}^{q} \lambda_{ij}^2 \implies \sqrt{\psi_i} = \sqrt{1 - \sum_{j=1}^{q} \lambda_{ij}^2}$$

2. second equation is just the standard deviation obtained at previous step

3. in the last (which is not totally clear) we started looking

$$\alpha_{ij} = -\frac{\lambda_{ij}}{\sqrt{\psi_i}} \implies -\lambda_{ij} = \alpha_{ij}\sqrt{\psi_i} \implies \lambda_{ij}^2 = \alpha_{ij}^2 \psi_i$$

   In setting the unitary variance we thus have

$$1 = \sum_{j=1}^{q} \lambda_{ij}^2 + \psi_i = \sum_{j=1}^{q} \psi_i \alpha_{ij}^2 + \psi_i = \psi_i \left(\sum_{j=1}^{q} \alpha_{ij}^2 + 1 \right)$$

   and thus

$$\psi_i = \frac{1}{\sum_{j=1}^{q} \alpha_{ij}^2 + 1} \implies \sqrt{\psi_i} = \frac{1}{\sqrt{\sum_{j=1}^{q} \alpha_{ij}^2 + 1}}$$

   So apparently we have that $sta_{ij} = \alpha_{ij} \cdot \sqrt{\psi_i}$ for whatever reason: α_{ij} can assume values in $-\infty, +\infty$, we standardize them in a way

3.4.1 Standard errors

The determination of exact standard errors for the parameter estimates is not possible and we use asymptotic variance-covariance matrix using the information matrix.
If we recall the set of parameters $\boldsymbol{\beta}$ we have

$$var(\hat{\boldsymbol{\beta}})^{-1} = E\left[-\frac{\partial^2 L}{\partial \beta_i \partial \beta_j} \right]$$

It can be approximated by using the observed matrix

$$var^*(\hat{\boldsymbol{\beta}})^{-1} = \left[\sum_{h=1}^{n} \frac{1}{f^2(\mathbf{x}_h)} \frac{\partial f(\mathbf{x}_h)}{\partial \beta_i} \frac{\partial f(\mathbf{x}_h)}{\partial \beta_j} \right]$$

3.5 Goodness of fit

TO assess goodness of fit:

- the null hypothesis is written

$$H_0 : \boldsymbol{\pi} = \boldsymbol{\pi}(\boldsymbol{\alpha})$$

  where $\boldsymbol{\pi}$ is the vector of true probability (associated with each response patters) and $\boldsymbol{\pi}(\boldsymbol{\alpha})$ is computed from/expressed as parameter of the model. So *we don't want to refuse.*

- In practice, the sample proportion $\mathbf{P}$ are compared with $\boldsymbol{\pi}(\hat{\boldsymbol{\alpha}})$ where $\hat{\boldsymbol{\alpha}}$ is the estimator with some method

- classical test statistics are the Pearson χ^2 statistic and the likelihood ratio test G^2; when sample size n is large and p is small both statistics follow a chi square distribution with degrees of freedom $2^p - p(q+1) - 1$ (our "n" here is 2^p, the number of response patterns, and the number of parameter estimated $p(q+1)$[9]). However when p increases (and n is not that high, sparse data), chi square distributing does not hold anymore.

- the chi-square statistic is:

$$\chi^2 = \sum_{r=1}^{2^p} \frac{(nP_r - n\pi_r(\hat{\boldsymbol{\alpha}}))^2}{n\pi_r(\hat{\boldsymbol{\alpha}})}$$

  where nP_r is the observed number of positive responses while $n\pi_r(\hat{\boldsymbol{\alpha}})$ the expected. It assumes the same shape of the chi-square test (expected - observed)

$$\chi^2 = \sum_{k=1}^{K} \frac{(E_k - O_k)^2}{E_k}$$

  where for us the k categories are the response patterns.

- the LRT is:

$$G^2 = 2\sum_{r=1}^{2^p} nP_r \log\left(\frac{nP_r}{n\pi_r(\hat{\boldsymbol{\alpha}})}\right)$$

  with the same role of nP_r observed successes vs $n\pi_r(\hat{\boldsymbol{\alpha}})$ based on the model

Important remark 19 (Problematic example). With $p = 10$ observed variables and $2^p = 1024$ response patterns and a sample size $n = 1000$, we expect many responses with $n\pi_i(\hat{\boldsymbol{\alpha}}) \leq 1$, so the two statistics will have very small denominators and very large values so it's very common to reject the null (not because the model is bad).
So the first thing to do when we use this test is to look at observed frequency ed expected frequency for each response pattern; if we see that there are a lot of expected frequency below 5 or 1 don't use these tests.

[9] All the alphas, one for each latent variable, plus intercept, per every observed variable.

Important remark 20 (Remedies).

In many cases we can't use the previous two tests. What can we do?

- Group the response patterns with expected frequencies less than 5. However there's risk to have degrees of freedom equal to 0.
- Empirical sampling distribution of the test statistics using the parametric bootstrap method (instead of assuming it's $\sim \chi^2$)
- Consider residuals calculated from marginal frequencies:
  - $P(x_i = 1)$ first-order margins (probability of individual responding 1 in any variable), contain no information about the dependencies among the x's;
  - $P(x_i = 1, x_j = 1)$ second-order margins (individual with two 1 in two variables), information about pairwise association
  - $P(x_i = 1, x_j = 1, x_k = 1)$ third-order margins.

  Second and third-order margins usually suffice. What we do is to compute
  - O: observed frequencies for any marginal probability
  - E: corresponding expected frequency

  and then calculate the

$$R = \frac{(O - E)^2}{E}$$

  a Large values of R for the second-order margins identify pairs of x's responsible of the bad fit.
  Rule of thumb: $R > 4$ indicates bad fit (similar to chisq with 1 df, for which critical value is 3.84)

3.6 Posterior analysis: latent variable estimation

NB: l'essere è il non essere non è, il software calcola

As in the general GLLVM, all the information about $\mathbf{y}$ is contained in the posterior distribution $h(\mathbf{y}|\mathbf{x})$.
In the case of the logit/normal model the distribution depends on $\mathbf{x}$ only through the q-variate sufficient statistic $\mathbf{X} = \mathbf{Ax}$ where $\mathbf{A} = \{\alpha_{ij}\}$.
Component scores are often used for scaling, expecially in measurement abilities. The posterior mean $E(y_j|\mathbf{x}_h)$ and the components give the same ranking to response patterns/individuals.

3.7 Example

3.7.1 Estimation

After having derived the frequency distribution of response patterns, and had a look at correlations among observed variables, parameter estimates for the example in table 3.3. Item are 0=agree and 1=disagree? We have estimates for the one-factor model:

Item	$\hat{\alpha}_{i0}$	s.e.	$\hat{\alpha}_{i1}$	s.e.	st$\hat{\alpha}_{i1}$	$\hat{\pi}_i(0)$
WomanDecide	-0.72	(0.33)	4.15	(0.85)	0.97	$0.33 = \frac{\exp(-0.72)}{1+\exp(-0.72)}$
CoupleDecide	1.11	(0.35)	4.50	(0.81)	0.98	0.75
NotMarried	2.18	(0.61)	6.21	(1.54)	0.99	0.90
CannotAfford	1.15	(0.28)	3.49	(0.50)	0.96	0.76

Table 3.3: Various estimates

- $\hat{\alpha}_{i0}$ are the intercepts unstardized loadings.
  Lowest of intercept is "woman decide" while highest is notmarried: the intercept is associated to the probability of positive response (disagree=antiabortion statement) for a median individual (having 0 on the latent variable). The associated estimated probability is in the last column.
- $\hat{\alpha}_{i1}$ are the load of the single latent variable on the observed ones and can be interpreted as discrimination parameter. An higher score increase the impact of latent variable on the probability of positive (antiabortist) response in a given item (items most affected by latent variable), and at the same time suggest which observed variable is mostly correlated/suggestive of latent variable of interest.
  Again item most affected is NotMarried which is thus the most discriminative one for the latent variable
- s.e. (in brackets) are all the standard error for the various loadings
- st$\hat{\alpha}_{i1}$ are the standardized alphas: bah just a normalized version (commentata velocemente)
- $\hat{\pi}_i(0)$ is the estimate of probability of disagreeing (antiabortion statement) for a median subject (having latent variable=0). THe probability of positive response for a median individual in WomanDecide is just 33%, while the highest is attained fo NotMarried (with 90%)

3.7.2 Goodness of fit

We have 300 individuals with 2^4 response patterns;

- we obtain that the test are largely statistically significant with $G^2 = 17.85$ and $\chi^2 = 15.09$ (with 3 degrees of freedom p-values are 0.0005 and 0.001) so according to these two tests we would reject the model. However we're not super sure if the rejection depends on the model bein bad or on data sparcity
- thus we look at residuals for second order margins in table 3.4: in the first row we see the subject which gave 0/0 to item 2 and 1. Actually 147 subject have this mini-response pattern where the expected were 143 and being very similar residual R (last column) is low 0.07, and especially $<$ 4, thus means we have no problem with this pattern.
  Same things done for the other rows suggest (all are < 4) that there are no problem with the models, which can be accepted (contrary to tests results, which rejected the model due to sparsity).

Being a good model we go with latent variable estimation.

Response	Item i	Item j	O	E	O-E	$(O-E)^2/E$
(0,0)	2	1	147	143.74	3.26	0.07
	3	1	131	133.17	-2.17	0.04
	3	2	117	119.69	-2.69	0.06
	4	1	129	133.68	-4.68	0.16
	4	2	114	116.09	-2.09	0.04
	4	3	116	111.79	4.21	0.16
(0,1)	2	1	7	11.30	-4.30	1.64
	3	1	7	5.94	1.06	0.19
	3	2	21	19.42	1.58	0.13
	4	1	16	11.99	4.01	1.34
	4	2	31	29.58	1.42	0.07
	4	3	29	33.88	-4.88	0.70
(1,0)	2	1	66	69.89	-3.89	0.22
	3	1	82	80.46	1.54	0.03
	3	2	37	35.35	1.65	0.08
	4	1	84	79.95	4.05	0.21
	4	2	40	38.95	1.05	0.03
	4	3	22	27.32	-5.32	1.04
(1,1)	2	1	159	154.07	4.93	0.16
	3	1	159	159.43	-0.43	0.00
	3	2	204	204.54	-0.54	0.00
	4	1	150	153.58	-3.38	0.07
	4	2	194	194.38	-0.38	0.00
	4	3	212	206.01	5.99	0.17

Table 3.4: Residual for second ordere margins

Response Pattern	Observed frequency	Expected frequency	$E(y\|\mathbf{x})$	$\sigma(y\|\mathbf{x})$	Component score	Total score
0000	103	100	-1.19	0.55	0.00	0
0001	13	16.6	-0.61	0.32	3.49	1
1000	1	1.7	-0.55	0.30	4.15	1
0100	9	9.1	-0.52	0.29	4.50	1
0010	10	12.3	-0.38	0.26	6.21	1
1001	0	1.3	-0.29	0.24	7.64	2
0101	6	7.4	-0.27	0.24	7.99	2
1100	3	1.0	-0.24	0.24	8.65	2
0011	21	14.8	-0.18	0.24	9.70	2
1010	0	2.0	-0.14	0.25	10.37	2
0110	7	2.0	-0.12	0.26	10.71	2
1101	3	1.9	-0.01	0.28	12.14	3
1011	6	6.2	0.14	0.32	13.86	3
0111	44	41.1	0.17	0.32	14.20	3
1110	12	7.2	0.24	0.34	14.87	3
1111	141	143.9	0.95	0.61	18.35	4
Total	379					

Table 3.5: component score estimate

3.7.3 Latent variable estimation

In table 3.5 we see

- the observed frequency of response patterns and expected (and here actually we see the sparsity of data in terms of expected frequency below 5);
- we have the expected value of the latent variable for each response pattern in $\mathbb{E}[y|\mathbf{x}]$, on which we can rank/sort individuals on the level of pro abortion (1111 pro, 0000 against)
- $\sigma(y|\mathbf{x})$ is the sd of the posterior distribution of y
- the column on component score (linear combination of loading multiplied by x) has the the *same ranking* of response patterns of $\mathbb{E}[y|\mathbf{x}]$
- total score is just the sum of 1 and we see it has roughly the ordering of expected y and component score (which can be more precise btw, ordering within patterns having the same total score)

Chapter 4

Latent class models for binary data

Remark 24. Binary observed, binary latent.

4.1 Introduction

Suppose that

- there are p observed binary variables denoted with $x_1, x_2, \ldots, x_p$ ($x_i = 0, 1$);
- the association of these variables is accounted for a single latent binary variable y.
- within groups (labelled 0 and 1) of latent variable the x's are mutually independent (that is all the correlation is due to the latent variable). *conditional independence* assumption

The:

- prior distribution of latent variable y is assumed to be a bernoulli:

$$h(1) = P(y = 1) = \eta$$
$$h(0) = 1 - h(1) = 1 - \eta$$

- conditional distribution of x_i given y is a Bernoulli variable

$$g(x_i|y) = \pi_{iy}^{x_i}(1 - \pi_{iy})^{1-x_i} \quad (x_i, y = 0, 1)$$

- the joint distribution of observed follows to be ($\sum$ (prior $\cdot$ likelihood)):

$$f(\mathbf{x}) = \eta \prod_{i=1}^{p} \pi_{i1}^{x_i}(1 - \pi_{i1})^{1-x_i} + (1 - \eta) \prod_{i=1}^{p} \pi_{i0}^{x_i}(1 - \pi_{i0})^{1-x_i}$$

  that is the sum of two terms,

1. the first of which is the probability of positive response for y multiplied by the product (for conditional independence) of each conditional probability $g(x_i|y)$
2. the second is product of probability of negative response for y multiplied by the product of the bernoulli distributions

4.2 After goodnes of fit tests

Important remark 21. After done the goodness of fit checks (to evaluate if the model has a good fit to the data):

- If the *model is adequate* we define a rule to allocate the individuals in the two latent classes based on the observed values (basically using the posterior distribution of the latent variables given the observed variables)
- If the *model is not adequate* we can:
  - extend the number of classes to a number of classes of the latent variables to $K \geq 3$ (not anymore dichotomous);
  - consider models that do not assume the conditional independence;
  - consider models with continuous latent variables

4.3 How to allocate units to latent variable groups

To allocate to the most likely group, we use the posterior probability of belonging to the two classes given the observed variables:

$$\begin{aligned} h(1|\mathbf{x}) = P(y=1|x_1\ldots,x_p) &= \frac{P(y=1,x_1\ldots,x_p)}{f(x_1\ldots,x_p)} \\ &= \frac{h(y=1)g(x_1\ldots,x_p|y=1)}{f(\mathbf{x})} \\ &= \frac{\eta\prod_{i=1}^{p}\pi_{i1}^{x_i}(1-\pi_{i1})^{1-x_i}}{\eta\prod_{i=1}^{p}\pi_{i1}^{x_i}(1-\pi_{i1})^{1-x_i}+(1-\eta)\prod_{i=1}^{p}\pi_{i0}^{x_i}(1-\pi_{i0})^{1-x_i}} \\ h(0|\mathbf{x}) &= \frac{(1-\eta)\prod_{i=1}^{p}\pi_{i0}^{x_i}(1-\pi_{i0})^{1-x_i}}{\eta\prod_{i=1}^{p}\pi_{i0}^{x_i}(1-\pi_{i0})^{1-x_i}+(1-\eta)\prod_{i=1}^{p}\pi_{i0}^{x_i}(1-\pi_{i0})^{1-x_i}} \end{aligned}$$

The allocation rule is the following (that is the use class with higher posterior probability):

- if $h(1|\mathbf{x}) > h(0|\mathbf{x})$ the unit is allocated in latent class 1;
- if $h(1|\mathbf{x}) \leq h(0|\mathbf{x})$ the unit is allocated in latent class 0.

4.4 Latent class models with K-class latent variables

The extension to a K-class model is almost immediate[1]. With:

[1] In an exploratory analysis K is a parameter to be determined, otherwise it is given

- π_{ij} probability of a positive response for variable i on category j (with $i = 1, \ldots, p$ and $j = 0, 1, \ldots, K-1$);
- K prior parameters η_j, that are prior probability that a randomly chosen individual is in class j (with $\sum_{j=0}^{K-1} \eta_j = 1$)
- assuming conditional independence

$$g(\mathbf{x}|j) = \prod_{i=1}^{p} g(x_i|j)$$

It follows that generalization of joint marginal distribution of x is as follows

$$f(\mathbf{x}) = \sum_{j=0}^{K-1} \eta_j \prod_{i=1}^{p} \pi_{ij}^{x_i} (1 - \pi_{ij})^{(1-x_i)}$$

where at the sum here instead of having 2 we have K sums members, other than that is the same.

Remark 25. One of our task will be to choose the number of classes; here we don't know K. We'll start with 2 and increase if the model is not good according to goodness of fit.

The posterior probability that an individual with response vector $\mathbf{x}$ belongs to category j is again applying Bayes:

$$h(j|\mathbf{x}) = \frac{\overbrace{\eta_j}^{\text{prior}} \overbrace{\prod_{i=1}^{p} \pi_{ij}^{x_i} (1 - \pi_{ij})^{1-x_i}}^{\text{likelihood}}}{f(\mathbf{x})}, \quad (j = 0, 1, \ldots, K-1)$$

Similary to the two class case, the posterior distribution can be used to construct an allocation rule where an individual is allocated in the class with the highest posterior probability.
The vector of the unknown parameters to be estimated[2] is $\theta = (\eta_0, \ldots, \eta_{K-1}, \pi_{10}, \ldots, \pi_{p,K-1})$.
With K classes the number of parameter to be estimated are K (priors) + $K \cdot p$ conditional probabilities, that is $K(p+1)$.

4.4.1 Maximum likelihood estimation

As before, we can use the E-M algorithm. For a random sample of size n the log-likelihood can be written as sum on all individuals of the logarithm of joint (splitted prior times likelihood)

$$l = \sum_{h=1}^{n} \log \left\{ \sum_{j=0}^{K-1} \overbrace{\eta_j}^{h(y)} \overbrace{\prod_{i=1}^{p} \pi_{ij}^{x_{ih}} (1 - \pi_j)^{1-x_{ih}}}^{g(\mathbf{x}|y)} \right\}$$

[2](differently from the previous model where we estimated the alphas and treshold here we estimate probability directly

Estimating η_j The loglikelihood is maximized subject to constraint $\sum_j \eta_j = 1$, so we find the maximum of

$$\phi = l + \gamma \sum_{j=0}^{K-1} \eta_j$$

(it's a lagrange/constraint maximization) with γ is an undetermined multiplier (it is equivalent to the Lagrange multiplier).
Finding partial derivatives of ϕ and nulling it:

$$\frac{\partial \phi}{\partial \eta_j} = \frac{\sum_{h=1}^{n} g(\mathbf{x}_h|j)}{f(\mathbf{x}_h)} + \gamma = 0$$

where the first term is the derivative of the likelihood while γ of the $\gamma \sum_{j=0}^{K-1} \eta_j$ component.
Remembering that EM depends on posterior expectation we have to derive quantities that depends on posterior density. Some algebrical manipulations suggest that:

$$h(j|\mathbf{x}_h) = \frac{\eta_j g(\mathbf{x}_h|j)}{f(\mathbf{x}_h)} \Rightarrow \frac{h(j|\mathbf{x}_h)}{\eta_j} = \frac{g(\mathbf{x}_h|j)}{f(\mathbf{x}_h)} \tag{4.1}$$

Thus it follows that we can substitute the expression above in the nulled derivative as follows

$$\frac{\partial \phi}{\partial \eta_j} = \sum_{h=1}^{n} \frac{h(j|\mathbf{x}_h)}{\eta_j} + \gamma = 0 \Rightarrow \sum_{h=1}^{n} h(j|\mathbf{x}_h) = -\gamma \eta_j \tag{4.2}$$

Summing up over the $K-1$ categories

$$\sum_{h=1}^{n} \underbrace{\sum_{j=0}^{K-1} h(j|\mathbf{x}_h)}_{=1} = -\gamma \underbrace{\sum_{j=0}^{K-1} \eta_j}_{=1} \Rightarrow n = -\gamma$$

and thus we obtain by substituting $-\gamma = n$ in 4.2

$$\hat{\eta}_j = \frac{\sum_{h=1}^{n} h(j|\mathbf{x}_h)}{n}$$

this $\hat{\eta}_j$, the ML estimator, is interpreted in terms of posterior expectation: it's the average of the posterior probabilities that the n-units belongs to the latent class j.

Estimating π**s** We've seen the estimator for η, what about the πs? In the same way doing partial derivative wrt π_{ij} and setting it to 0 we have

$$\frac{\partial \phi}{\partial \pi_{ij}} = \sum_{h=1}^{n} \eta_j \frac{\partial \phi}{\partial \pi_{ij}} \frac{g(\mathbf{x}_h|j)}{f(\mathbf{x}_h)} = \frac{\eta_j}{\pi_{ij}(1-\pi_{ij})} \sum_{h=1}^{n} (x_{ih} - \pi_{ij}) \frac{g(\mathbf{x}_h|j)}{f(\mathbf{x}_h)} = 0$$

By similarly substituting 4.1 we end up with:

$$\frac{\eta_j}{\pi_{ij}(1-\pi_{ij})}\sum_{h=1}^{n}(x_{ih}-\pi_{ij})\frac{h(j|\mathbf{x}_h)}{\eta_j}=0$$
$$\sum_{h=1}^{n}x_{ih}h(j|\mathbf{x}_h)=\pi_{ij}\sum_{h=1}^{n}h(j|\mathbf{x}_h)$$

from which

$$\hat{\pi}_{ij}=\frac{\sum_{h=1}^{n}x_{ih}h(j|\mathbf{x}_h)}{\sum_{h=1}^{n}h(j|\mathbf{x}_h)} \tag{4.3}$$

where

$$h(j|\mathbf{x}_h)=\frac{\eta_j\prod_{i=1}^{p}\pi_{ij}^{x_{ih}}(1-\pi_{ij})^{1-x_{ih}}}{\sum_{k=0}^{K-1}\eta_j\prod_{i=1}^{p}\pi_{ik}^{x_{ih}}(1-\pi_{ik})^{1-x_{ih}}}$$

In 4.3 we can intepret the probability of positive answer on the i-th element coming from j-th class as a probability (at the numerato x_{ih} that works as a switch (0-1) and at the denominator the maximum possible which are sum of posterior probability given data) and is a posterior things which is suitable for EM algorithm

4.4.2 Maximum likelihood estimation: EM algorithm

Both formulas of estimator depends on h that we don't know (depending on eta and pis). We use EM to estimate iteratively eta pis and h [3].
The steps of the E-M algorithm are the following:

1. choose an initial set of posterior probabilities $h(j|\mathbf{x}_h)$
2. E-step: compute a first approximation of $\hat{\eta}_j$ and $\hat{\pi}_{ij}$
3. M-step: obtain improved estimates of $h(j|\mathbf{x}_h)$, by replacing $\hat{\eta}_j$ and $\hat{\pi}_{ij}$ of the previous step
4. return to Step 2 and continue until convergence

Remark 26. In the latent class model there is the problem of likelihood convergence into *local maxima*

- The risk increases as K increases the number of observed variables is large and the sample size is small.
- a strategy can be using many different starting point (eg 10) and choosing the solution with the highest loglikelihood

[3] in this case we've no problem of integrals but have close expression of parameter estimates as seen before

4.4.3 Goodness of fit

Story is the same of latent trait model. Given that observed variables are binary we consider also in this case the Pearson χ^2 statistic and the likelihood ratio test G:

$$\chi^2 = \sum_{r=1}^{2^p} \frac{(nP_r - n\hat{\pi}_r)^2}{n\hat{\pi}_r)}$$
$$G^2 = 2\sum_{r=1}^{2^p} nP_r \log\left(\frac{nP_r}{n\hat{\pi}_r}\right)$$

formula above are exactly the same but in this case changes the way in which we estimate $\hat{\pi}_r$, that in this case is obtained as marginal distribution of each response pattern

$$\hat{\pi}_r = f(\mathbf{x}_r) = \sum_{j=0}^{K-1} \hat{\eta}_j \prod_{i=1}^{p} \hat{\pi}_{ij}^{x_{ir}} (1 - \hat{\pi}_{ij})^{(1-x_{ir})}$$

Under regular conditions, χ^2 and G^2 converge to a chi square with $2^p - (K-1) - Kp - 1$ degrees of freedom (equivalent of n is again the response patterns while the numbers of parameter to be estimated will be $K-1$ pis, last is determined, and Kp conditional probabilities).
Again in presence of sparse data we use the low-order residuals as before.

4.5 Latent class model vs latent trait model

Remember:

- latent class has binary latent variable
- latent trait has continuous latent variable

What is interesting is that latent class model is special case of latent trait model in which prior distribution consists of discrete probability masses: Considering for example the case of $K = 2$, we have that in the logit/normal model:

$$\text{logit}(\pi_i(y)) = \alpha_{i0} + \alpha_{i1} y$$

where y has 0 mean and unit standard deviation. If in these latent trait model we assume that y is discrete (instead of continuous), taking two values $\sqrt{(1-\eta)/\eta}$ and $-\sqrt{\eta/(1-\eta)}$ with probabilities η and $1-\eta$ respectively, we obtain exactly that $E(y) = 0$ and $Var(y) = 1$ and furthermore

$$\text{logit}(\pi_{i0}) = \alpha_{i0} - \alpha_{i1}\sqrt{\frac{\eta}{1-\eta}}$$
$$\text{logit}(\pi_{i1}) = \alpha_{i0} + \alpha_{i1}\sqrt{\frac{1-\eta}{\eta}}$$

In this case the two parametrization are equivalent; the authors show an example where they estimate a latent class model (classicaly) and a latent trait with

Table 6.1 Parameter estimates for the two-class model fitted to the Law School Admission Test VI data together with a comparison of the estimates of $\{\alpha_{i0}\}$ and $\{\alpha_{i1}\}$ obtained from (6.4) and (6.6) and from the logit/normal model (from Table 4.1).

			Logit/normal estimates		Latent class estimates	
i	$\hat{\pi}_{i0}$	$\hat{\pi}_{i1}$	$\hat{\alpha}_{i0}$	$\hat{\alpha}_{i1}$	$\hat{\alpha}_{i0}$	$\hat{\alpha}_{i1}$
1	0.852	0.965	2.77	0.83	2.75	0.74
2	0.528	0.812	0.99	0.72	0.97	0.64
3	0.301	0.693	0.25	0.89	0.22	0.79
4	0.611	0.849	1.28	0.69	1.27	0.61
5	0.776	0.924	2.05	0.66	2.03	0.59

$\hat{\eta} = 0.638$

Figure 4.1: Model comparison

dichotomized (I guess). In table 4.1 a model comparison on estimates on the same dataset The differences among the alphas are very low. The example is another example which suggest that the prior distribution of y is irrelevant (as we saw at the beginning).
If aim is to

- **classify**, assume dichotomic latent variable
- **rank**, assume continuous latent variable

4.6 Examples

4.6.1 Example 1

We apply the latent class model to the attitude to abortion data (on which now we assume a binary latent variable and we want to classify individuals). For simplicity we assume a *two-class* model;

- estimated conditional probabilities and prior probabilities ($\hat{\eta}_j$) are reported in table 4.1
  - the prior $\hat{\eta}_j$ of belonging to class 1 is lower 0.39 than that to the class 2 (0.61); we should label which are class 1 and class 2. the conditional probability helps us in doing that
  - the conditional probability of answering positively given latent class are the pis (that is give).

  For woman decide, the probability of positive response in the first class is quite low (0.01); the same occurs for the other 3. THese are *individual against abortion.*
  In the second class otherwise the probability of positive response is higher for all the items (going from 71% to 96%); these should be favourable to abortion.
  Before predicting/assigning individual, however, we have to check model goodness of fit

Item	$\hat{\pi}_{i1}$	$\hat{\pi}_{i2}$
WomanDecide	0.01 (0.01)	0.71 (0.03)
CoupleDecide	0.09 (0.03)	0.91(0.02)
NotMarried	0.12 (0.04)	0.96 (0.02)
CannotAfford	0.15 (0.04)	0.91 (0.02)
$\hat{\eta}_j$	0.39 (0.03)	0.61 (0.03)

Table 4.1: Example 1: two class model

- regarding goodness of fit $\chi^2 = 44.81, df = 6$ $G^2 = 37.02, df = 4$ we largely regect the model; as already seen these data are affected by sparsicity, so we better look at residuals similarly;
- for the two class model for the second order margins in table 4.2 the observed are the same of the example with the previous model, but the expected changes (now computed with this model).
  - we have no expected frequency below 5 but
  - Looking at the last column we have only one chi-squared residuals greater than 4; just 1 residual > 4 can be considered acceptable

  So we conclude that model is a good model and we can perform posterior estimation of probabilities
- estimated posterior probabilities of class membership in table 4.3
  - surely patern 0000 (disagree on all the items) will be allocated to class 1, while 1111 (and others with three positive response) to class 2
  - regarding the other ones we have to check the higher posterior probability

  So final result of this kind of analysis is that we have a clustering/label to each group of the individuals given their responses.

4.6.2 Example 2

An example regarding attitude to science and technology data:

- 7 items originally measured on a four point scale, dichotomised as follows:
  - 'strongly disagree' and 'disagree to some extent' are coded as 0
  - 'agree to some extent' and 'strongly agree' are coded as 1.
- items Environment, Technology and Industry are negatively expressed. They are recoded so that a *high score corresponds to positive attitude towards science* and technology (otherwise it would be difficult to interpret the solutions)
- number of respondents is 392
- $2^7 = 128$ possible response patterns

	Items	O	E	$O-E$	$(O-E)^2/E$
Response (0,0)	2,1	147	137.79	9.21	0.62
	3,1	131	130.16	0.84	0.05
	3,2	117	117.58	-0.58	0.00
	4,1	129	129.12	-0.12	0.00
	4,2	114	114.61	-0.61	0.00
	4,3	116	109.97	6.03	0.33
Response (0,1)	1,2	66	75.21	-9.21	1.13
	1,3	82	82.84	-0.84	0.01
	1,4	84	83.88	-0.12	0.00
	2,1	7	16.21	-9.21	5.24
	2,3	37	36.42	0.58	0.01
	2,4	40	39.89	0.61	0.01
	3,1	7	7.84	-0.84	0.09
	3,2	21	20.42	0.58	0.02
	3,4	22	28.03	-6.03	1.30
	4,1	16	15.88	0.12	0.00
	4,2	31	30.39	0.61	0.01
	4,3	29	35.03	-6.03	1.04
Response (1,0)	2,1	159	149.79	9.21	0.57
	3,1	159	158.16	0.84	0.01
	3,2	204	204.58	-0.58	0.00
	4,1	150	150.12	-0.12	0.00
	4,2	194	194.61	-0.61	0.00
	4,3	212	205.97	6.03	0.18

Table 4.2: Example 1: residuals

Response Pattern	$\hat{h}(j=1\|\mathbf{x})$	$\hat{h}(j=2\|\mathbf{x})$	Class allocation
0000	1.00	0.00	1
0001	0.99	0.01	1
0010	0.96	0.04	1
0100	0.98	0.02	1
1000	0.95	0.05	1
0011	0.31	0.69	2
0101	0.45	0.55	2
0110	0.21	0.79	2
1100	0.16	0.84	2
0111	0.00	1.00	2
1011	0.00	1.00	2
1101	0.00	1.00	2
1110	0.00	1.00	2
1111	0.00	1.00	2

Table 4.3: Example 1: class prediction

- many patterns do not occur

We fit a two-class model.

- Estimated conditional probabilities and prior probabilities for the two-class model in table 4.4.
  The prior probability of belonging to the second class 0.79, while for the first one is 0.21.
  To interpret thses two class:
  - the probability of positive response in the second class is almost everywhere higher than probability of positive response from the first class but in a few cases (Environment, Technology). In general we can interpret the second class as those which are pro-science
  - in the first column we have high probability on almost all but to Future, Work and Benefit: seems these are in favour of tech/science (high percentages after all) but are skeptical about social benefit of it and for future generations. So it's not against tech advantages but has some doubts
- goodness of fit again refuse null with $\chi^2 = 118.48, df = 4\, G^2 = 109.03, df = 4$, but there may be problems of approximation/sparsity
- chi-squared residuals in table 4.5 greater than 4 for the second order margins, for a lot of couples.
  In this case almost all items present bad fit. One thing we can do is to increase the number of classes, estimating a new model with single latent variable having 3 groups
- Estimated conditional probabilities and prior probabilities for the **three-class** model in tab 4.6 (author dont show the residuals but they say they are ok):
  - If compared, the prior probability of the last class is the same (0.79) the *old first class is splitted in the new first and second class*;
  - similarly the conditional probability in the third class are almost equal to the previous one
  - the new first class (9% of prior prob): high probability of positive answer for Environment, tehchnology industry, but low (less than .50) on (Social) Benefit work and Future. Again this is people not against science but with some doubts on specific aspects
  - the same applies for the second group which is generally positive about science but skeptical regarding technology and environment

  So basically this splits the skeptical but general favourable to science in two better distinct groups

Item	$\hat{\pi}_{i1}$	$\hat{\pi}_{i2}$
Comfort	0.75 (0.06)	0.95 (0.02)
Environment	0.76 (0.06)	0.68 (0.03)
Work	0.36 (0.09)	0.75 (0.04)
Future	0.20 (0.18)	0.94 (0.04)
Technology	0.74 (0.06)	0.72 (0.03)
Industry	0.84 (0.05)	0.86 (0.02)
Benefit	0.41 (0.09)	0.77 (0.03)
$\hat{\eta}_j$	0.21 (0.07)	0.79 (0.07)

Table 4.4: Example 2: probabilities

	Items	O	E	$O-E$	$(O-E)^2/E$
Response (0,0)	5,1	17	10.05	6.95	4.81
	5,2	52	33.18	18.82	10.67
	6,1	10	5.56	4.44	3.54
	6,2	31	17.23	13.77	11.00
	6,5	26	15.83	10.17	6.53
Response (0,1)	2,5	67	85.82	-18.82	4.13
	5,2	57	75.82	-18.82	4.67
	6,2	26	39.77	-13.77	4.77
	6,5	31	41.17	-10.17	2.51
Response (1,0)	2,5	57	75.82	-18.82	4.67
	2,6	26	39.77	-13.77	4.77
	5,2	67	85.82	-18.82	4.13
	5,6	31	41.17	-10.17	2.51

Table 4.5: Example 2: residuals

Item	$\hat{\pi}_{i1}$	$\hat{\pi}_{i2}$	$\hat{\pi}_{i3}$
Comfort	0.63	0.80	0.95
Environment	0.74	0.29	0.76
Work	0.28	0.92	0.67
Future	0.16	0.95	0.82
Technology	0.74	0.00	0.83
Industry	0.74	0.55	0.92
Benefit	0.00	0.72	0.77
$\hat{\eta}_j$	0.09	0.12	0.79

Table 4.6: Example 2: estimated probabilities

Chapter 5

Latent trait poly ord

Look at models for

- categorical unordered observed variable (with more than two categories)
- categorical ordered observed variable

5.1 Latent trait model for polytomous data

Let's assume we have

- p observed variable x_i with $i = 1, \dots, p$
- each has c_i number of categories labelled $s = 0, \dots, c_i - 1$. We recode x_i as follows (basically recoding an observed variable in a vector)

$$x_i(s) = \begin{cases} 1, & \text{if the response falls in category } s; \\ 0, & \text{otherwise} \end{cases}$$

- we can collect a single response of an individual on an item in a c_i-length vector is denoted by $\mathbf{x}_i$

$$\mathbf{x}_i = \begin{bmatrix} x_i(0) \\ x_i(1) \\ \dots \\ x_i(c_i) \end{bmatrix}$$

  with $\sum_s x_i(s) = 1$; if $c_i = 2$ we have the binary case (which can be simplified to $\mathbf{x}_i = x_i(1)$)

- full response pattern to all the p variable for an individual can be collected in a set of vectors $\mathbf{x}^\mathsf{T} = (\mathbf{x}_1^\mathsf{T}, \mathbf{x}_2^\mathsf{T}, \dots, \mathbf{x}_p^\mathsf{T})$

NB: c'è qualcosa di sbagliato nelle trasposizioni dei vettori? sarebbe una riga? comunque poi non utilizzato

- as in the binary case we define the response function as

$$P(x_i(s) = 1|\mathbf{y}) = \pi_{is}(\mathbf{y}), \quad \text{with } s = 0, \dots, c_i - 1 \text{ and } i = 1, \dots, p.$$

  so we have as many probability as positive response as categories are, with the constraint $\sum_s \pi_{is}(\mathbf{y}) = 1$ (prob of positive response in at least one category for each item is 1).

In the classical GLLVM x_i and θ_i are supposed to be scalars. In this case they are vectors; thus GLLVM generalise to provide a vector-valued GLLVM. We obtain for:

- **random part**: the conditional distribution of $\mathbf{x}_i$ given $\mathbf{y}$ is taken from the *multinomial* distribution (basically the proper probability selected by the exponent switch)

$$\begin{aligned} g_i(\mathbf{x}_i|\mathbf{y}) &= \prod_{s=0}^{c_i-1} \pi_{is}(\mathbf{y})^{x_{i(s)}} \\ &\overset{(1)}{=} \prod_{s=0}^{c_i-1} \pi_{i0}(\mathbf{y})^{x_{i(s)}} \left(\frac{\pi_{is}(\mathbf{y})}{\pi_{i0}(\mathbf{y})}\right)^{x_{i(s)}} \\ &\overset{(2)}{=} \pi_{i0}(\mathbf{y}) \prod_{s=0}^{c_i-1} \left(\frac{\pi_{is}(\mathbf{y})}{\pi_{i0}(\mathbf{y})}\right)^{x_{i(s)}} \\ &\overset{(3)}{=} \pi_{i0}(\mathbf{y}) \exp \sum_{s=0}^{c_i-1} x_{i(s)} \log \left(\frac{\pi_{is}(\mathbf{y})}{\pi_{i0}(\mathbf{y})}\right) \\ &\overset{(4)}{=} \pi_{i0}(\mathbf{y}) \exp \boldsymbol{\theta}_i^{\mathsf{T}} \mathbf{x}_i \end{aligned}$$

  where

  - (1) we want to put the formula in the exponential distributions/GLM-style (I guess) and so start by multiplying and divide by $\pi_{i0}(\mathbf{y})^{x_{i(s)}}$; by
  - (2) the first term within product $\pi_{i0}(\mathbf{y})^{x_{i(s)}}$ will be $\pi_{i0}(\mathbf{y})$ only in correspondance of the category for which there was a positive response and 1 otherwise; so basically does not depend on s and we can take it out of sum
  - (3) I do the exp/log trick to remove the productory
  - (4) we just rewrite as vector product with

$$\boldsymbol{\theta}_i^{\mathsf{T}} = \left(0 = \log \frac{\pi_{i0}(\mathbf{y})}{\pi_{i0}(\mathbf{y})}, \log \frac{\pi_{i1}(\mathbf{y})}{\pi_{i0}(\mathbf{y})}, \log \frac{\pi_{i2}(\mathbf{y})}{\pi_{i0}(\mathbf{y})}, \ldots, \log \frac{\pi_{i(c_i-1)}(\mathbf{y})}{\pi_{i0}(\mathbf{y})}\right)$$

  So here theta is no more a single param but a vector of params/log-odds, where the first category of the polytomous variable is arbitrarily selected to be the reference category.

- **systematic part**: the linear predictor is also a vector and is combination of the latent variables $(y_1, \ldots, y_q)$ times the effects $\boldsymbol{\alpha}$ of each latent variable on each category's probability:

$$\boldsymbol{\eta}_i = \boldsymbol{\alpha}_{i0} + \boldsymbol{\alpha}_{i1} y_1 + \ldots + \boldsymbol{\alpha}_{iq} y_q$$

  where $\boldsymbol{\alpha}_{ij} = (\alpha_{ij}(0) = 0, \alpha_{ij}(1), \ldots, \alpha_{ij}(c_i - 1))$

- **link function**: logit (in this case $\boldsymbol{\theta}_i$ has already the log-odd included so we don't see it below)

$$\boldsymbol{\theta}_i = \boldsymbol{\alpha}_{i0} + \boldsymbol{\alpha}_{i1} y_1 + \ldots + \boldsymbol{\alpha}_{iq} y_q$$

Moreover as in binary case we can express the model in terms of probability of responding the s category to the i-th question as

$$\pi_{is}(\mathbf{y}) = \frac{\exp(\alpha_{i0}(s) + \sum_{j=1}^{q} \alpha_{ij}(s) y_j)}{\sum_{r=0}^{c_i - 1} \exp(\alpha_{i0}(r) + \sum_{j=1}^{q} \alpha_{ij}(r) y_j)}$$

Remark 27. Some remarks:

- also in this case it is easily to show that the components are sufficient statistics for the latent variables and are given by

  $$X_j = \sum_{i=1}^{p} \sum_{s=0}^{c_i - 1} \alpha_{ij}(s) x_i(s)$$

  where X_j is the total score for an individual and is sufficient statistics for the j-th latent variable.
  Here we cycle on the s categories of the observed variables and $x_i(s)$ works as a switch (sum the alphas for the category which are observed)

- for interpretations:
  - the interpretation of π's is probabilities for the median individual (i-th item, s category).
  - the α's may be interpreted as in the binary case (the higher the loading the more is the effect of the latent variable on the observed) but, separately, as functions of s and i (we have many loadings as the categories are).
    If loading for the first category are much smaller than loadings for the second category it means that the latent variables tends to be associated with the first category credo. Also if the loadings of one latent variable are higher than the ones from another latent variables, the item will be better discriminating the first latent variable (eg tipically one can consider the average of loadings)

- in the polytomous case the equivalence between UVA (underlying variable approach) and RF does not hold anymore.

5.1.1 Model estimation

Parameter to be estimated are the intercept and the loadings for each category. Even in this case we use the E-M algorithm, which is easily generalised from binary to the polytomous case.

- we start with loglik

  $$l = \sum_{h=1}^{n} \log g(\mathbf{x}_h | y_h)$$

- we define the E-step in terms of $E(L|\mathbf{x})$ and we end up to an expression that depends on posterior probability of y

  $$r_{it}(s) = \sum_{h=1}^{n} x_{ih}(s) h(y_t | \mathbf{x}_h)$$

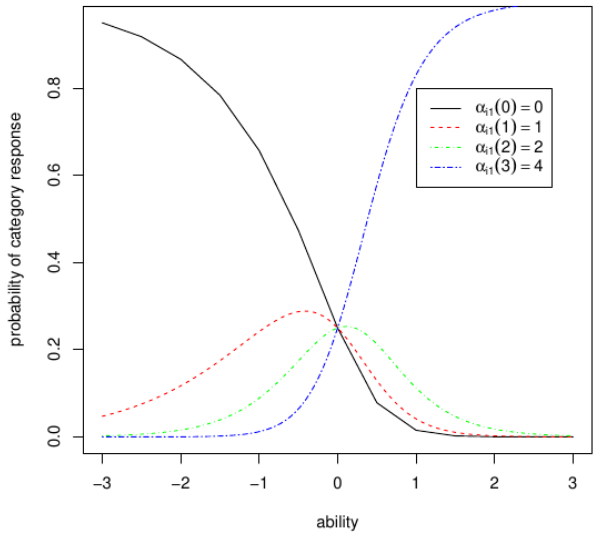


Figure 5.1: probabilities of response

where as in the binary case we have

$$n_t = \sum_{h=1}^{n} h(y_t|\mathbf{x}_h)$$

In the e-step we compute these quantities above given initial assumpion on h(y|x) coming from setup or previous iterations

- in the M-step we maximize the $E(l|\mathbf{x})$ with respect to the parameters. The procedure is the same as in the binary case.
- repeat until convergence

5.1.2 Probabilities of response categories

In figure 5.1 the probabilities of response in a certain category where categories here are level of correctness (0=completely wrong, 3=absolute right).
For a given item we have a curve for each category, due to a lodings of 0,1,2,4 for the category in order (we fixed the intercept to 0 for all the categories $\alpha_{i0}(s) = 0, \quad s = 0, \ldots, 3$). The idea is that

- probability of all correct (blue line) is high with high ability
- while prob of responding in the first category (completely wrong, black line) will be higher for lowest value of ability

5.1.3 Example

Example on attitudes toward the environment:

- $n = 291$ individuals
- $p = 6$ items which are
  - Lead from petrol [LeadPetrol]
  - River and sea pollution [RiverSea]
  - Transport and storage of radioactive waste [RadioWaste]
  - Air pollution[AirPollution]
  - Transport and disposal of poisonous chemicals [Chemical]
  - Risks from nuclear power station [Nuclear]
- possible responses: 'very concerned','slightly concerned', 'not very concerned', 'not at all concerned' (we ignore ordinality of data now). Items are assumed to be indicators of individuals' attitudes towards environmental issues.
- since the proportion of individuals falling into the 'not very concerned' is less than 10%, categories 'not very concerned' and 'not al all concerned' were collapsed. Thus we have 3 categories for each items, the first category is assumed as reference.

The results are shown in table 5.1:

- for each observed variable we see have a row for each category (but the reference one) with the corresponding coefficients (one intercept and one slope associated to the single latent variable, with their standard errors within parentheses). In general what we comment here is the value of the loadings
- the highest value of the loadings correspond to RiverSea category 3 (5.06), RadioWaste cat 3, Airpollution cat 3 and Chemicals cat 3 (4.75); these are the items that more contribute to the measure of attitude toward environment (one should compute the mean I guess within same category)
- another thing is the difference of loadings between category within the same item: the highest diff (3) is attained with RiverSea (5.06 - 2.11) and airpollution. For these items we have more variability/separation on the latent space and thus the item should be more predictive of the latent variable
- **gof**: we dont' have software to compute godness of fit for polytomous model, lol

5.2 Ordinal variables

5.2.1 Response function model

Remark 28. Most of the cases in social sciences we have ordinal categories. Examples of ordered variables are often attitudinal statements with response alternative such as 'strongly disagree', 'disagree', 'agree' and 'strongly agree' (Likert scale).

Items	Category	$\alpha_{i0}(s)$	$\alpha_{i1}(s)$
LeadPetrol	2	-0.75 (0.18)	1.34 (0.27)
	3	-3.06 (0.41)	2.06 (0.45)
RiverSea	2	-2.33 (0.37)	2.11 (0.44)
	3	-8.87 (2.60)	5.06 (1.74)
RadioWaste	2	-2.57 (0.50)	3.31 (0.73)
	3	-6.00 (1.16)	5.14 (1.12)
AirPollution	2	-1.39 (0.35)	3.10 (0.69)
	3	-8.53 (1.62)	6.52 (1.24)
Chemicals	2	-2.83 (0.62)	3.61 (0.95)
	3	-5.43 (1.08)	4.75 (1.20)
Nuclear	2	-0.33 (0.18)	1.54 (0.33)
	3	-1.95 (0.38)	2.82 (0.51)

Table 5.1: attitude toward environment example

Similarly to the previously we consider the case in which:

- each item i has c_i ordered categories
- categories have response probabilities $\pi_{is}(\mathbf{y}), \quad (s = 1, \ldots, c_i)$, that is the probability that, given $\mathbf{y}$, a response falls in category s for the variable i.
- $\sum_s \pi_{is}(\mathbf{y}) = 1$ (summing these probabilities along categories has to sum up to 1)

We want to generalize/reuse the results of the binary case, we split the categories into two groups (to make it similar to the binary case):

1. Group 1: categories $1, \ldots, s$
2. Group 2: categories $s+1, \ldots, c_i$

We want to apply the binary logit model: at this aim we define the *cumulative probabilities*:

NB: here the label for whatever reason are reverted fuck SAS

$$\gamma_{is}(\mathbf{y}) = P(x_i \leq s|\mathbf{y}) = \pi_{i1}(\mathbf{y}) + \pi_{i2}(\mathbf{y}) + \ldots + \pi_{is}(\mathbf{y}) \quad \textit{(label 1)}$$
$$1 - \gamma_{is}(\mathbf{y}) = P(x_i > s|\mathbf{y}) = \pi_{is+1}(\mathbf{y}) + \pi_{i2}(\mathbf{y}) + \ldots + \pi_{ic_i}(\mathbf{y}) \quad \textit{(label 0)}$$

The GLLVM is defined as follows:

- **random part**: the conditional distribution of $\mathbf{x}_i$ given $\mathbf{y}$ is taken equally from the *multinomial distribution* but in this case we can make further substitution using cumulative probability defined above:

$$g(\mathbf{x}_i|\mathbf{y}) = \prod_{s=1}^{c_i} \pi_{is}(\mathbf{y})^{x_{is}} = \prod_{s=1}^{c_i} (\gamma_{is}(\mathbf{y}) - \gamma_{i,s-1}(\mathbf{y}))^{x_{is}}$$

where similarly $x_{is} = 1$ if a randomly selected individual responds to category s of the item i-th and $x_{is} = 0$ otherwise.

- **systematic component**: we have a linear predictor that is a system of equation (many items), for the s-th category:

$$\eta_{is} = \alpha_{i0(s)} - \sum_{j=1}^{q} \alpha_{ij} y_j$$

we have

  - an intercept for each category/which depends on the category
  - a linear combination of *loadings which are the same/does not depend on category* (are not category specific). This is an **assumption made** (and different from the polytomous case where each category has its alphas)
  - particularly the sign minus in front of linear combination: as latent variable increase and coefficient increase the linear predictor decrease

- **Link function**: logit as in binary

$$\log\left(\frac{\gamma_{is}(\mathbf{y})}{1-\gamma_{is}(\mathbf{y})}\right) = \text{logit}(\gamma_{is}(\mathbf{y})) = \eta_{is} = \alpha_{i0(s)} - \sum_{j=1}^{q} \alpha_{ij} y_j \qquad (5.1)$$

where

  - we're modelling the log odd of responding in category s or lower over responding above
  - the minus in the linear predictor depends on how we constructed the logit on lhs (putting at the numerator the probability of lowest class and denominator highest): by doing so, an *increase* in the factor loadings or in the latent variable, diminish the probability of responding at category s or lower and increase the probability of *responding over category s* (which is at the denominator).[1]
    The α_{ij} are interpreted as factor loadings. The negative sign in front of the factor loadings indicates that, as y_j increases, the response on the observed item x_i is more likely to fall at the high end of the scale.
  - The factor loadings remain the same across categories of the same variable, that is the discriminating power of the item does not depend on the category
  - The intercept parameters are category dependent. This reflect the fact that as the threshold increases for the response, we assume the difficulty will be also increasing. Hence

$$\alpha_{i0(1)} \leq \alpha_{i0(2)} \leq \ldots \leq \alpha_{i0(c_i)} = +\infty$$

    In figure 5.2 the relation between ability and cumulative response probability. It's displayed the effect of increasing intercept for categories 1 to 4 (from -0.5 to 3.5), fixing a common loading $\alpha_{i1} = 1.0$. As intercept increase the curve moves to the left: considering a fixed

[1] Typically this is the parametrization done but in other books the model could be specified by reverting the ratio and putting a + into the linear combination.

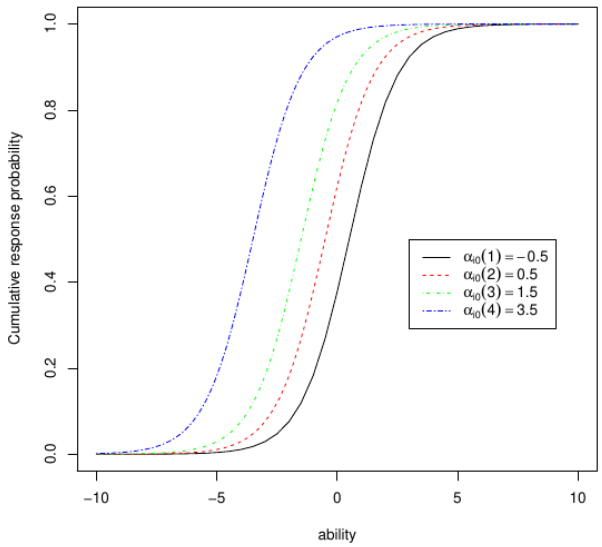


Figure 5.2: cumulative response probabilities for varying intercept

median level 0 of ability, as intercept increase the cumulative response probability (dovrebbe essere γ_{is}) that is probaiblity of responding in lower than s increases (because basically we're adding categories/probabilities). Eg going from first category to second the probability increase because we have both the probability of the first and the second category

TODO: comunque non mi convice perché allora la quarta non dovrebbe essere sempre a 100?

Important remark 22. This model is called the *proportional odds model*. This name comes from the fact that, in the one-factor case, the difference between two cumulative logits, that is, the left side of (5.1), for two persons with factor scores y_1 and y_2 is proportional to $y_2 - y_1$.

Remark 29. Model estimation can be performed via the E-M algorithm. The steps are the same as before.

5.2.1.1 Goodness of fit

Again the Pearson χ^2 and the likelihood ratio G^2 in this case defined as:

$$\chi^2 = \sum_{r=1}^{C} \frac{(nP_r - n\pi_r(\hat{\boldsymbol{\alpha}}))^2}{n\pi_r(\hat{\boldsymbol{\alpha}})}$$

$$G^2 = 2\sum_{r=1}^{C} nP_r \log\left(\frac{nP_r}{n\pi_r(\hat{\boldsymbol{\alpha}})}\right)$$

where

- observed frequency are nP_r (P_r observed proportion) while expected $n\pi_r(\hat{\boldsymbol{\alpha}})$;

- for the *number of response pattern* above the sum, C, we sum up to $C = \prod_{j=1}^{p} c_j$ (instead of up to 2^p, since each items can have more than two categories). In this case C is the number of possible response patterns;
- under regular conditions, the two statistics are asymptotically distributed as a chi-square with $C - \sum_j c_j - p - pq - 1$ degrees of freedom (C number of response patterns, $\sum_j c_j + p + pq$ parameters where $\sum_j c_j - p$ are the number of tresholds/intercepts (for each item one treshold less than the number of categories, pq are number of loadings)

Also in this case the sparse problem is very frequent: bivariate residuals

- is defined as:

$$GF_{fit}^{(ij)} = n \sum_{a=1}^{c_i} \sum_{b=1}^{c_j} \frac{(P_{ab} - \pi_{ab})^2}{\pi_{ab}}$$

  where P_{ab} and π_{ab} are the observed and expected bivariate residuals
- rule of thumb: $GF_{fit}^{(ij)} > 4c_i c_j$ indicates bad fit between item i and item j. (must be 4 for each single couple of category a, b in the sum of the definition);
- if bad fit occurred for many item increase the number of factors.

5.2.2 Underlying variable model

Remark 30. In the polytomous-unordered case the UVA approach is not equivalent to the response function approach. However if the variable are ordinal we can use the equivalence between UVA and response function approach like in the binary case. Very often researcher prefers UVA approach.

To define the UVA approach in the ordinal case we **assume** that:

- the i-th observed variable, with c_i ordered categories, is the realization of an underlying standard normal ξ_i for which are defined categorizing ticks, τs, such that:

$$x_i = s \quad \text{if} \quad \tau_{i,s-1} \leq \xi_i \leq \tau_{i,s} \quad (s = 1, \ldots, c_i)$$

  where $\tau_{i0} = -\infty, \tau_{ic} = \infty$. That is
  - between $-\infty$ and τ_{i1} we have that $x_i = 1$
  - between τ_{i1} and τ_{i2} we have that $x_i = 2$
  - between τ_{i,c_i-1} and $+\infty$ we have that $x_i = c_i$
- as in the binary case, that holds a standard linear factor model, for which the underlying variable $\boldsymbol{\xi}$ (which is p-dimensional, that is for every observed item we have an underlying variable) is the byproduct of our latent variables $\mathbf{y}$:

$$\boldsymbol{\xi} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{y} + \boldsymbol{\varepsilon}$$

  We assume that the underlying bivariate distributions are normal.

Regarding **estimation** we do it as in the binary case:

- we can fit the factor model provided we can estimate the correlation coefficients;
- estimates of this coefficients are called *polychoric correlations* (equivalent of pearson correlation, which was tetracoric correlation in binary case). They are estimated using the maximum likelihood method (first we estimate the threshold, the three step approach...);
- once estimated the correlation we estimate the factor model using classical methods as in binary case. Instead of starting from the pearson correlation matrix with start from the polychoric correlation matrix (as done in classical linear factor analysis)

Important remark 23. Again if we use pearson correlation instead of polychoric/tetracoric we start with an underestimated correlation (pearson underestimate correlation)

Important remark 24 (Equivalence). The equivalence found in the binary case holds also in the ordinal case between UVA and response function approach. So one can estimate the parameters using the UVA approach (which is always feasible) and obtain the estimates produced by the other in a simple way.
In particular can be proved that (if we use the same link, logit or probit)

$$\alpha_{i0(s)} = \frac{\tau_{i,s-1} - \mu_i}{\sqrt{\psi}}, \quad \alpha_{ij} = \frac{\lambda_{ij}}{\sqrt{\psi}}$$

TO go from one parametrization to the other similarly we have to multiply this above by 1.8?? The equivalence holds if we use the same link function (wheter it is logit or probit): for the probit model and for the logit model in so far as the logit is a good approximation of the probit.
As before we obtain for standardized loadings:

$$st\alpha_{ij} = \frac{\alpha_{ij}}{\sqrt{1 + \sum_j \alpha_{ij}^2}}$$

5.2.3 Parameter interpretation in proportional odds model

Important remark 25.

- for the proportional odds model the sufficiency principle does not hold: we cannot compute the components as proxy of the latent variables; what we can do is to use posterior expectation of y|x to have an estimate of factor/scores/latent variables;
- the parameter π_{is} is still interpreted as the probability that the median individual falls in category s on the variable i;
- the α's cannot be interpreted as category scores or as weights in a component; are interpreted as discrimination parameters/loadings;
- the posterior density of $\mathbf{y}$ is no longer dependent on a linear function of $\mathbf{x}$.

5.2.4 Example of proportional odds model

Attitude to science and technology analyzed treating observed variables as ordered:

- $n = 392$ respondents
- 7 items
  - Science and technology are making our lives healthier, easier and more comfortable. [Comfort]
  - Scientific and technological research cannot play an important role in protecting the environment and repairing it. [Environment]
  - The application of science and new technology will make work more interesting [Work]
  - Thanks to science and technology, there will be more opportunities for the future generations. [Future]
  - New technology does not depend on basic scientific research. [Technology]
  - Scientific and technological research do not play an important role in industrial development. [Industry]
  - The benefits of science are greater than any harmful effects it may have. [Benefit]
- Possible response categories are 'strongly disagree', 'disagree to some extent', 'agree to some extent', 'strongly agree'.

First analysis It was done by focusing on positive worded Comfort, Work, Future, Benefit only, responses considered indicators of attitude towards science and technology. A *one factor model* was estimated. In table:

- 5.2 Estimated factor loadings with standard errors in brackets and standardized loadings. (the threshold are not important here, the loadings are of interest)
  We see that all the loadings are significant (considered the ratio between loading and standard error); all these items significantly contributes in the measure of the attitude toward science/technology and the most important (item with the highest loading) is Future; we can see it better from the standardized coefficient as well were Future has the highest value (0.92, where the *maximum is 1*). So the latent variable has *positive effect* on all these observed variables with higher effect on Future and lowest effect on comfort
- 5.3, for goodness of fit, the sums of chi-square residuals for pairs of items from two-way margins for the one-factor model (we didn't report chisq or lrt because 4 categories, a lot of response patterns and possible spars data).
  Since all the items has 4 possible response we have to check that the cells are less than $4 \cdot c_i \cdot c_j = 4 \cdot 4 \cdot 4 = 64$, which is fullfilled for the table (all the GF_{fit} are ok), so we think is a reasonable model.

	α_{i1}	$st\alpha_{i1}$
Comfort	1.04 (0.16)	0.72
Work	1.23 (0.19)	0.77
Future	2.28 (0.32)	0.92
Benefit	1.10 (0.16)	0.74

Table 5.2: Estimated factor loadings

	Work	Future	Benefit
Comfort	25.54	11.98	27.23
Work		9.21	23.27
Future			17.41

Table 5.3: Sums of chi-square residuals for pairs of items from two-way margins for the one-factor model

We could be more restrictive by imposing a threshold of 3 for each couple of item, that is to consider a cell below $3 \cdot 4 \cdot 4 = 48$ as ok, and even in this case we're good to go

- 5.4, for residuals again, thechi-squared residuals for the two-way marginals of items *Comfort and Work*, that are the residual for each pair of item of Comfort and Work.
  - these are the terms within the sum of GF_fit;
  - for example the item at the first row and second column is the residual for the category 2 of comfort and 1 of work;
  - if we do this analysis unpacked. all the cells here must be < 4 (or < 3): even if overvall the GF_{fit} is within threshold, we observe that for a couple of items (1-4) $11.5 > 4$. However for just one of this and with overall GF_{fit} ok we consider this a good model.
- we conclude that a one factor model for this four items (Comfort, Work, Future, Benefit) is a good model.

A second analysis We consider all the seven items (reversing the item which are needed) and fit a one-factor model at first. In table:

- 5.5, for goodnes of fit, sums of chi-squared residuals for pair of items from the two-way margins for the one-factor model.

Categories	1	2	3	4
1	0.87	1.79	0.44	11.50
2	2.43	0.01	0.14	2.09
3	0.02	0.04	0.45	2.39
4	0.66	0.26	1.51	0.94

Table 5.4: Chi-squared residuals for the two-way marginals of items Comfort and Work

Here there are a lot of residuals that are above the threshold (be it 64 or 48) so we connclude 1 factor model for the 7 items is not that good. At this point we can choose to

  - delete some items (eg going back to the previous 4) eg especially the one that are responsible for the bad fit, in this case item 2, 5, and 6
  - increasing the number of factors going to a two factor model

- 5.6 the estimates for the two-factor model.
  In the table are reported all the tresholds $\alpha_{i0}(s)$ and also the estimation of the loadings for the two factors (α_{i1} and α_{i2}).
  For each item we have three threshold becaus the category are four (number of threshold estimated is number of categories-1) and by definition the threshold are in increasing order. What info the treshold gives?

  - if the distance between adjacent tresholds is large there's a big difference between proportion of people that answer to one category or the other;
  - otoh if the distance between the tresholds is almost the same it means that the proportion of people that reply to all category are almost the same

  Usually we're more interested in the value of the loadings which allows us to interpret the factors:

  - the first factor has high loadings in corresponding to environment (1.61) technology (1.71) and industry (1.55). The second factor has high loading for confort (1.16) work (1.2) future (2.16) and benefit (1.12)
    We could interpret the first factor regarding the belief that science/technology plays an important role for science *technology/industry point of view itself* (see questions)); the other factor is more related to the *social benefit* of science/technology
    Note: in general some clustering of variables could be due as well to the fact that some items were reversed in meaning and people tend to respond more consistently just for this reason. This could occurr in this case (she says) because the first factor is mostly composed by items (environment, technology, industry) which have reversed meaning

- 5.7 residuals for pair of items from the two-way margins for the two-factor model. All residuals are < 64 so model is good: 1 factor was not enough to explain all the associations among observed variables, but including 1 more latent variable solves our problem

Comparison Pearson correlation and polychoric In figure 5.3 the correlation between pearson and polychoric correlation;

- pearson is systematically lower than polychoric

Items	2	3	4	5	6	7
1	13.33	26.20	12.41	18.93	9.21	24.51
2		27.54	24.09	98.36	90.76	23.52
3			10.28	21.08	35.78	23.11
4				23.23	26.90	17.17
5					103.24	17.09
6						20.28

Table 5.5: Second analysis: sums of chi-squared residuals for pair of items from the two-way margins for the one-factor model.

Items	Category	$\alpha_{i0}(s)$	α_{i1}	α_{i2}
Comfort	1	-5.00 (2.09)	0.27 (0.20)	1.16 (0.18)
	2	-2.74 (1.77)		
	3	1.53 (0.33)		
Environment	1	-3.45 (1.20)	1.61 (0.35)	0.09 (0.22)
	2	-1.26 (0.87)		
	3	0.99 (0.71)		
Work	1	-2.95 (0.86)	-0.39 (0.35)	1.20 (0.23)
	2	-0.90 (0.83)		
	3	2.30 (0.45)		
Future	1	-5.05 (1.92)	-0.30 (0.28)	2.16 (0.34)
	2	-2.13 (1.43)		
	3	1.90 (0.62)		
Technology	1	-4.17 (1.60)	1.71 (0.36)	0.08 (0.24)
	2	-1.49 (1.04)		
	3	1.07 (0.70)		
Industry	1	-4.71 (1.60)	1.55 (0.31)	0.55 (0.25)
	2	-2.53 (1.30)		
	3	0.45 (0.59)		
Benefit	1	-3.38 (1.46)	-0.08 (0.00)	1.12 (0.20)
	2	-1.00 (0.72)		
	3	1.71 (0.39)		

Table 5.6: Second analysis: fit a second-factor model.

Items	2	3	4	5	6	7
1	12.36	25.14	11.99	18.80	7.37	24.58
2		21.64	23.29	20.94	31.86	22.85
3			9.44	15.31	30.54	23.64
4				21.93	26.71	17.23
5					34.87	16.67
6						20.78

Table 5.7: Second analsysis: Sums of chi-squared residuals for pair of items from the two-way margins for the two-factor model.

Table 8.10 *Correlation matrices for science and technology data*

	Pearson correlations				Polychoric correlations			
	Comfort	Work	Future	Benefit	Comfort	Work	Future	Benefit
Comfort	1.00				1.00			
Work	0.15	1.00			0.20	1.00		
Future	0.28	0.40	1.00		0.35	0.48	1.00	
Benefit	0.33	0.17	0.31	1.00	0.41	0.21	0.38	1.00

Figure 5.3: Correlations comparisons

Table 8.11 *UV approach, estimated factor loadings with standard errors in brackets and chi-squared measures of fit, science and technology data (four items)*

	ML $\hat{\alpha}^*_{i1}$	WLS $\hat{\alpha}^*_{i1}$
Comfort	0.48 (0.06)	0.57 (0.07)
Work	0.55 (0.06)	0.56 (0.06)
Future	0.79 (0.06)	0.78 (0.06)
Benefit	0.51 (0.06)	0.58 (0.06)
chi-squared	28.77	10.21
p-value	0.00	0.0061

Figure 5.4: ML vs WLS

- the smaller the number of categories the more biased are the results, eg with binary is the worst (with 10-15 categories the results are almosts the same)

Comparison ML vs WLS In table 5.4 solutions with underlying variable approach is reported alongside (only loadings): ML is what we've seen, in `lavaan` only WLS is available:

- There are some differences between estimators; WLS is like an OLS with weight matrices that allows to analyze ordinal data
- WLS is the best method for UVA

exam:

- classical linear factor model
- binary data
- **not for ordinal data**?

5.3 Uso di R

```
library(ltm)
library(lavaan)

## The dataset on science and technology in included in the \texttt{ltm} package

head(Science)

##          Comfort       Environment           Work         Future     Technology
## 1 strongly agree    strongly agree strongly agree          agree strongly agree
## 2          agree    strongly agree          agree          agree          agree
## 3          agree          disagree       disagree       disagree strongly agree
## 4          agree             agree       disagree       disagree strongly agree
## 5          agree strongly disagree strongly agree strongly agree       disagree
## 6 strongly agree             agree strongly agree          agree          agree
##         Industry           Benefit
## 1          agree          disagree
## 2          agree             agree
## 3 strongly agree             agree
## 4 strongly agree             agree
## 5          agree strongly disagree
## 6 strongly agree             agree

str(Science)

## 'data.frame': 392 obs. of  7 variables:
##  $ Comfort    : Factor w/ 4 levels "strongly disagree",..: 4 3 3 3 3 4 3 3 3 4 ...
##  $ Environment: Factor w/ 4 levels "strongly disagree",..: 4 4 2 3 1 3 2 2 3 3 ...
##  $ Work       : Factor w/ 4 levels "strongly disagree",..: 4 3 2 2 4 4 2 2 3 3 ...
##  $ Future     : Factor w/ 4 levels "strongly disagree",..: 3 3 2 2 4 3 3 3 4 3 ...
##  $ Technology : Factor w/ 4 levels "strongly disagree",..: 4 3 4 4 2 3 4 3 4 3 ...
##  $ Industry   : Factor w/ 4 levels "strongly disagree",..: 3 3 4 4 3 4 4 4 4 3 ...
##  $ Benefit    : Factor w/ 4 levels "strongly disagree",..: 2 3 3 3 1 3 4 4 2 3 ...

######## Proportional Odds Model Estimation ######

## - grm command permit toestimate the proportional odd model (name comes from "gradu
##   response model", an alternate name for proportional odd model)
## - limitation of this model is that we can estimate a model with
##   just 1 latent variable so we select the 4 variable from the
##   first example (so we dont specify the number of latent varible)
## - the command just take the data: we don't have to specify any number of
##   latent variable because it's 1

sel_vars <- c("Comfort","Work","Future","Benefit")
science_sel <- Science[sel_vars]

# model 1 on only 4 variables
(POM <- ltm::grm(science_sel))

##
```

```
## Call:
## ltm::grm(data = science_sel)
##
## Coefficients:
##          Extrmt1  Extrmt2  Extrmt3  Dscrmn
## Comfort   -4.672   -2.536    1.408   1.041
## Work      -2.385   -0.735    1.849   1.226
## Future    -2.281   -0.965    0.856   2.299
## Benefit   -3.060   -0.906    1.543   1.094
##
## Log.Lik: -1609

# The
# - loadings are in the Dscrmn column which are exactly solutions in table 1.2
# - tresholds are reported in Extrmt1  Extrmt2  Extrmt3 and we see that are
#   ordered (within item in row)
#
# bivariate margins as in table 1.3
margins(POM, print=3)

##
## Call:
## ltm::grm(data = science_sel)
##
## Fit on the Two-Way Margins
##
##         Comfort Work  Future Benefit
## Comfort -       25.48 12.01  27.36
## Work            -      9.20  23.21
## Future                 -     17.48
## Benefit                      -

# model with 1 latent variable on all the dataset
POM2 <- ltm::grm(Science)
## to see the bad fit from table 1.5 is below
margins(POM2, print=3)

##
## Call:
## ltm::grm(data = Science)
##
## Fit on the Two-Way Margins
##
##             Comfort Environment Work  Future Technology Industry Benefit
## Comfort     -       13.35       37.59 49.42  20.53       7.52     87.41
## Environment         -           23.92 24.90  20.49      31.73     25.29
## Work                            -     98.81  17.60      30.44     33.52
## Future                          ***   -      24.43      27.15     77.57
## Technology                                 -          34.24     18.98
## Industry                                                -         21.40
```

```
## Benefit     ***                         ***                         -
##
## '***' denotes pairs of items with lack-of-fit

## ====================================================
## We use lavaan to apply the underlying variable approach
#
# just for curiosity some variable may be categorical and other continuous;
# lavaan permits this and we only need to specify which one are ordered.  with
# continuous data it will do pearson correlation, with ordinal and binary data
# will consider tetracoric/polychoric correlation and for continuos-categorical
# correlations it's used so called _polyserial_correlations_

# UVA Model: 1 factor
# -------------------
# - difference with linear factor models is the specification of the parameter
#   ordered: WHEN WE HAVE BINARY OR ORDERED DATA put in the ordered parameters
#   the variables names with binary/ordered data, so that lavaan knows it has to
#   apply underlying variable approach
# - if we don't put std.lv we'll have loadings fits 1
# - fit.measures = TRUE in summary shows other measures than chi-square
model.f1 <- 'f1 =~ Comfort + Work + Future + Benefit'
fit <- cfa(model.f1, data = science_sel, ordered = sel_vars, std.lv = TRUE)
summary(fit, fit.measures = TRUE)

## lavaan 0.7-2 ended normally after 8 iterations
##
##   Estimator                                         DWLS
##   Optimization method                           NLMINB
##   Number of model parameters                        16
##
##   Number of observations                           392
##
## Model Test User Model:
##                                                   Standard      Scaled
##   Test Statistic                                  15.702      22.940
##   Degrees of freedom                                   2           2
##   P-value (Unknown)                                   NA       0.000
##   Scaling correction factor                                    0.690
##   Shift parameter                                              0.183
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                                 337.723     276.078
##   Degrees of freedom                                   6           6
##   P-value                                             NA       0.000
##   Scaling correction factor                                    1.228
##
## User Model versus Baseline Model:
```

```
##
##   Comparative Fit Index (CFI)                    0.959       0.922
##   Tucker-Lewis Index (TLI)                       0.876       0.767
##
##   Robust Comparative Fit Index (CFI)                         0.892
##   Robust Tucker-Lewis Index (TLI)                            0.675
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.132       0.164
##   90 Percent confidence interval - lower         0.077       0.108
##   90 Percent confidence interval - upper         0.197       0.227
##   P-value H_0: RMSEA <= 0.050                    0.009       0.001
##   P-value H_0: RMSEA >= 0.080                    0.940       0.992
##
##   Robust RMSEA                                               0.186
##   90 Percent confidence interval - lower                     0.122
##   90 Percent confidence interval - upper                     0.258
##   P-value H_0: Robust RMSEA <= 0.050                         0.000
##   P-value H_0: Robust RMSEA >= 0.080                         0.996
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.061       0.061
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.983
##   90 Percent confidence interval - lower         0.963
##   90 Percent confidence interval - upper         0.994
##
##   Robust GFI                                                 0.967
##   90 Percent confidence interval - lower                     0.937
##   90 Percent confidence interval - upper                     0.985
##
## Parameter Estimates:
##
##   Parameterization                                 Delta
##   Standard errors                            Robust.sem
##   Information                                  Expected
##   Information saturated (h1) model        Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)
##   f1 =~
##     Comfort           0.515    0.057    9.118    0.000
##     Work              0.526    0.049   10.723    0.000
##     Future            0.777    0.047   16.384    0.000
##     Benefit           0.531    0.051   10.510    0.000
```

```
##
## Thresholds:
##                    Estimate  Std.Err  z-value  P(>|z|)
##     Comfort|t1       -2.234    0.172  -12.960    0.000
##     Comfort|t2       -1.314    0.088  -14.952    0.000
##     Comfort|t3        0.749    0.070   10.652    0.000
##     Work|t1          -1.377    0.091  -15.155    0.000
##     Work|t2          -0.428    0.066   -6.536    0.000
##     Work|t3           1.079    0.079   13.693    0.000
##     Future|t1        -1.803    0.119  -15.091    0.000
##     Future|t2        -0.774    0.071  -10.937    0.000
##     Future|t3         0.691    0.069    9.981    0.000
##     Benefit|t1       -1.611    0.105  -15.415    0.000
##     Benefit|t2       -0.500    0.066   -7.531    0.000
##     Benefit|t3        0.845    0.072   11.685    0.000
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)
##    .Comfort           0.734
##    .Work              0.723
##    .Future            0.396
##    .Benefit           0.718
##     f1                1.000
#
## In order
## - Estimation is WLS (ML is not available for estimation)
## - then we have the chisquare test; however if the smaple size is LARGE we cannot
##   rely on chisquare when using lavaan
## - we look at indices CFI and TLI which should be both greater than 0.95
##   (here not a properly good model: CFI acceptable, TLI no)
## - looking at RMSE is high (0.13) not good
## - so according to lavaan this factor model is not a good model (differently
##   from the response function approach in which we have margins that were good)
##
## however let's look at parameters estimates:
##
## - foreach item we have the three treshold estimates in Thresholds (taus, 3
##   for 4 categories), while the common alphas are in Latent Variables. Future
##   is the item associated with the highest loading
##
## - finally we have the estimated varainces of the error terms associated to
##   each item (because we're estimating a linear factor model). So for example
##   for confort the variance of the error term is 0.734
##
## - we can apply/check relation between estimations with response function approach
##   the UVA: eg we see that 0.515 is UVA estimate for comfort (should be lambda
##   in our notation). We need to divide it by the square root of uniqueness
##   and multiplying by 1.8 (because we're using for the response function the
##   logit and for UVA the probit) and check if the two values are similar
```

```
##   > (0.515 / sqrt(0.734))*1.8
##   [1] 1.082
##   if we go to before in Column Dscrmn we get 1.041 for Comfort which is
##   similar
##
## What is more interesting is that with lavaan we can fit even a two factor
## model

# UVA Model 2 factors
# -------------------
#
# don't forget that with lavaan we can fit only CONFIRMATORY factor models,
# that means that we have to give to lavaan a model we have in mind.  so we
# take the result from the exploratory factor model presented in the
# theoretical part and split the variable in the two factors accordingly.
# Also important to specify that all variables are ordered.
## Basically with model.f2 below we imply to fix the loading of the non used
# variable names to 0: eg Environment for f2 is set to 0

## model.f2 <-'f1 =~ Comfort + Work + Future + Benefit
## f2 =~ Environment + Technology + Industry'
model.f2 <- '
f1 =~ Environment + Technology + Industry
f2 =~ Comfort + Work + Future + Benefit
'
fit2 <- cfa(model.f2, data = Science, ordered = names(Science), std.lv = TRUE)
summary(fit2, fit.measures=TRUE)
```

```
## lavaan 0.7-2 ended normally after 11 iterations
##
##   Estimator                                         DWLS
##   Optimization method                           NLMINB
##   Number of model parameters                        29
##
##   Number of observations                           392
##
## Model Test User Model:
##                                                Standard      Scaled
##   Test Statistic                                 46.333      53.886
##   Degrees of freedom                                 13          13
##   P-value (Unknown)                                  NA       0.000
##   Scaling correction factor                                   0.891
##   Shift parameter                                             1.904
##     simple second-order correction
##
## Model Test Baseline Model:
##
##   Test statistic                               683.313     529.063
##   Degrees of freedom                                 21          21
##   P-value                                            NA       0.000
```

```
##   Scaling correction factor                                  1.304
##
## User Model versus Baseline Model:
##
##   Comparative Fit Index (CFI)                    0.950       0.920
##   Tucker-Lewis Index (TLI)                       0.919       0.870
##
##   Robust Comparative Fit Index (CFI)                         0.909
##   Robust Tucker-Lewis Index (TLI)                            0.853
##
## Root Mean Square Error of Approximation:
##
##   RMSEA                                          0.081       0.090
##   90 Percent confidence interval - lower         0.057       0.066
##   90 Percent confidence interval - upper         0.107       0.115
##   P-value H_0: RMSEA <= 0.050                    0.020       0.004
##   P-value H_0: RMSEA >= 0.080                    0.555       0.761
##
##   Robust RMSEA                                               0.091
##   90 Percent confidence interval - lower                     0.064
##   90 Percent confidence interval - upper                     0.120
##   P-value H_0: Robust RMSEA <= 0.050                         0.008
##   P-value H_0: Robust RMSEA >= 0.080                         0.770
##
## Standardized Root Mean Square Residual:
##
##   SRMR                                           0.068       0.068
##
## Goodness of Fit Index:
##
##   Goodness of Fit Index (GFI)                    0.976
##   90 Percent confidence interval - lower         0.959
##   90 Percent confidence interval - upper         0.988
##
##   Robust GFI                                                 0.970
##   90 Percent confidence interval - lower                     0.949
##   90 Percent confidence interval - upper                     0.985
##
## Parameter Estimates:
##
##   Parameterization                                   Delta
##   Standard errors                             Robust.sem
##   Information                                   Expected
##   Information saturated (h1) model          Unstructured
##
## Latent Variables:
##                    Estimate  Std.Err  z-value  P(>|z|)
##   f1 =~
##     Environment       0.658    0.051   12.821    0.000
```

```
##     Technology        0.694    0.047   14.698    0.000
##     Industry          0.631    0.049   12.901    0.000
##   f2 =~
##     Comfort           0.525    0.056    9.356    0.000
##     Work              0.520    0.049   10.548    0.000
##     Future            0.773    0.047   16.448    0.000
##     Benefit           0.533    0.050   10.582    0.000
##
## Covariances:
##                   Estimate  Std.Err  z-value  P(>|z|)
##   f1 ~~
##     f2                0.045    0.072    0.621    0.535
##
## Thresholds:
##                   Estimate  Std.Err  z-value  P(>|z|)
##     Environment|t1   -1.447    0.094  -15.311    0.000
##     Environment|t2   -0.514    0.067   -7.729    0.000
##     Environment|t3    0.450    0.066    6.835    0.000
##     Technology|t1    -1.686    0.110  -15.343    0.000
##     Technology|t2    -0.589    0.068   -8.715    0.000
##     Technology|t3     0.464    0.066    7.034    0.000
##     Industry|t1      -1.951    0.134  -14.547    0.000
##     Industry|t2      -1.056    0.078  -13.531    0.000
##     Industry|t3       0.219    0.064    3.428    0.001
##     Comfort|t1       -2.234    0.172  -12.960    0.000
##     Comfort|t2       -1.314    0.088  -14.952    0.000
##     Comfort|t3        0.749    0.070   10.652    0.000
##     Work|t1          -1.377    0.091  -15.155    0.000
##     Work|t2          -0.428    0.066   -6.536    0.000
##     Work|t3           1.079    0.079   13.693    0.000
##     Future|t1        -1.803    0.119  -15.091    0.000
##     Future|t2        -0.774    0.071  -10.937    0.000
##     Future|t3         0.691    0.069    9.981    0.000
##     Benefit|t1       -1.611    0.105  -15.415    0.000
##     Benefit|t2       -0.500    0.066   -7.531    0.000
##     Benefit|t3        0.845    0.072   11.685    0.000
##
## Variances:
##                   Estimate  Std.Err  z-value  P(>|z|)
##    .Environment       0.567
##    .Technology        0.518
##    .Industry          0.601
##    .Comfort           0.724
##    .Work              0.730
##    .Future            0.402
##    .Benefit           0.716
##     f1                1.000
##     f2                1.000
## looking at summary of fit,
```

```
## - again the chisquare is rejected but if we look at CFI and TLI, the TLI is
##   better than previous' one, but still lower than 0.95
## - the mean square error is on the boundary being 0.081 (usually this is
##   acceptable if < 0.08)
## - it's not the best model we can have
##
## but looking at the solutions anyway: for each factor we have the parameter
## estimates and all of them are significantly related to the factor
## (significant p).
##
## the correlation between the two factor in this case is not significant, so
## this could be another warning on the fact that people replied on different
## items differently just because items were reverted. We could expect these
## two factors to be related/correlated.
##
## the treshold are all significant and estimation of the variances: variances
## of the latent variable are 1 because we fitted the solution with variance=1
## (credo con std.lv=TRUE) in order to have standardized latent variables
##
## we could check the corrispondance of loadings but actually this won't work
## since this is a confirmatory model (with coefficient set to 0) while the
## other is an exploratory model


## se ci sono cose che non tornano è perché potremmo non avere fatto
## l'inversione btw

## ------------------------------------
## Example attitude toward environment
## ------------------------------------
## (Paused) A: 00:31:30 / 00:34:54 (90%)
## manca dataset
```

Chapter 6

Latent class models with polytomous manifest variables

Both (single) latent variable and manifest variables are categorical with multiple categories, I guess

6.1 Introduction

6.1.1 Unordered polytomous manifest variables

If the observerd variables are categorical with unordered response, as before,

- when x_i has c_i categories we code

$$x_i(s) = \begin{cases} 1 & \text{if the response falls in category } s \\ 0 & otherwise \end{cases}$$

- c_i vector is denoted by $\mathbf{x}_i$ and $\sum_s x_i(s) = 1$
- full response pattern $\mathbf{x} = (\mathbf{x}_1', \mathbf{x}_2', \ldots, \mathbf{x}_p')$
- $\pi_{ij}(s)$ probability of an individual in class j (on latent variable y) is in category s on observed variable i:

We express marginal distribution of $\mathbf{x}$ as the sum on the all (latent) categories of the prior probability η_j times the likelihood of x given y:

$$f(\mathbf{x}) = \sum_{j=0}^{K-1} \eta_j \prod_{i=1}^{p} \prod_{s=0}^{c_i-1} \pi_{ij}(s)^{x_i(s)}$$

this is similar to the binary case but only difference here is the multinomial $\pi_{ij}(s)^{x_i(s)}$.
With bayes we can have the posterior probability for class j given $\mathbf{x}$ with:

$$h(j|\mathbf{x}) = \frac{\eta_j \prod_{i=1}^{p} \prod_{s=0}^{c_i-1} \pi_{ij}(s)^{x_i(s)}}{f(\mathbf{x})}$$

with η_j prior.

6.1.2 Ordered polytomous manifest variables

If the observed variables are ordinal what changes? To preserve the ordinality property of the variable x_i, as before we define cumulative probabilities (this time going from s upward)

$$\gamma_{ijs} = \pi_{ij}(s) + \pi_{ij}(s+1) + \ldots + \pi_{ij}(c_i - 1)$$

to rewrite our marginal with them:

$$\begin{aligned} f(\mathbf{x}) &= \sum_{j=0}^{K-1} \eta_j \prod_{i=1}^{p} \prod_{s=0}^{c_i-1} \pi_{ij}(s)^{x_i(s)} \\ &= \sum_{j=0}^{K-1} \eta_j \prod_{i=1}^{p} \prod_{s=0}^{c_i-1} (\gamma_{ij}(s) - \gamma_{ij}(s+1))^{x_i(s)} \end{aligned}$$

6.2 ML estimation

EM algorithm is used here as well: we maximize the uncomplete loglikelihood using two sets of constraints.
The uncomplete loglikelihood is written as

$$L = \sum_{h=1}^{n} \log f(\mathbf{x}_h)$$

The two sets of constraints regards the various probability:

$$\begin{aligned} \sum \eta_j &= 1 \\ \sum_{s=0}^{c_i-1} \pi_{ijs} &= 1 \end{aligned}$$

The function to be maximised is (constrainted with lagrange) the likelihood where we add constraints:

$$\phi = L + \gamma \sum_{j=0}^{K-1} \eta_j + \sum_{j=0}^{K-1} \sum_{i=1}^{p} \beta_{ij} \sum_{s=0}^{c_i-1} \pi_{ijs}$$

After some steps we get the estimator (same formula got for latent class binary data, that is prior distribution for each latent class is an average of the posterior distribution and the probability for each category is an expectede value of an x weighted by posterior distribution)

$$\begin{aligned} \hat{\eta}_j &= \frac{1}{n} \sum_{h=1}^{n} h(j|\mathbf{x}_h) \\ \hat{\pi}_{ijs} &= \frac{\sum_{h=1}^{n} h(j|\mathbf{x}_h) x_{ihs}}{n\hat{\eta}_j} \end{aligned}$$

6.3 Identifiability

One problem is that with many parameters the model could be not identifiable,

- that is more than one point in the parameter space can yields the same likelihood
- we could end up in a local maxima point so we need to check different initialization point

In particular:

- if we consider the contingency table with $c_1 \times c_2 \times \ldots \times c_p$ cells, the independent probabilities are $\prod_{i=1}^{p} c_i - 1$.
  This is the amount/number of information we have
- the number of the model parameters to be estimated are

$$K \sum_{i=1}^{p} (c_i - 1) + K - 1$$

  where $K - 1$ are the number of prior probability (with K number of category of latent variable) and $K \sum_{i=1}^{p} (c_i - 1)$ are the number of conditional probabilities. These are a lot of parameters, this is a drawback of latent class analysis
- the model is identified if the amount of information is greater than parameters to be stimated.
  OTOH if the model parameters are greater than the independent cell probabilities then we have identification problem. Particularly the model is unidentifiable if

$$\prod_{i=1}^{p} c_i - 1 < K \sum_{i=1}^{p} (c_i - 1) + K - 1$$

  Note that this is a necessary but not sufficient condition (probabilities are subject to other constraints) $\Rightarrow$ local identifiability.

6.4 Example

- data from the 1975 General Household Survey with a sample size of 1472
- Clogg (1979) who fitted a three-class latent variable model to it
- three questions about degree of satisfaction with family (F), hobbies (H) and residence (R)
- Answers to each question were classified as *low*, *medium* or *high* (so ordinal data)

So

Class		I	II	III
Family	low	0.48	0.06	0.03
	medium	0.28	0.36	0.02
	high	0.24	0.58	0.95
Hobbies	low	0.85	0.18	0.07
	medium	0.00	0.50	0.14
	high	0.15	0.32	0.79
Residence	low	0.53	0.21	0.06
	medium	0.31	0.53	0.22
	high	0.16	0.26	0.72
	$\hat{\eta}_j$	0.04	0.41	0.55

Figure 6.1: Parameter estimates $\hat{\pi}$s of three-class latent variable models for life satisfaction data

- clogg fitted a model with 3 classes for the latent variable [1]. The parameter estimates are given in Table 6.1. this gives the probability of responding in each category for each item (π_{ijs}).
  The first class of latent variable is the class with lower (prior) percentage of people (0.04) while the third class with higher percentage (0.55).
  How are these class profiled:
  - the first class is composed by individual with low satisfaction higher in all domains (0.48 in family, 0.85 in hobbies and 0.53 in residence)
  - the second class is mid tier
  - the third class has the highest satisfactions (0.95 in family, 0.79 in hobbies, 0.72 in residence)

  This is the way we interpret latent class model with ordinary data
- goodness of fit is good:
  - Observed and expected frequencies are in table 6.2: the response pattern are in row so for example the first is 0 satisfaction for all the domain (F/H/R). In general observed and expected frequencies are very similar.
  - value of $\chi^2 = 2.36$ which, with $27-18-2-1 = 6$ degrees of freedom, indicates an extremely good fit.
- resto non commentato

[1] to choose the number of classes for the latent variable we can go from 2 upward using BIC as a criteriod

Table 6.9 Observed and expected frequencies for life satisfaction data ($n = 1472$).

Response pattern			Observed frequency	Expected frequency	Response pattern			Observed frequency	Expected frequency
F	H	R			F	H	R		
0	0	0	15	14.1	2	1	0	45	43.3
0	0	1	11	11.3	2	1	1	117	116.4
0	0	2	7	6.8	2	1	2	126	123.0
1	0	0	16	15.6	0	2	0	5	5.7
1	0	1	26	25.3	0	2	1	14	11.7
1	0	2	12	13.2	0	2	2	16	17.5
2	0	0	23	22.8	1	2	0	18	16.6
2	0	1	49	48.9	1	2	1	38	40.4
2	0	2	54	57.0	1	2	2	27	27.8
0	1	0	3	4.0	2	2	0	64	61.2
0	1	1	12	10.3	2	2	1	191	193.6
0	1	2	5	7.2	2	2	2	466	466.9
1	1	0	23	23.0					
1	1	1	58	58.1	Total			1472	1472.0
1	1	2	31	29.9					

Figure 6.2: Observed and expected frequencies

Table 6.10 Body satisfaction: log-likelihood, BIC and number of parameters.

Model	Log-likelihood	BIC	Number of parameters
2-Class	−1747.5965	3686.6454	36
3-Class	−1643.7688	3521.5349	44
4-Class	−1593.8066	3464.1554	52
5-Class	−1555.1662	3429.4195	60
6-Class	−1534.3650	3430.3622	68

Figure 6.3: Body satisfaction: log-likelihood, BIC and number of parameters.

Table 6.11 Body satisfaction: estimated category and class probabilities.

Item	Class 1	Class 2	Class 3	Class 4	Class 5
Thighs					
1	0.0000	0.0088	0.1422	0.8251	0.0000
2	0.0157	0.3815	0.7355	0.1737	0.0000
3	0.2883	0.5191	0.1197	0.0012	0.0066
4	0.6769	0.0904	0.0025	0.0000	0.5135
5	0.0191	0.0002	0.0000	0.0000	0.4799
Body build					
1	0.0000	0.0001	0.0101	0.3690	0.0000
2	0.0095	0.0546	0.5189	0.6154	0.0000
3	0.1704	0.3627	0.4005	0.0155	0.0005
4	0.6902	0.5446	0.0698	0.0001	0.1011
5	0.1299	0.0380	0.0006	0.0000	0.8984
Buttocks					
1	0.0003	0.0128	0.1331	0.7734	0.0000
2	0.0370	0.2846	0.6143	0.2213	0.0014
3	0.3341	0.5264	0.2366	0.0053	0.0624
4	0.5240	0.1692	0.0158	0.0000	0.4755
5	0.1045	0.0069	0.0001	0.0000	0.4606
Hips					
1	0.0000	0.0017	0.1049	0.8794	0.0000
2	0.0166	0.1654	0.7184	0.1200	0.0001
3	0.2805	0.5702	0.1714	0.0006	0.0118
4	0.6162	0.2554	0.0053	0.0000	0.3439
5	0.0868	0.0073	0.0000	0.0000	0.6443
Legs					
1	0.0001	0.0090	0.0391	0.8309	0.0000
2	0.0253	0.3410	0.5891	0.1678	0.0010
3	0.2396	0.4700	0.3233	0.0012	0.0429
4	0.6131	0.1750	0.0479	0.0000	0.5013
5	0.1219	0.0051	0.0006	0.0000	0.4547
Figure					
1	0.0000	0.0012	0.0538	0.5476	0.0000
2	0.0043	0.0815	0.5169	0.4272	0.0000
3	0.1251	0.4241	0.3706	0.0249	0.0000
4	0.8102	0.4869	0.0586	0.0003	0.0321
5	0.0604	0.0064	0.0001	0.0000	0.9679
Weight					
1	0.0011	0.0236	0.1682	0.4275	0.0001
2	0.0239	0.1600	0.3864	0.4055	0.0033
3	0.2021	0.4211	0.3445	0.1493	0.0687
4	0.5372	0.3479	0.0964	0.0173	0.4472
5	0.2358	0.0475	0.0045	0.0003	0.4808
$\hat{\eta}_j$	0.3316	0.3269	0.1959	0.0875	0.0581

Figure 6.4: Body satisfaction: estimated category and class probabilities